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TECHNICAL APPENDICES

VERSION 6.2

INTRODUCTION

Here I present the living master record of Stack Theory associated with my thesis "How To Build Conscious Machines", and my papers. It is primarily a theory of intelligence and consciousness. This document contains mathematical proofs and experimental results for generalisation-optimal learning. These results have implications for a variety of physical and philosophical topics from rigorous formal notions of artificial general intelligence, to theoretical neurobiology and philosophy of mind. While some of this material has appeared in other publications, substantial improvements have been made to both the formalism and the proofs, along with error corrections. Hence it is both a self-contained paper and a compilation of material.

1. Everything is of course in the main thesis[1] to which this file is an appendix. This includes results like the novelty selection model behind *Why Is Anything Alive?* and The Temporal Gap which appeared in the thesis, weren't published as separate papers before then. Many of the other core Stack Theory ideas like weakness were introduced and developed over many of my earliest papers.

2. The relations between ethics, symbol construction and generalisation-optimal learning were first explored in my earliest papers[2]. That was then combined with compression to offer an explanation for the Fermi paradox in[3], and to offer a specification of an artificial scientist which became a target for this thesis[4].

[1] Bennett, M. T. (2025b). *How To Build Conscious Machines*. PhD thesis, The Australian National University

[2] Bennett, M. T. (2022c). Symbol emergence and the solutions to any task. In *Artificial General Intelligence*, Lecture Notes in Computer Science, pages 30–40, Cham. Springer; and Bennett, M. T. and Maruyama, Y. (2022b). Philosophical specification of empathetic ethical artificial intelligence. *IEEE Transactions on Cognitive and Developmental Systems*, 14(2):292–300

[3] Bennett, M. T. (2022a). Compression, the fermi paradox and artificial super-intelligence. In *Artificial General Intelligence*. Springer Nature

[4] Bennett, M. T. and Maruyama, Y. (2022a). The artificial scientist: Logicist, emergentist, and universalist approaches to artificial general intelligence. In *Artificial General Intelligence*, Lecture Notes in Computer Science, pages 45–54, Cham. Springer

3. Then I first formalised weakness and ran experiments[5]. This is what led me to conclude weakness and simplicity are distinct, adopting a position against Ockham's Razor.

4. The subsequent 2023 AGI conference papers began my main thrust into generalisation-optimal learning[6], which produces a hierarchy of selves that explain aspects of consciousness[7], and linguistic meaning[8].

5. In 2025 I expanded this program to formalise an upper bound on intelligence in terms of adaptability[9], examine the difference between computational and biological machinery[10], and to establish why Ockham's Razor *sort of* works but is certainly not optimal[11].

6. The origins and nature of consciousness were addressed in[12].

7. I then spent some time writing a survey for my thesis chapter of the same name[13], criticised the orthogonality thesis in[14].

8. Other papers such as[15] served to inspire or refine these various ideas.

[5] Bennett, M. T. (2022b). Computable Artificial General Intelligence. *Under Review*

[6] Bennett, M. T. (2023c). The optimal choice of hypothesis is the weakest, not the shortest. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 42–51. Springer

[7] Bennett, M. T. (2023a). Emergent causality and the foundation of consciousness. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 52–61. Springer. OUCI metadata page: https://ouci.dntb.gov.ua/en/works/7BoXJMW4/

[8] Bennett, M. T. (2023b). On the computation of meaning, language models and incomprehensible horrors. In *Artificial General Intelligence*. Springer Nature

[9] Bennett, M. T. (2024b). Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

[10] Bennett, M. T. (2024a). Are biological systems more intelligent than artificial intelligence? In press, 2026, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds

[11] Bennett, M. T. (2024c). Is complexity an illusion? In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

[12] Bennett, M. T., Welsh, S., and Ciaunica, A. (2024). *Why Is Anything Conscious?* Preprint

[13] Bennett, M. T. (2025e). What is artificial general intelligence? In *Artificial General Intelligence*, volume 16057 of *Lecture Notes in Computer Science*, pages 30–42. Springer. Originally circulated as "What the F*ck Is Artificial General Intelligence?"

[14] Bennett, M. T. (2025c). Lies, damned lies, and the orthogonality thesis. *Preprint*

[15] Ganapathy, A. and Bennett, M. T. (2021). Cybernetics and the future of work. In *2021 IEEE 21CW*; Bennett, M. T. (2025d). Optimal policy is weakest policy. In *Artificial General Intelligence*, volume 16057 of *Lecture Notes in Computer Science*, pages 43–56. Springer; and Bennett, M. T. (2025a). A formal theory of optimal learning with experimental results. *IJCAI*, pages 10967–10968

9. The Temporal Gap was properly formalised in the workshop paper[16].

10. Likewise another workshop paper added a few formal results for the hierarchy of selves[17].

11. Spacetime and capacity bounds are formally proven in[18].

12. Time and identity issues for language model agents are explored in[19].

13. A more aggressive warning label about language model agents is[20].

14. Stack Theory was extended to cognition spaces and hybrid morphospaces in[21].

[16] Bennett, M. T. (2026a). A mind cannot be smeared across time. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

[17] Bennett, M. T. (2026b). No selves, no consciousness. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

[18] Bennett, M. T. (2026d). Spacetime bounds on consciousness

[19] Elija Perrier, M. T. B. (2026). Time, identity and consciousness in language model agents. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

[20] Perrier, E. and Bennett, M. T. (2025). Position: Stop acting like language model agents are normal agents

[21] Solé, R., Seoane, L. F., Pla-Mauri, J., Bennett, M. T., Hochberg, M. E., and Levin, M. (2026). Cognition spaces: natural, artificial, and hybrid

STACK THEORY

CORE DEFINITIONS

Definitions 1 to 4 are taken from this reference[22].

[22] Bennett, M. T. (2024b). Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

Definition 1 (Environment, programs, and vocabularies)**.**
An environment *is a nonempty set* Φ *of mutually exclusive states. It is assumed infinite by default. A* (declarative) program *is any set of states* $p \subseteq \Phi$*. For any set X, write* 2^X *for the set of all subsets of X. In particular, let* $\mathcal{P} := 2^\Phi$ *denote the set of all programs (which may be understood as* potential *facts).*

A fact *about a state is a program that contains that state. An* aspect *of the environment is a set x of programs such that* $x \in \{ l \subseteq \mathcal{P} \mid \bigcap_{p \in l} p \neq \emptyset \}$*. I say an aspect exists given a state if it is a set of only facts. An aspect is something that can exist. In contrast, a* vocabulary *is any set of programs* $\mathfrak{v} \subseteq \mathcal{P}$*. Unless specified otherwise, vocabularies and aspects are by default assumed finite.*

Intuition. Think of Φ as the space of all conceivable or possible objective realities. In physics terms, think of each state as a **microstate** for the whole universe. Each state is one reality, and is mutually exclusive with other states. However I don't assign states any specific ontological content. Rather, each state represents a change or difference of reality from all the other states. Because of the mutual exclusivity this equates time with change, so a state is just a point of change and a possible timeline or world is just a sequence of states. A program is just a constraint (a property that can hold in some of those configurations). An aspect of the environment is anything that can exist. A vocabulary represents a set of programs from which aspects are formed, namely the positions a body can occupy. A vocabulary represents what something that exists can do, but all those things don't necessarily exist together. I begin with these definitions because they are general enough to describe every possible world, and so I can find invariants among those worlds.

Definition 2 (Abstraction layer, statements, and truth sets).
A vocabulary $\mathfrak{v}$ induces an embodied language

$$L_{\mathfrak{v}} = \{\, l \subseteq \mathfrak{v} \mid \bigcap_{p \in l} p \neq \emptyset \,\}.$$

*Elements $l \in L_{\mathfrak{v}}$ are called **statements**. Unless specified otherwise, statements are assumed finite. This is because, unless stated otherwise, the vocabulary is finite.*

*Statements behave like conjunctions of vocabulary programs. The **truth set** of l is*

$$T(l) := \bigcap_{p \in l} p \subseteq \Phi,$$

*so l is **true** at $\phi \in \Phi$ iff $\phi \in T(l)$, with the convention $T(\emptyset) := \Phi$.*

Intuition. A system is represented by an abstraction layer. A statement l is a physical configuration the system can be in. For example, my body makes a statement when I raise my arm. The truth set $T(l)$ of l is the region of Φ where every ingredient in l is true, so you can think of a statement made in an abstraction layer as a constraint on environment states.

Definition 3 (Completions, extensions, and equivalence).
*A **completion** of a statement $x \in L_{\mathfrak{v}}$ is any statement $y \in L_{\mathfrak{v}}$ such that $x \subseteq y$. The **extension** of x is the set of its completions:*

$$\mathrm{Ext}(x) = \{\, y \in L_{\mathfrak{v}} \mid x \subseteq y \,\}.$$

*For $X \subseteq L_{\mathfrak{v}}$, write $\mathrm{Ext}(X) := \bigcup_{x \in X} \mathrm{Ext}(x)$. I say x and y are **equivalent** iff $\mathrm{Ext}(x) = \mathrm{Ext}(y)$ (they support exactly the same refinements within the vocabulary).*

Intuition. Completions add further constraints while staying compatible with what was already said. A coarse description can be completed by adding detail. What this means is if I raise my arm, this constrains what my body can do next. If I then raise my other arm, I add a further constraint. There are degrees of freedom in what I don't specify. Raising my arm is a more coarse-grained description than raising both my arms. The extension of a statement is the set of ways the system could refine that statement later, given its vocabulary. I formalised this because extensions are my bookkeeping device for how much uncertainty one has not ruled out. Weakness, generalisation probability, and stack bottlenecks are all functions of $\mathrm{Ext}(x)$. So to summarise what I have so far.

- Φ is an environment.

- $\mathcal{P} = 2^{\Phi}$ is the set of programs. In physical terms, think of a program as a property of the microstates it contains. In epistemological terms, a program is a potential fact, and it is true of the states it contains.
- $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary. It is finite by default, and it is only otherwise when specifically stated for the purpose of proving various bounds.
- $L_{\mathfrak{v}} \subsetneq 2^{\mathfrak{v}}$ is the satisfiable fragment (in general).

In any real practical setting, $L_{\mathfrak{v}}$ is an extremely small subset of $2^{\mathfrak{v}}$ because $L_{\mathfrak{v}}$ represents the configurations a body can be in, and while a body might do many things it usually cannot do them all at the same time. Most statements tend to be mutually exclusive. The extension $\text{Ext}(x)$ of a statement x is all the things the body can do while still doing x.

Definition 4 (Weakness). *Fix a vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ and a statement $l \in L_{\mathfrak{v}}$. Recall (Definition 3) that the* extension *of l is*

$$\text{Ext}(l) = \{ y \in L_{\mathfrak{v}} \mid l \subseteq y \}.$$

*The **weakness** of l is the cardinality of its extension:*

$$w(l) := |\text{Ext}(l)|.$$

*When $w(l_1) > w(l_2)$ I say that l_1 is **weaker** than l_2 **in the weakness sense**.*

Remark 1 (Weakness does not imply inclusion). *Weakness compares only* cardinalities. *In general, $w(l_1) \geq w(l_2)$ does* not *imply $l_1 \subseteq l_2$, nor does it imply* $\text{Ext}(l_1) \supseteq \text{Ext}(l_2)$. *However, if $l_1 \subseteq l_2$ (i.e. l_2 is a completion of l_1), then* $\text{Ext}(l_2) \subseteq \text{Ext}(l_1)$ *and hence $w(l_1) \geq w(l_2)$.*

Intuition. Weakness $w(l)$ counts how many distinct completions remain open after asserting l in the system's language. Put plainly, it measures how much stuff the body can do and still do l. It is degrees of freedom that remain open. It measures how non-committal the statement is inside that vocabulary. A larger extension means there are more ways to add detail later without contradiction, so the statement is harder to falsify by future evidence. Weakness compares sizes, not inclusion, so you can have two statements where one generalises better even though neither contains the other.

Remark 2 (Implementation warning). *Be careful presenting these definitions to large language models (LLMs). $L_{\mathfrak{v}}$ represents everything the body can do. LLMs have a tendency to set $L_{\mathfrak{v}}$ to $2^{\mathfrak{v}}$, which is absurd. It* *is like claiming the body can do everything, everywhere all at once. LLMs appear to do this because it collapses weakness into a pure length proxy, which conforms to popular misconception that simplicity is the best proxy. In that incorrect setting, if $\mathfrak{v}$ is finite, then for any $\pi \subseteq \mathfrak{v}$ I would have $w(\pi) = |\text{Ext}(\pi)| = 2^{|\mathfrak{v}|-|\pi|}$, so weakness would depend only on the number of ingredients in π. This destroys the intended semantic dependence on $T(\cdot)$. In any realistic scenario, $L_{\mathfrak{v}}$ is a very small subset of $2^{\mathfrak{v}}$ because a body can do many things, and most of them not at the same time. Statements are often mutually exclusive.*

Definition 5 (Abstractor function)**.** *An* ***abstractor function*** *f is defined on pairs $(\mathfrak{v}, l)$ where $\mathfrak{v} \subseteq \mathcal{P}$ is a vocabulary (finite unless stated otherwise) and $l \in L_{\mathfrak{v}}$ is a statement. It returns the (possibly infinite) sub-vocabulary*

$$f(\mathfrak{v}, l) \ := \ \{ \, T(o) \mid o \in \text{Ext}(l) \, \} \ \subseteq \ \mathcal{P}.$$

In words, $f(\mathfrak{v}, l)$ is the set of truth sets of all completions of l. If $\mathfrak{v}$ is finite, then $L_{\mathfrak{v}}$ (hence $\text{Ext}(l)$) is finite, so $f(\mathfrak{v}, l)$ is finite. If $\mathfrak{v}$ is infinite, $f(\mathfrak{v}, l)$ may be infinite.

Intuition. The abstractor takes a policy statement l and lists every completion o it still allows. It maps each completion to its truth set $T(o)$ and uses those truth sets as the next vocabulary. This is the formal bridge from what you leave open to what you can represent next. The next layer's primitives are the macro regularities that survive the current policy's constraints.

SPACE AND TIME

This module integrates and corrects material from *A Mind Cannot Be Smeared Across Time*[23] and *Spacetime Bounds on Consciousness*[24]. The main correction is semantic. A *windowing map* should pick out objective time indices. The induced *window trajectory* should be the sequence of states on those indices. That separates the combinatorics of time from the ontology of states.

[23] Bennett, M. T. (2026a). A mind cannot be smeared across time. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

[24] Bennett, M. T. (2026d). Spacetime bounds on consciousness

Definition 6 (Objective time and trajectories)**.** *Objective time is indexed by $u \in \mathbb{N}$. An* ***objective trajectory*** *is a function $\tau : \mathbb{N} \to \Phi$ such that $\tau(u+1) \neq \tau(u)$ for all u.*

Notation. $\mathbb{N}_{>0} = \mathbb{N} \setminus \{0\}$ denotes the positive integers.

Intuition. Objective time is just the counter on physical transitions. A trajectory is one possible history of the environment. A possible *timeline*. The transition from environment state to environment state preserves some aspects of the environment, and destroys others. The condition $\tau(u+1) \neq \tau(u)$ matches the intended reading of a tick as a change.

Definition 7 (Encoding map)**.** *Fix a vocabulary $\mathfrak{v} \subseteq \mathcal{P}$. Define the **encoding map** $\mathrm{Enc}_{\mathfrak{v}} : \Phi \to L_{\mathfrak{v}}$ by*

$$\mathrm{Enc}_{\mathfrak{v}}(\phi) \;:=\; \{\, p \in \mathfrak{v} \mid \phi \in p \,\}.$$

Intuition. $\mathrm{Enc}_{\mathfrak{v}}(\phi)$ is the complete list of vocabulary programs that are true at ϕ. It is the most informative statement the vocabulary can make about that state.

Lemma 1 (Encoding is the maximal true statement)**.** *For every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(\phi)$ is true at ϕ. If $l \in L_{\mathfrak{v}}$ is also true at ϕ, then $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$.*

Proof sketch. Every ingredient $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$ contains ϕ, so $\phi \in T(\mathrm{Enc}_{\mathfrak{v}}(\phi))$. If l is true at ϕ, then each ingredient $p \in l$ must contain ϕ, hence $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$[25].

[25] Full proof is in Theorems and Proofs.

Definition 8 (Realised encoding set)**.** *Fix a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ on environment Φ. Define the **realised encoding set** as the image of the encoding map*

$$\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}}) \;:=\; \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in \Phi\} \;\subseteq\; L_{\mathfrak{v}}.$$

Intuition. $\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}})$ is the set of distinct maximal descriptions that the vocabulary actually realises in the environment. If states are microstates, then these maximal descriptions are the maximally specific macrostates that abstraction layer can express.[26]

[26] $\mathrm{Im}(\mathrm{Enc}_v) \subseteq L_v$, but the reverse inclusion can fail regardless of whether Φ is infinite.

Why the inclusion holds one way. For any $\varphi \in \Phi$, $\mathrm{Enc}_v(\varphi)$ is a statement in L_v because every program in it contains φ, so the intersection is non-empty (Lemma 1).

Why it can fail the other way. Here *maximal* means maximal with respect to set inclusion among true statements at a given state: $\mathrm{Enc}_v(\varphi)$ is the largest statement in L_v that is true at φ, in the sense that every other true statement at φ is a subset of it (Lemma 1). No program can be added to it while keeping it true. $\mathrm{Im}(\mathrm{Enc}_v)$ contains only *maximal* true statements—the fullest description the vocabulary can make at each state. But L_v contains *all* satisfiable subsets of v, including non-maximal ones.

Counterexample. A *witness* is simply a specific element whose existence demonstrates a claim. Here, a witness $\varphi \in p$ is a concrete state certifying that p is non-empty and that $\{p\}$ is satisfiable. Let $v = \{p, q\}$ with $p \subsetneq q$ (both non-empty). Then $\{p\} \in L_v$ since $p \neq \emptyset$. But for any witness $\varphi \in p$, since $p \subset q$ I have $\varphi \in q$ too, so $\mathrm{Enc}_v(\varphi) = \{p, q\} \neq \{p\}$. And for $\varphi \in q \setminus p$, $\mathrm{Enc}_v(\varphi) = \{q\}$. So $\{p\}$ is never realised as an encoding, and $L_v \neq \mathrm{Im}(\mathrm{Enc}_v)$.

Definition 9 (Encoding capacity)**.** *For a finite vocabulary $\mathfrak{v}$, define its **encoding capacity** in bits by*

$$\mathsf{I}(\mathfrak{v}) \;:=\; \log_2 \big|\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}})\big|.$$

Intuition. $\mathsf{I}(\mathfrak{v})$ is the log count of distinct realised encodings. It is a hard ceiling on how many cases that vocabulary can represent. Encoding capacity is needed for the spacetime bound results in[27].

Definition 10 (Patch capacity ceiling)**.** *A bounded physical region $\mathcal{R}$ admits at most $N_{\max}(\mathcal{R})$ reliably distinguishable internal states. Define its bit ceiling as*

$$\mathsf{I}_{\max}(\mathcal{R}) \;:=\; \log_2 N_{\max}(\mathcal{R}).$$

A vocabulary is realisable inside $\mathcal{R}$ only if $\mathsf{I}(\mathfrak{v}) \leq \mathsf{I}_{\max}(\mathcal{R})$.

Intuition. If the patch cannot support enough distinguishable internal states, then the joint description you want is not physically instantiable there. More computation does not create more distinguishable memory states.

[27] Bennett, M. T.

Definition 11 (Join vocabulary)**.** *For vocabularies $\mathfrak{v}_1$ and $\mathfrak{v}_2$ on the same environment, define the join*

$$\mathfrak{v}_1 \vee \mathfrak{v}_2 \;:=\; \mathfrak{v}_1 \cup \mathfrak{v}_2.$$

Lemma 2 (Join capacity is subadditive)**.** *For finite vocabularies $\mathfrak{v}_1$ and $\mathfrak{v}_2$,*

$$\mathsf{I}(\mathfrak{v}_1 \vee \mathfrak{v}_2) \leq \mathsf{I}(\mathfrak{v}_1) + \mathsf{I}(\mathfrak{v}_2).$$

Proof. For $c \in \mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1 \vee \mathfrak{v}_2})$ define

$$r(c) := (c \cap \mathfrak{v}_1, c \cap \mathfrak{v}_2) \in \mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1}) \times \mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2}).$$

The map r is injective. So

$$\left|\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1 \vee \mathfrak{v}_2})\right| \leq \left|\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_1})\right| \left|\mathrm{Im}(\mathrm{Enc}_{\mathfrak{v}_2})\right|$$

and taking $\log_2$ gives the result. □

Intuition. The join cannot have more distinct encodings than the number of pairs of encodings its parts can realise. So joint descriptions can overflow a finite patch even when each partial description fits on its own.

Lemma 3 (Decoder keys are hidden entropy)**.** *Let $\mathcal{M}$ be a* ***finite*** *microstate*[28] *set with $|\mathcal{M}| = 2^S$, where S is the number of bits needed to index the microstate set. Let $\mathcal{D}$ be a finite family of operationally distinct decoders. For each $D \in \mathcal{D}$, let $\mathcal{M}^{(D)} \subseteq \mathcal{M}$ be the set of microstates on which D succeeds uniformly at some fixed accuracy. Assume a uniform coverage bound*

$$|\mathcal{M}^{(D)}| \leq d_{\max} \qquad \textit{for every } D \in \mathcal{D}.$$

If the protocol must succeed for every microstate, meaning $\bigcup_{D \in \mathcal{D}} \mathcal{M}^{(D)} = \mathcal{M}$, then

$$|\mathcal{D}| \geq \frac{|\mathcal{M}|}{d_{\max}} = \frac{2^S}{d_{\max}}.$$

Therefore any control register that must select among $\mathcal{D}$ at verification time must support at least $|\mathcal{D}|$ reliably distinguishable control states. Equivalently any key vocabulary $\mathfrak{v}_K$ representing that register must satisfy

$$\mathsf{I}(\mathfrak{v}_K) \geq \log_2 |\mathcal{D}| \geq S - \log_2 d_{\max}.$$

Proof. The covering condition gives

$$|\mathcal{M}| \leq \sum_{D \in \mathcal{D}} |\mathcal{M}^{(D)}| \leq |\mathcal{D}|\, d_{\max}.$$

Rearrange to get the lower bound on $|\mathcal{D}|$. A selector register that can implement any $D \in \mathcal{D}$ needs at least one reliably distinguishable control state per operationally distinct decoder. So the register has at least $|\mathcal{D}|$ internal states and any representing vocabulary satisfies $\mathsf{I}(\mathfrak{v}_K) \geq \log_2 |\mathcal{D}|$. □

[28] $\mathcal{M}$ is finite meaning it is not a complete description of the universe's microstate like $\phi \in \Phi$ would be.

Intuition. If each decoder label leaves at most $d_{\max}$ microstates unresolved, then the label must carry the rest of the micro information. So the instruction budget is a hidden entropy term.

Definition 12 (Induced layer trajectory and layer time)**.** *Fix a vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ and an objective trajectory $\tau : \mathbb{N} \to \Phi$. Define the induced* ***layer trajectory***

$$\tau^{(\mathfrak{v})}(u) \ := \ \mathrm{Enc}_{\mathfrak{v}}(\tau(u)) \ \in \ L_{\mathfrak{v}}.$$

Define the ***layer tick map*** $\kappa_{\mathfrak{v}} : \mathbb{N} \to \mathbb{N}$ *by* $\kappa_{\mathfrak{v}}(0) = 0$ *and, for all* u*,*

$$\kappa_{\mathfrak{v}}(u+1) \ := \ \kappa_{\mathfrak{v}}(u) + \begin{cases} 1 & \text{if } \tau^{(\mathfrak{v})}(u+1) \neq \tau^{(\mathfrak{v})}(u) \\ 0 & \text{otherwise.} \end{cases}$$

The induced ***layer time set*** *is the range*

$$\mathbb{N}^{\langle \mathfrak{v} \rangle} \ := \ \kappa_{\mathfrak{v}}(\mathbb{N}) \ \subseteq \ \mathbb{N}.$$

The induced ***layer time trajectory*** *is the map*

$$\tau^{\langle \mathfrak{v} \rangle} : \mathbb{N}^{\langle \mathfrak{v} \rangle} \to L_{\mathfrak{v}}, \qquad \tau^{\langle \mathfrak{v} \rangle}(t) \ := \ \tau^{(\mathfrak{v})}(\min\{u \in \mathbb{N} : \kappa_{\mathfrak{v}}(u) = t\}).$$

Intuition. $\tau^{(\mathfrak{v})}$ is what the vocabulary can actually see along the trajectory. If the environment changes in a way the vocabulary cannot distinguish, then $\tau^{(\mathfrak{v})}$ does not change. The tick map $\kappa_{\mathfrak{v}}$ counts only those changes. The induced layer time $\tau^{\langle \mathfrak{v} \rangle}$ is the compressed movie where you delete all repeated frames at that vocabulary. The range $\mathbb{N}^{\langle \mathfrak{v} \rangle}$ can be a strict subset of $\mathbb{N}$. If the encoding stops changing, then layer time stops too.

Proposition 1 (Layer time can be arbitrarily sparse)**.** *For any $M \in \mathbb{N}_{>0}$ there exist an environment Φ, a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, and an objective trajectory $\tau : \mathbb{N} \to \Phi$ such that the induced layer trajectory $\tau^{(\mathfrak{v})}$ changes at most once during the first M objective ticks.*

Proof sketch. Let $\Phi = \{\phi_0, \ldots, \phi_M\}$ and let $\mathfrak{v} = \{p\}$ with $p = \Phi$. Then $\mathrm{Enc}_{\mathfrak{v}}(\phi_i) = \{p\}$ for every i. Choose $\tau(u) = \phi_u$ for $u = 0, \ldots, M$ and extend arbitrarily. Then $\tau^{(\mathfrak{v})}$ is constant on the first M ticks[29].

[29] Full proof is in Theorems and Proofs.

Definition 13 (Window environments and sliding windows)**.** *Fix a horizon $\Delta \in \mathbb{N}_{>0}$. Define the* ***window environment***

$$\Phi_\Delta \ := \ \Phi^{\Delta+1}.$$

The ***sliding window map*** *is the function* $W_\Delta : \mathbb{N} \to 2^{\mathbb{N}}$ *given by*

$$W_\Delta(u) \ := \ \{u, u+1, \ldots, u+\Delta\}.$$

Given an objective trajectory $\tau : \mathbb{N} \to \Phi$, the induced ***window trajectory*** *is*

$$\tau_\Delta : \mathbb{N} \to \Phi_\Delta, \qquad \tau_\Delta(u) \ := \ \big(\tau(u), \tau(u+1), \ldots, \tau(u+\Delta)\big).$$

Intuition. A window is a short slice of objective time. The map $W_\Delta(u)$ selects the $(\Delta+1)$ consecutive indices starting at u, and $\tau_\Delta(u)$ records the corresponding $(\Delta+1)$-tuple of states. Successive windows shift by one objective tick, so they overlap. If you want to augment this definition with a stride s for whatever reason, just sample the window trajectory on the arithmetic progression $u = 0, s, 2s, \ldots$. However all Temporal Gap results below are pointwise in a window, so this stride choice never enters the proofs and so I removed it.

Definition 14 (Ingredient-wise occurrence and co-instantiation)**.** *Fix a horizon Δ and a window $\sigma = (\phi_0, \ldots, \phi_\Delta) \in \Phi_\Delta$. Let $l \in L_{\mathfrak{v}}$ be a statement.*

*I say that l is **ingredient-wise satisfied** (or **occurs**) on σ if*

$$\forall\, p \in l\ \exists\, k_p \leq \Delta \text{ such that } \phi_{k_p} \in p.$$

*I say that l is **co-instantiated** on σ if*

$$\exists\, k \leq \Delta \text{ such that } \phi_k \in T(l) = \bigcap_{p \in l} p.$$

Intuition. Ingredient-wise satisfaction is the weak condition. Each ingredient shows up somewhere in the window, but not necessarily at the same time. Co-instantiation is the strong condition. All ingredients are true at one and the same objective tick.

Definition 15 (Temporal lifts)**.** *Fix $\Delta \in \mathbb{N}_{>0}$ and write Φ_Δ for the window environment. For a program $p \subseteq \Phi$, define its **existential** and **universal** temporal lifts by*

$$\Diamond_\Delta p \ := \ \{\, \sigma \in \Phi_\Delta \mid \exists k \leq \Delta\ \sigma_k \in p \,\}, \qquad \Box_\Delta p \ := \ \{\, \sigma \in \Phi_\Delta \mid \forall k \leq \Delta\ \sigma_k \in p \,\}.$$

For a statement $l \in L_{\mathfrak{v}}$, define the lifted statement

$$\Diamond_\Delta l \ := \ \{\, \Diamond_\Delta p \mid p \in l \,\}, \qquad \Box_\Delta l \ := \ \{\, \Box_\Delta p \mid p \in l \,\}.$$

Intuition. $\Diamond_\Delta p$ means p is true at least once in the window. $\Box_\Delta p$ means p is true at every tick in the window. Lifting a whole statement l ingredient-wise makes a new statement in the window environment. This is the clean way to formalise the Temporal Gap as a quantifier order issue.

Satisfaction notation. If $\sigma \in \Phi_\Delta$ and $P \subseteq \Phi_\Delta$, write $\sigma \models P$ to mean $\sigma \in P$. In particular, $\sigma \models \Diamond_\Delta p$ means p holds at least once in σ, and $\sigma \models \Box_\Delta p$ means p holds at every position in σ.

Lemma 4 ($\Box_\Delta$ commutes with conjunction)**.** *For all statements $l \in L_{\mathfrak{v}}$,*

$$\Box_\Delta T(l) \ = \ T(\Box_\Delta l).$$

Proof sketch. Both sides say that every ingredient is true at every position in the window[30].

[30] Full proof is in Theorems and Proofs.

Lemma 5 ($\Diamond_\Delta$ does not commute with conjunction)**.** *For all statements $l \in L_{\mathfrak{v}}$,*

$$\Diamond_\Delta T(l) \;\subseteq\; T(\Diamond_\Delta l).$$

If $\Delta \geq 1$ and l has at least two ingredients, then the inclusion can be strict.

Proof sketch. Co-instantiation implies ingredient-wise occurrence, so the inclusion always holds. Strictness is witnessed by a two step window where the ingredients occur at different steps[31].

[31] Full proof is in Theorems and Proofs.

Definition 16 (Within-window persistence)**.** *Fix a trajectory $\tau : \mathbb{N} \to \Phi$ and a horizon $\Delta \in \mathbb{N}_{>0}$. A program $p \subseteq \Phi$ is (τ, Δ)-**persistent** if for every objective time $u \in \mathbb{N}$ and all $0 \leq k \leq k' \leq \Delta$,*

$$\tau(u+k) \in p \implies \tau(u+k') \in p.$$

Intuition. Persistence says that once an ingredient becomes true inside a window, it stays true until the window ends. Persistence closes the Temporal Gap. If the ingredients cannot wink in and out, then seeing each ingredient at least once forces a moment of co-instantiation.

Proposition 2 (Persistence collapses the Temporal Gap)**.** *Fix $\Delta \in \mathbb{N}_{>0}$ and a trajectory τ. Let $l \in L_{\mathfrak{v}}$ be a statement whose every ingredient $p \in l$ is (τ, Δ)-persistent. Then for every objective time $u \in \mathbb{N}$,*

$$\tau_\Delta(u) \models T(\Diamond_\Delta l) \quad\implies\quad \tau_\Delta(u) \models \Diamond_\Delta T(l).$$

Proof sketch. If each ingredient occurs somewhere in the window, let k be the latest of the first occurrence times. By persistence, every ingredient is still true at time k, so the whole conjunction is true at k[32].

[32] Full proof is in Theorems and Proofs.

Definition 17 (Contributor model)**.** *Let X be any set and $n \geq 1$. Define the **contributor environment***

$$\mathfrak{C} \;:=\; X \times 2^{[n]},$$

where $[n] = \{1, \dots, n\}$ and the second component $A \subseteq [n]$ records the set of active contributors. For each contributor $i \in [n]$, define the program

$$p_i \;:=\; \{(x, A) \in \mathfrak{C} : i \in A\}.$$

Let $l_n := \{p_1, \dots, p_n\}$, so $T(l_n) = \{(x, [n]) : x \in X\}$.

Intuition. The contributor model records which contributors are active at each time step. A conjunction that mentions many contributors can only be true when all of those contributors are active at the same time step. I use this to help turn the vague story about integration into a constraint.

Definition 18 (Concurrency capacity)**.** *An architecture has* ***concurrency capacity*** $c \in \{1, \ldots, n\}$ *if its admissible trajectories* $\tau : \mathbb{N} \to \Phi$ *satisfy*

$$|A_u| \le c \quad \textit{for all objective times } u,$$

where $\tau(u) = (x_u, A_u)$*.*

Intuition. Capacity c is the maximum number of contributors that can be active at one objective tick. If some grounded content requires more than c contributors at once, then co-instantiation is structurally impossible at that time scale.

Theorem 1 (Synchronous threshold for co-instantiation)**.** *Fix* $n \ge 2$ *and the conjunction* l_n *from Definition 17. If an architecture has concurrency capacity* $c < n$*, then along every admissible trajectory* τ *and for every horizon* Δ*,*

$$\tau_\Delta(u) \not\models \Diamond_\Delta T(l_n) \quad \textit{for all } u.$$

However, if each contributor $i \in [n]$ *is active at least once within some horizon* Δ *window meaning there exists* u *such that for every* i *there exists* $k_i \le \Delta$ *with* $i \in A_{u+k_i}$ *then*

$$\tau_\Delta(u) \models T(\Diamond_\Delta l_n).$$

So the architecture admits ingredient-wise satisfaction but forbids co-instantiation.

Proof sketch. If $c < n$, no objective state ever lies in $T(l_n)$, so no window can satisfy $\Diamond_\Delta T(l_n)$. If each ingredient occurs somewhere in the window, then the window satisfies $\Diamond_\Delta p_i$ for every i, hence it satisfies $T(\Diamond_\Delta l_n) = \bigcap_i \Diamond_\Delta p_i$[33].

[33] Full proof is in Theorems and Proofs.

SPACETIME BOUNDS ON CONSCIOUS UNITY

The above time semantics treat a moment as a windowed sequence and asks whether ingredients overlap inside that window. They do not yet ask whether the ingredients can *talk to each other* quickly enough to count as one causally unified event.

This is an issue for later when I discuss consciousness. A subjective conscious moment *feels* unified, which is to say it feels like it is co-instantiated in objective time, and that all the parts of a subjective moment connected together. This is called "unity". However a bunch of ingredients could all be co-instantiated in time and disconnected in space. So, to properly discuss questions of unity, we need to understand space as well as time, and formalise speed limits. Hence here is a spacetime bound adding one extra postulate which is needed to understand conscious unity[34]. A unified subjective moment is not only co-instantiated somewhere in its own window. It

[34] Bennett, M. T. (2026d). Spacetime bounds on consciousness

is also causally closed within that window. At minimum, the ingredients must exchange influence by two-way signalling along some integration architecture. If the substrate has a speed ceiling v, this turns unity into a size constraint.

Definition 19 (Window occurrence and co-instantiation for grounded moments)**.** *Fix a statement ℓ^m at some layer m and let $g^0 \in L_{\mathfrak{v}^0}$ be its grounding at the base layer, typically $g^0 = \mathrm{Ground}_{0 \leftarrow m}(\ell^m)$ as in Definition 39. For a trajectory τ, horizon Δ, and window start $u \in \mathbb{N}$, define*

$$\mathrm{Occur}_{W_\Delta}(\ell^m, \tau, u) \iff \tau_\Delta(u) \models T(\Diamond_\Delta g^0)$$

and

$$\mathrm{CoInst}_{W_\Delta}(\ell^m, \tau, u) \iff \tau_\Delta(u) \models \Diamond_\Delta T(g^0).$$

Intuition. $\mathrm{Occur}_{W_\Delta}$ is the weak condition. Every grounded ingredient shows up somewhere in the window. $\mathrm{CoInst}_{W_\Delta}$ is the strong condition. The full grounded conjunction is true at one objective tick. This repackages the Temporal Gap in a form that can be attached to later claims about conscious moments[35].

[35] Bennett, M. T. (2026a). A mind cannot be smeared across time. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press; and Elija Perrier, M. T. B. (2026). Time, identity and consciousness in language model agents. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

Definition 20 (Chord and Arpeggio postulates)**.** *Fix a windowing horizon Δ and a predicate $\mathrm{PhenReal}(\ell^m, \tau, u)$ meaning that the layer-m statement ℓ^m is phenomenally real for trajectory τ at window index u. I say the trajectory satisfies the* ***Chord*** *postulate for ℓ^m iff*

$$\forall u \in \mathbb{N} \qquad \mathrm{PhenReal}(\ell^m, \tau, u) \Longrightarrow \mathrm{CoInst}_{W_\Delta}(\ell^m, \tau, u).$$

I say the trajectory satisfies the ***Arpeggio*** *postulate for ℓ^m iff two conditions hold.*

1. *for every $u \in \mathbb{N}$,*

$$\mathrm{PhenReal}(\ell^m, \tau, u) \Longrightarrow \mathrm{Occur}_{W_\Delta}(\ell^m, \tau, u),$$

2. *there exists at least one $u^\star \in \mathbb{N}$ such that*

$$\mathrm{PhenReal}(\ell^m, \tau, u^\star) \wedge \mathrm{Occur}_{W_\Delta}(\ell^m, \tau, u^\star) \wedge \neg\mathrm{CoInst}_{W_\Delta}(\ell^m, \tau, u^\star).$$

Intuition. Chord says a conscious moment is like a musical chord. The ingredients have to sound together. Arpeggio says a conscious moment can instead be smeared across a window so long as the ingredients each occur somewhere inside it. The two postulates are different regimes, not a weak and strong version of the same claim[36].

[36] Bennett, M. T. (2026a). A mind cannot be smeared across time. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press; and Bennett, M. T. (2026d). Spacetime bounds on consciousness

Definition 21 (Weak and strong persistence scores)**.** *Fix a finite evaluation set $U \subseteq \mathbb{N}$ of window indices. For a trajectory τ, a statement ℓ^m, and horizon Δ, define*

$$P_{\mathrm{weak}}(\tau, \ell^m; U, \Delta) := \frac{1}{|U|} \sum_{u \in U} \mathbf{1}\left[\mathrm{Occur}_{W_\Delta}(\ell^m, \tau, u)\right]$$

and

$$P_{\text{strong}}(\tau, \ell^m; U, \Delta) := \frac{1}{|U|} \sum_{u \in U} \mathbf{1}\left[\text{CoInst}_{W_\Delta}(\ell^m, \tau, u)\right].$$

Intuition. These scores make the postulates operational. Weak persistence asks how often the grounded ingredients show up somewhere in the window. Strong persistence asks how often they are jointly active at one tick. This is the same measurement split used in the language model agent paper[37].

[37] Elija Perrier, M. T. B. (2026). Time, identity and consciousness in language model agents. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

Proposition 3 (Strong persistence is bounded by weak persistence)**.** *For every trajectory τ, statement ℓ^m, finite evaluation set U, and horizon Δ,*

$$P_{\text{strong}}(\tau, \ell^m; U, \Delta) \le P_{\text{weak}}(\tau, \ell^m; U, \Delta).$$

Proof sketch. Co-instantiation implies ingredient-wise occurrence window by window. Averaging the indicator functions preserves that inequality. A full proof is in the Theorems and Proofs chapter.

Corollary 1 (Sequential architectures separate Chord from Arpeggio)**.** *Fix grounded content $g^0 = l_n$ from Definition 17 with $n \ge 2$ contributors. If an architecture has concurrency capacity $c < n$, then no phenomenally real moment with grounded content g^0 can satisfy Chord. However if each contributor occurs somewhere in the window, Arpeggio can still be satisfied.*

Proof sketch. Theorem 1 says $c < n$ forbids co-instantiation while still allowing ingredient-wise occurrence. Chord requires the former. Arpeggio only requires the latter at least somewhere.

Remark 3 (How the two postulates use the later bounds)**.** *Under Chord, Theorem 1 and Theorem 2 are both necessary for a unified moment. The concurrency result constrains how many grounded ingredients can be active together. The spacetime bound constrains how far apart they can be. Under Arpeggio, neither constraint is forced in the same way. This is why Arpeggio is much more permissive*[38].

[38] Bennett, M. T. (2026d). Spacetime bounds on consciousness

Definition 22 (Diameter of a finite set of sites)**.** *Let (X, d) be a metric space and let $U \subseteq X$ be a finite set of sites. Define its* ***diameter*** *by*

$$\text{diam}(U) \ := \ \max_{u, u' \in U} d(u, u').$$

Intuition. $\text{diam}(U)$ is just the largest physical separation between any two sites that carry grounded ingredients. If you want one moment to span a huge system, you are asking for $\text{diam}(U)$ to be large.

Definition 23 (Edge-wise round trip integration graph)**.** *Fix a duration $\theta > 0$, a speed ceiling $v > 0$, and a finite set of contributor sites U in a metric space (X, d). Fix a connected undirected graph $G = (U, E)$. I say* *that U is G-**integrated within** θ if for every edge $\{u, u'\} \in E$ there exist travel times $\eta_{u\to u'}, \eta_{u'\to u} \geq 0$ such that*

$$\eta_{u\to u'} + \eta_{u'\to u} \leq \theta$$

and the speed ceiling applies in both directions

$$d(u, u') \leq v\,\eta_{u\to u'} \qquad \text{and} \qquad d(u, u') \leq v\,\eta_{u'\to u}.$$

Intuition. G is a wiring diagram for who must exchange influence within one moment. The condition is deliberately minimal. Each required pair must be able to send a signal out and receive a signal back inside the same window. If you cannot afford a round trip, you do not have causal closure. You have a distributed system that is still mid-conversation when the moment ends.

Definition 24 (Hop distance and hop diameter)**.** *Let $G = (U, E)$ be a connected undirected graph. Define the **hop distance** $d_G(u, u')$ as the length of the shortest path between u and u' in G. Define the **hop diameter** by*

$$h(G) \;:=\; \max_{u,u' \in U} d_G(u, u').$$

Intuition. $d_G(u, u')$ is how many relays you need if information can only travel along required edges. $h(G)$ is the worst case hop count across the architecture. It is a pure architecture number. It ignores geometry and only remembers how many handoffs the integration demand forces you to use.

Theorem 2 (Graph diameter spacetime bound)**.** *Assume propagation speed at most v. If U is G-integrated within duration θ, then for all $u, u' \in U$,*

$$d(u, u') \leq \frac{v\theta}{2}\, d_G(u, u').$$

In particular,

$$\mathrm{diam}(U) \leq \frac{v\theta}{2}\, h(G).$$

Proof sketch. Fix $u, u' \in U$ and take a shortest path $u = u_0, u_1, \ldots, u_k = u'$ in G. Every edge must support a round trip inside θ, so at least one direction on that edge has travel time at most $\theta/2$. The speed ceiling then forces each hop distance to satisfy $d(u_i, u_{i+1}) \leq v\theta/2$. Summing along the path and using the triangle inequality gives the bound[39].

[39] Full proof is in Theorems and Proofs.

Intuition. This is just the triangle inequality plus the fact that spacetime has a speed limit. If one moment must be stitched together by within-window round trips, then each hop can only span so much distance. Architectures that require many hops make the size limit tighter.

Remark 4 (Architecture factor examples). *If G is a star graph centred at a hub, then $h(G) = 2$ and the bound becomes* $\mathrm{diam}(U) \leq v\theta$. *If G is complete, then $h(G) = 1$ and the bound becomes* $\mathrm{diam}(U) \leq v\theta/2$. *In general, the architecture factor is $h(G)/2$.*

Intuition. A hub can integrate distant sites in two hops. Everyone talks to the hub and the hub talks back. All-to-all exchange is stricter. Every pair must complete a round trip directly, so the farthest pair must fit inside half a window in one direction.

PANCOMPUTATIONAL ENACTIVISM

Definition 25 ($\mathfrak{v}$-task). *Let $\mathfrak{v} \subseteq \mathcal{P}$ be a vocabulary*[40]. *A* ***$\mathfrak{v}$-task*** *is a pair $\alpha = \langle I_\alpha, O_\alpha \rangle$ where*

[40] This vocabulary will be used to represent an embodied agent or learner.

$$I_\alpha \subseteq L_\mathfrak{v} \qquad \textit{and} \qquad O_\alpha \subseteq \mathrm{Ext}(I_\alpha)\,.$$

So I_α is the set of admissible inputs, and O_α is the set of correct outputs. I sometimes assume the task is ***total****, meaning that for every input $i \in I_\alpha$ there exists an output $o \in O_\alpha$ with $i \subseteq o$.*

Given tasks $\alpha = \langle I_\alpha, O_\alpha \rangle$ and $\omega = \langle I_\omega, O_\omega \rangle$, I call α a ***child*** *of ω, written $\alpha \sqsubset \omega$, if $I_\alpha \subsetneq I_\omega$ and $O_\alpha \subseteq O_\omega$. I write $\alpha \sqsubseteq \omega$ if $\alpha = \omega$ or $\alpha \sqsubset \omega$.*

If $\mathfrak{v}$ is finite then $L_\mathfrak{v}$ is finite and the set of $\mathfrak{v}$-tasks is finite.

In that finite-vocabulary case I define the ***level*** *(or* ***height****) of a task ω, written $\ell(\omega)$, as the maximum length of a strict child-chain starting at ω*

$$\ell(\omega) \;:=\; \max\Big\{ k \in \mathbb{N} \;\Big|\; \exists\, \langle \alpha_0, \alpha_1, \ldots, \alpha_k \rangle \textit{ with } \alpha_0 = \omega \textit{ and } \alpha_{i+1} \sqsubset \alpha_i \textit{ for } i = 0, \ldots, k-1 \Big\}.$$

Definition 26 (Task spaces). *For a vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, let $\Gamma_\mathfrak{v}$ denote the set of all $\mathfrak{v}$-tasks. Let*

$$\Gamma \;:=\; \bigcup_{\mathfrak{v} \subseteq \mathcal{P}} \Gamma_\mathfrak{v}$$

denote the class of all tasks across all vocabularies.

Intuition. $\Gamma_\mathfrak{v}$ is the task space available at a fixed abstraction layer, meaning a fixed vocabulary. Γ ranges over all abstraction layers by taking the union over vocabularies. This keeps two questions separate. What you can even ask depends on representation. What you can solve depends on the policy you choose inside that representation.

Machine learning is typically divided into learning and inference. To learn a task, an organism learns a policy (strictly speaking, an organism *is* a policy). By constraining how I complete inputs, a policy allows us to do abductive inference (much as a constraint in a SAT solver would). Inference requires a **policy** and learning a policy requires a **proxy**, the definitions of which follow. Stack Theory is the idea that everything is an infinite stack of abstraction layers. Pancomputational Enactivism is the formalisation of enactivism within Stack Theory.

Definition 27 (Inference and correct policies)**.**
Fix a vocabulary $\mathfrak{v} \subseteq \mathcal{P}$ and a $\mathfrak{v}$-task $\alpha = \langle I_\alpha, O_\alpha \rangle$.

*A **policy** for α is any statement $\pi \in L_{\mathfrak{v}}$. I interpret π as a constraint on how I complete inputs.*

*I say that π is a **correct policy** for α if its compatible completions on admissible inputs are exactly the correct outputs*

$$\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) \;=\; O_\alpha.$$

*The set of all correct policies for α is denoted Π_α. Given an input $i \in I_\alpha$, **inference under** π means selecting any output*

$$e \in \mathrm{Ext}(i) \cap \mathrm{Ext}(\pi)\,.$$

If $\pi \in \Pi_\alpha$ then every such e lies in O_α.

Definition 28 (λ-tasks)**.** *Fix a $\mathcal{P}$-task ρ. An **uninstantiated task** (or λ**-task**) associated with ρ is a function λ_ρ whose domain contains all* finite *vocabularies $\mathfrak{v}' \subseteq \mathcal{P}$ (and may contain additional, possibly infinite vocabularies when explicitly specified) and such that $\lambda_\rho(\mathfrak{v}')$ is a $\mathfrak{v}'$-task satisfying $\lambda_\rho(\mathfrak{v}') \sqsubseteq \rho$ and*

$$\ell\big(\lambda_\rho(\mathfrak{v}')\big) \;=\; \max\Big\{\, \ell(\omega) \;\Big|\; \omega \text{ is a } \mathfrak{v}'\text{-task and } \omega \sqsubseteq \rho \,\Big\}.$$

When $\mathfrak{v}'$ is finite (hence $L_{\mathfrak{v}'}$ is finite), the maximum exists. When comparing λ-tasks I fix a reference vocabulary $\mathfrak{v}_{\mathrm{ref}}$ in the common domain (often the "full" vocabulary $\mathcal{P}$ when $\mathcal{P}$ is finite) and define

$$\lambda_1 \sqsubset \lambda_2 \quad\Longleftrightarrow\quad \lambda_1(\mathfrak{v}_{\mathrm{ref}}) \sqsubset \lambda_2(\mathfrak{v}_{\mathrm{ref}}).$$

Unless explicitly stated otherwise, all later optimisation results involving ϵ, the Law of the Stack, and the unagi construction are applied only to finite vocabularies in the domain of λ_ρ.

Learning is the process by which a system generalises from one task to another. A task has been **learned** by a system that embodies a correct policy for that task. Humans typically learn from **examples**. Here I represent examples using child tasks, and the goal of the learner is to generalise to some unknown parent task. The learner has *learned* a task ω from α if they embody a policy $\pi \in \Pi_\alpha$ that also belongs to Π_ω.

Definition 29 (Learning)**.** *Assume there exists a system (such as an organism) represented by an abstraction layer with vocabulary $\mathfrak{v}$. I call this system the* learner. *Let α be a $\mathfrak{v}$-task and Π_α denote its nonempty set of correct policies*[41]*. Suppose a learner restricts attention to Π_α as the candidate set of policies. Now assume there is a second task ω such that $\alpha \sqsubset \omega$. The learner's goal is to* learn *a policy $\pi \in \Pi_\omega$, by singling out a member of Π_α. To do this it uses a binary relation $<$ on Π_α as a preference proxy (I read $l_1 < l_2$ as "l_2 is preferred to l_1"). I say that the learner* ***learns*** *ω under proxy $<$ if it outputs a $<$-maximal correct policy, i.e., some $\pi^\star \in \Pi_\alpha \cap \Pi_\omega$ such that*

$$\forall\, \pi \in \Pi_\alpha \cap \Pi_\omega : \quad \neg(\pi^\star < \pi).$$

I say the learner or the chosen $\pi^\star$ generalises to ω if the learner learns ω. Two proxies used throughout[42] *are:*

- ***Weakness proxy*** *$<_w$: $l_1 <_w l_2$ iff $|\text{Ext}(l_1)| < |\text{Ext}(l_2)|$. Thus, $<_w$-maximal elements are those with maximal completion-set cardinality $|\text{Ext}(l)|$.*
- ***Simplicity proxy*** *$<_s$: $l_1 <_s l_2$ iff $|l_1| > |l_2|$. Thus, $<_s$-maximal elements are those with minimal description length $|l|$.*

[41] α is like a training set, and Π_α the set of possible model parameters which would reproduce that training set.

[42] Not an exhaustive list of possible proxies, just my proposed proxy and the orthodox proxy popularised by Ockham and his acolytes.

Generalisation results for the weakness proxy. The following propositions formalise the sense in which maximising weakness (extension cardinality) is optimal for generalisation under a maximally uninformative prior, and show why description length can fail as a proxy for generalisation.

Proposition 4 (Sufficiency of weakness under a maximally uninformative prior)**.** *Let $\mathfrak{v}$ be finite and let α and ω be $\mathfrak{v}$-tasks with $\alpha \sqsubset \omega$. Write $U := L_\mathfrak{v} \setminus \text{Ext}(I_\alpha)$ for the set of* unseen *outputs, i.e., statements that never arise as completions of the child inputs I_α. Because $\mathfrak{v}$ is finite, U is finite, so the uniform distribution on 2^U is well-defined. Assume a maximally uninformative model of how the parent extends the child. Conditional on α, the parent selects an additional set $S \subseteq U$ of correct outputs chosen uniformly at random from 2^U, and its overall correct-output set is $O_\omega = O_\alpha \cup S$. Then, among all $\pi \in \Pi_\alpha$, the probability that π is also correct for ω*[43] *is maximised by choosing a $<_w$-maximal policy (equivalently, by maximising $|\text{Ext}(\pi)|$).*

[43] Formally, the random-set model tracks *compatibility* with unseen task growth: $S \subseteq \text{Ext}(\pi) \cap U$ means the policy survives all future requirements in the unseen region. In the specific extension model used here, correctness on the seen region is held fixed and the parent differs from the child only by adding unseen outputs, so this compatibility event coincides with correctness for the generated parent task. For more general parent-growth models, the same counting argument should be read as a compatibility result

Proof sketch. I show that under the maximally uninformative extension model, is that the weakest policy that is correct on the observed history α maximises the probability of being correct on the true parent task. This works because the history α constrains only finitely many input output cases, so many parent tasks are still compatible. A policy fails to generalise only by ruling out some completion that the true parent will later require. A policy with a larger extension rules out fewer completions, so it is compatible with more possible parents. Under an ignorance prior over which parent is real, that counting advantage turns into a probability advantage[44].

[44] Full proof is in Theorems and Proofs.

Proposition 5 (Necessity of weakness under the same prior). *Under the assumptions of Proposition 4, any policy $\pi \in \Pi_\alpha$ that maximises $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ must be $<_w$-maximal in Π_α. In particular, any non-$<_w$-maximal correct policy is strictly suboptimal for maximising this generalisation probability.*

Proof sketch. I show that weakness maximisation is not just sufficient for optimal generalisation under the same prior. It is necessary. Any rule that sometimes picks a non-$<_w$-maximal correct policy will do strictly worse on some parent distributions. This works because if you ever pick a correct policy that is not among the weakest correct ones, then there exists another correct policy that leaves more completions open. The only way a policy fails is by forbidding a completion that the true parent later demands. A more restrictive choice is therefore always vulnerable to being surprised by some parent that a weaker correct policy would survive[45].

[45] Full proof is in Theorems and Proofs.

Notation. For any statement or policy $\pi \in L_\mathfrak{v}$, its extension is denoted $\mathrm{Ext}(\pi)$. I also write $\mathrm{Ext}(\pi)$ for the same object. So $\mathrm{Ext}(\pi) = \mathrm{Ext}(\pi)$.

Theorem 3 (Weakness maximisation beyond the uniform prior). *Let $\mathfrak{v}$ be finite and let α and ω be $\mathfrak{v}$-tasks with $\alpha \sqsubset \omega$. Write $U := L_\mathfrak{v} \setminus \mathrm{Ext}(I_\alpha)$ for the set of unseen outputs. Assume the parent extends the child by choosing a random set $S \subseteq U$ and setting $O_\omega = O_\alpha \cup S$. Let P be the conditional distribution of S given α, with no restriction to uniformity.*

*For each $\pi \in \Pi_\alpha$, define its P-**weakness** by*

$$w_P(\pi) := P\big(S \subseteq \mathrm{Ext}(\pi) \cap U\big) = \sum_{S \subseteq \mathrm{Ext}(\pi) \cap U} P(S).$$

Then

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) = w_P(\pi).$$

Consequently, a correct policy $\pi^\star \in \Pi_\alpha$ maximises the generalisation probability $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ if and only if it maximises $w_P(\pi)$ over Π_α.

*Assume in addition that P is **exchangeable** over U. That is, for every permutation $\sigma : U \to U$ and every subset $S \subseteq U$, I have $P(S) = P(\sigma(S))$. Then there exists a nondecreasing function $f : \{0, 1, \ldots, |U|\} \to [0, 1]$ such that*

$$w_P(\pi) \;=\; f\big(|\mathrm{Ext}(\pi) \cap U|\big)$$

for all $\pi \in \Pi_\alpha$. Therefore any $<_w$-maximal policy $\pi^\star \in \Pi_\alpha$ maximises $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ under P.

If also $P(|S| = 1) > 0$, then f is strictly increasing on $\{0, 1, \ldots, |U| - 1\}$. In that nondegenerate exchangeable case, every maximiser of $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ is $<_w$-maximal.

Proof sketch. I show you do not need a uniform prior over unseen outputs. For a general prior P, the optimal rule is still to maximise the P mass of completions you do not rule out. That is $w_P(\pi)$, a prior weighted version of weakness. This works because generalisation probability is exactly the probability that the parent task does not demand any completion your policy forbids. Under exchangeability, P depends on unseen outputs only through their count, so $w_P(\pi)$ collapses to a monotone function of $|\mathrm{Ext}(\pi) \cap U|$. That brings you back to ordinary weakness maximisation[46].

[46] Full proof is in Theorems and Proofs.

Remark 5 (Why Is Anything Alive? formalism and the SI results)**.** *The thesis chapter Why Is Anything Alive? and the unreleased standalone paper version[47] use a compact set model of novelty selection. It is the same model as Proposition 4 and Theorem 3. I record the notation here because readers often arrive from that paper rather than from Stack Theory.*

[47] Bennett, M. T. (2026e). Stack theory suite

Fix a finite statement space $L = L_{\mathfrak{v}}$. Let $\alpha \subseteq L$ be the set of outcomes observed so far. Let $U := L \setminus \alpha$ be the unobserved region. Let Π be the set of policies compatible with α. This means $\alpha \subseteq \mathrm{Ext}(\pi)$ for every $\pi \in \Pi$.

Let $S \subseteq U$ be the novelty set of future required outcomes. Assume S is drawn from a prior P over subsets of U. Define the buffer of a policy π as $B_\pi := \mathrm{Ext}(\pi) \cap U$. A policy survives the novelty event exactly when $S \subseteq B_\pi$. Define its degree of future function as

$$F(\pi) \;:=\; P(S \subseteq B_\pi).$$

Intuition. *The buffer is the part of the future you have not ruled out yet. A bigger buffer means you can tolerate more surprises.*

Under uniform ignorance P is uniform on 2^U. Then $F(\pi) = 2^{|B_\pi| - |U|}$. This is the survival law proved inside Proposition 4. If P is exchangeable, $F(\pi)$ is a nondecreasing function of $|B_\pi|$. This is the exchangeability clause of Theorem 3.

Functional information is defined inside the currently viable set Π. De-

fine

$$M(\pi) \;:=\; \{\, \pi' \in \Pi : F(\pi') \geq F(\pi) \,\}, \qquad I(\pi) \;:=\; -\log_2\!\left(\frac{|M(\pi)|}{|\Pi|}\right).$$

Intuition. *$I(\pi)$ measures how rare it is, within the policies that fit the past, to find policies that leave at least as much future open.*

The SI theorem called the law of increasing functional information is then

$$\mathbb{E}\big[I(\pi) \mid S \subseteq B_\pi\big] \;\geq\; \mathbb{E}\big[I(\pi)\big],$$

with strict inequality whenever $|B_\pi|$ is not constant over Π. In this appendix, that same statement is Proposition 20 after identifying Π with Π_α and B_π with $\mathrm{Ext}(\pi) \cap U$.

Proposition 6 (Simplicity can be sub-optimal)**.** *The Simplicity proxy $<_s$ is neither sufficient nor necessary for maximising the generalisation probability in Proposition 4. In particular, there exist tasks for which the $<_s$-maximal correct policy is* not *$<_w$-maximal, and therefore is strictly suboptimal under the maximally uninformative prior.*

Proof sketch. I show minimum description length can pick a worse generalising policy than weakness maximisation. In this framework, committalness is determined by the extension $\mathrm{Ext}(\pi)$. Description length is a property of the chosen encoding. By constructing a vocabulary where a very specific policy has an extremely short name while a strictly weaker policy has a longer description, you decouple length from extension. MDL can then prefer the over-specific policy even when a weaker correct alternative exists[48].

[48] Full proof is in Theorems and Proofs.

Proposition 7 (Subjectivity of description length)**.** *If there is no abstraction in the sense that the vocabulary equals the full program set ($\mathfrak{v} = \mathcal{P} = 2^\Phi$), then every statement is semantically representable with description length 1. In particular, for any $l \in L_\mathcal{P}$, the singleton statement $\{T(l)\}$ satisfies $T(\{T(l)\}) = T(l)$. Consequently, description length is highly representation dependent: in the extreme case $\mathfrak{v} = \mathcal{P}$, minimizing $|l|$ does not meaningfully distinguish among policies by semantic content.*

Proof sketch. I prove that when there is no abstraction, meaning $\mathfrak{v} = \mathcal{P}$, description length becomes irrelevant to generalisation. If every truth set is already a primitive element of the vocabulary, any policy's extension can be represented by a singleton statement without changing what completions it permits. You can always compress the syntax after the fact without affecting weakness. In that setting, sample efficiency is fully determined by extension size, so simplicity cannot be the driver [49].

[49] Full proof is in Theorems and Proofs.

Proposition 8 (Complexity confounding). *If the vocabulary $\mathfrak{v}$ is finite, then*

1. *semantically similar constraints need not admit equally short descriptions, and*
2. *description length can become spuriously correlated with generalisation, not because it is causal, but because it is confounded by weakness.*

Proof sketch. I prove simplicity can look predictive of sample efficiency even when it is not doing the causal work. Weakness is the true driver of sample efficiency. The syntax you use to write policies is a design choice. In some vocabularies, weak policies happen to have short descriptions. That creates a correlation between description length and extension size. The apparent power of simplicity is then confounding. It is weakness all the way down[50].

[50] Full proof is in Theorems and Proofs.

Definition 30 (Stack). *A* ***stack***[51] *of depth n is specified by the following data*[52].

1. *a sequence of uninstantiated tasks $\langle \lambda^0, \lambda^1, \dots, \lambda^n \rangle$ with $\lambda^{i+1} \sqsubset \lambda^i$ for $i = 0, \dots, n-1$*
2. *a sequence of vocabularies $\langle \mathfrak{v}^0, \mathfrak{v}^1, \dots, \mathfrak{v}^n \rangle$*
3. *a sequence of policies $\langle \pi^0, \pi^1, \dots, \pi^n \rangle$ with $\pi^i \in L_{\mathfrak{v}^i}$ for each i*

For each $i = 0, \dots, n-1$,

$$\mathfrak{v}^{i+1} = f(\mathfrak{v}^i, \pi^i).$$

I call a stack ***overconstrained*** *if there exists some layer i with $\Pi_{\lambda^i(\mathfrak{v}^i)} = \varnothing$.*

I call a stack an ***MCL architecture*** *(multi-layer causal learning) if it is not overconstrained and, for each layer i, the realised policy π^i is a $<_w$-maximal element of $\Pi_{\lambda^i(\mathfrak{v}^i)}$.*

[51] Bennett, M. T. (2024a). Are biological systems more intelligent than artificial intelligence? In press, 2026, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds

[52] A stack is also called an MLA or multi-layer-architecture in some of my publications.

Definition 31 (Utility of intelligence).
Every task $\gamma \in \Gamma$ has a "utility of intelligence"[53]. *It is the slack left open by the weakest correct policies for γ. Define*

$$\epsilon(\gamma) = \begin{cases} 0 & \text{if } \Pi_\gamma = \varnothing \\ \sup_{\pi \in \Pi_\gamma} |\mathrm{Ext}(\pi) \setminus O_\gamma| & \text{otherwise.} \end{cases}$$

For any $\pi \in \Pi_\gamma$ I have $O_\gamma \subseteq \mathrm{Ext}(\pi)$, so $\mathrm{Ext}(\pi) \setminus O_\gamma$ is the set of additional completions compatible with π beyond those demanded by correctness. In the common finite vocabulary case, the supremum is a maximum and $\epsilon(\gamma) \in \mathbb{N}$. Maximisation of utility means maximising $\bullet \overset{\epsilon}{<} \bullet$, which returns true iff $\epsilon(\alpha) < \epsilon(\omega)$.

[53] Assuming you accept that intelligence is the ability to generalise, then I can measure the utility of selecting policies in accord with Bennett's Razor by measuring the weakness of the weakest policy for a task. Tasks with weaker policies make more use of intelligence.

UTILITY

Theorem 4 (The Law of the Stack (bottleneck form)). *Consider a stack with adjacent layers i and i+1 and suppose*

$$\mathfrak{v}^{i+1} = f(\mathfrak{v}^i, \pi^i) = \{\, T(o) \mid o \in \mathrm{Ext}\Big(\pi^i\Big) \,\}.$$

Let $\gamma^{i+1} := \lambda^{i+1}(\mathfrak{v}^{i+1})$ be the induced layer-$(i+1)$ task.

Then the achievable behavioural flexibility at layer $i+1$ is bottlenecked by the weakness of the realised policy at layer i in the following cardinal sense:

$$\sup_{\pi \in \Pi_{\gamma^{i+1}}} |\mathrm{Ext}(\pi)| \ \leq\ 2^{|\mathrm{Ext}(\pi^i)|}.$$

In particular,

$$\epsilon(\gamma^{i+1}) \ \leq\ 2^{|\mathrm{Ext}(\pi^i)|}.$$

If $|\mathrm{Ext}\big(\pi^i\big)\,| < \infty$, then $\mathfrak{v}^{i+1}$ and $L_{\mathfrak{v}^{i+1}}$ are finite and $\epsilon(\gamma^{i+1}) \in \mathbb{N}$.

In that finite case the familiar bottleneck form holds

$$\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| \ \leq\ 2^{|\mathrm{Ext}(\pi^i)|}.$$

Equivalently,

$$|\mathrm{Ext}\Big(\pi^i\Big)\,| \ \geq\ \log_2\big(\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}|\big).$$

So increasing $\epsilon(\gamma^{i+1})$ requires increasing $|\mathrm{Ext}\big(\pi^i\big)\,|$, which means weakening the layer-i policy.

Proof sketch. A layer i policy π^i leaves a set of completions $\mathrm{Ext}\big(\pi^i\big)$ open. Those completions induce the next layer vocabulary $\mathfrak{v}^{i+1}$ as their truth sets, so $|\mathfrak{v}^{i+1}| \ \leq\ |\mathrm{Ext}\big(\pi^i\big)\,|$. Any layer $i+1$ statement is a subset of $\mathfrak{v}^{i+1}$. So the entire layer-$i+1$ language has size at most $2^{|\mathfrak{v}^{i+1}|}$, which is at most $2^{|\mathrm{Ext}(\pi^i)|}$. Every extension set $\mathrm{Ext}(\pi)$ at layer $i+1$ is a subset of that language, so it inherits the same bound. In the finite case, taking base-2 logs gives the familiar lower bound on $|\mathrm{Ext}\big(\pi^i\big)\,|$[54].

[54] Full proof is in Theorems and Proofs.

EXTENDING WEAKNESS TO CONTINUOUS DOMAINS

The preceding definitions assume a finite vocabulary, so counting measure is canonical. In genuinely continuous domains, cardinality stops separating policies because many different extension sets are equinumerous. The clean repair is to equip the embodied language with a finite reference measure and to compare correct policies by the measure of the unseen buffer they leave open. On correct policies, this agrees with ordinary μ-weakness up to an additive constant. The full proofs are in the Theorems and Proofs chapter[55].

[55] Full proofs are in Theorems and Proofs, Continuous Domains.

Definition 32 (Measured embodied language)**.** *A **measured embodied language** is a triple* $(\mathfrak{v}, \mathcal{A}_L, \mu)$ *where*

1. $\mathfrak{v} \subseteq \mathcal{P}$ *is a vocabulary (possibly infinite),*
2. $\mathcal{A}_L$ *is a* σ*-algebra on* $L_{\mathfrak{v}}$*, and*
3. μ *is a finite measure on* $(L_{\mathfrak{v}}, \mathcal{A}_L)$*.*

I assume that $\mathrm{Ext}(l) \in \mathcal{A}_L$ *for every statement* $l \in L_{\mathfrak{v}}$*.*

Intuition. The measure μ plays the same role in the infinite case that counting measure plays in the finite case. In particular, if $L_{\mathfrak{v}}$ is finite and μ is counting measure, then every definition below reduces to the earlier finite one.

Definition 33 (Continuous weakness (μ-weakness))**.** *Fix a measured embodied language* $(\mathfrak{v}, \mathcal{A}_L, \mu)$ *and a statement* $l \in L_{\mathfrak{v}}$*. Its **μ-weakness** is*

$$w_\mu(l) := \mu(\mathrm{Ext}(l)).$$

When $w_\mu(l_1) > w_\mu(l_2)$ *I say that* l_1 *is **μ-weaker** than* l_2*.*

Definition 34 (Continuous extension model)**.** *Fix a measured embodied language* $(\mathfrak{v}, \mathcal{A}_L, \mu)$ *and a* $\mathfrak{v}$*-task* α*. Let*

$$U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)\,,$$

and assume $U \in \mathcal{A}_L$ *with* $0 < \mu(U) < \infty$*.*

*A **continuous extension model** is a probability space* $(\Omega, \mathcal{F}, P)$ *together with a random measurable set* $S \subseteq U$ *such that, for every measurable* $B \subseteq U$*, the event* $\{S \subseteq B\}$ *is* $\mathcal{F}$*-measurable.*

For a correct policy $\pi \in \Pi_\alpha$*, define its unseen buffer by*

$$B_\pi := \mathrm{Ext}(\pi) \cap U,$$

and its generalisation score under P *by*

$$G(\pi, P) := P(S \subseteq B_\pi).$$

Intuition. S is the additional future demand and B_π is the unseen part of the future that π still permits. The policy survives exactly when every future demand lands inside that buffer.

Assumption 1 (Atomless standard unseen region)**.** *Throughout this subsection, assume that* $(U, \mathcal{A}_L|_U, \mu|_U)$ *is an atomless standard finite measure space.*

Intuition. This is the clean repair of the exchangeability argument. Without atomlessness, equal-measure sets can have different atomic structure; without standardness, there need not be enough measure-preserving relabellings to identify equal-measure sets. Under the stated assumption, equal-measure buffers are interchangeable up to measure-preserving bijection.

Definition 35 (μ-exchangeability)**.** *A continuous extension model P is **μ-exchangeable** if for every measure-preserving bijection $\sigma : U \to U$, the pushforward law of $\sigma(S)$ equals the law of S.*

Definition 36 (Nondegeneracy)**.** *A μ-exchangeable model P is **nondegenerate** if for every measurable $A \subseteq U$ with $\mu(A) > 0$,*

$$P\big(|S| = 1 \text{ and } S \subseteq A\big) > 0.$$

Lemma 6 (Buffer decomposition and equivalence to μ-weakness)**.** *Let $\pi \in \Pi_\alpha$ be correct. Then*

$$\mathrm{Ext}(\pi) = O_\alpha \sqcup B_\pi.$$

Consequently, if $O_\alpha \in \mathcal{A}_L$ and $\mu(O_\alpha) < \infty$, then

$$w_\mu(\pi) = \mu(O_\alpha) + \mu(B_\pi).$$

Hence, on the set of correct policies Π_α, maximising μ-weakness is equivalent to maximising $\mu(B_\pi)$.

Proof sketch. The language splits as $L_{\mathfrak{v}} = \mathrm{Ext}(I_\alpha) \sqcup U$. Intersect with $\mathrm{Ext}(\pi)$. Correctness gives $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$, and the unseen part is exactly B_π.

Theorem 5 (Continuous sufficiency under Poisson demands)**.** *If future demands are generated by a Poisson point process on U with intensity $\lambda\mu$, then for every correct policy $\pi \in \Pi_\alpha$,*

$$G(\pi, P) = \exp\big(-\lambda[\mu(U) - \mu(B_\pi)]\big).$$

In particular, $G(\pi, P)$ is strictly increasing in $\mu(B_\pi)$ and therefore, by Lemma 6, strictly increasing in $w_\mu(\pi)$ on Π_α.

Proof sketch. The event $S \subseteq B_\pi$ is exactly the event that the Poisson process places no points in $U \setminus B_\pi$. Apply the void-probability formula[56].

[56] Full proof is in Theorems and Proofs, Continuous Domains.

Theorem 6 (Continuous sufficiency under atomless μ-exchangeable demands)**.** *Assume 1. If P is μ-exchangeable, then there exists a nondecreasing function*

$$f_P : [0, \mu(U)] \to [0, 1]$$

such that for every correct policy $\pi \in \Pi_\alpha$*,*

$$G(\pi, P) = f_P\big(\mu(B_\pi)\big).$$

Equivalently, on correct policies, $G(\pi, P)$ *is a nondecreasing function of* μ*-weakness.*

Proof sketch. In an atomless standard finite measure space, equal-measure measurable sets are interchangeable under measure-preserving relabellings. So μ-exchangeability forces equal-measure buffers to have the same score. If $\mu(B_1) \le \mu(B_2)$, map B_1 into a subset of B_2 of the same measure. The event $\{S \subseteq B_1\}$ is then carried into a subset of $\{S \subseteq B_2\}$, giving monotonicity[57].

[57] Full proof is in Theorems and Proofs, Continuous Domains.

Theorem 7 (Continuous strict necessity)**.** *Assume 1. If P is* μ*-exchangeable and nondegenerate, and if two correct policies satisfy*

$$\mu(B_{\pi_1}) > \mu(B_{\pi_2}),$$

then

$$G(\pi_1, P) > G(\pi_2, P).$$

By Lemma 6, this is equivalent to strict ordering by μ*-weakness on* Π_α*.*

Proof sketch. Match as much of B_{π_2} as possible inside B_{π_1} by a measure-preserving relabelling. The unmatched remainder of B_{π_1} has positive measure. Nondegeneracy assigns positive probability to a singleton demand landing in that leftover region, which is caught by π_1 but not by π_2[58].

[58] Full proof is in Theorems and Proofs, Continuous Domains.

Theorem 8 (Unified optimality of μ-weakness)**.** *Assume 1, and let* $\mathcal{E}$ *be the class of all* μ*-exchangeable continuous extension models on* U*. For correct policies* $\pi_1, \pi_2 \in \Pi_\alpha$*:*

1. *if* $w_\mu(\pi_1) \ge w_\mu(\pi_2)$*, then* $G(\pi_1, P) \ge G(\pi_2, P)$ *for every* $P \in \mathcal{E}$*;*
2. *if* $w_\mu(\pi_1) > w_\mu(\pi_2)$ *and* $P \in \mathcal{E}$ *is nondegenerate, then* $G(\pi_1, P) > G(\pi_2, P)$*.*

Consequently, among correct policies, maximising μ*-weakness is equivalent to maximising the generalisation score for every nondegenerate model in* $\mathcal{E}$*. The* optimal value *of* μ*-weakness is unique, although the maximising policies need not be unique as sets.*

Theorem 9 (Minimax optimality of μ-weakness)**.** *Assume 1. If a* μ*-weakest correct policy* $\pi^* \in \Pi_\alpha$ *exists, then*

$$\max_{\pi \in \Pi_\alpha} \inf_{P \in \mathcal{E}} G(\pi, P) = \inf_{P \in \mathcal{E}} \max_{\pi \in \Pi_\alpha} G(\pi, P),$$

and both sides are achieved by π^**.*

Intuition. A μ-weakest correct policy is pointwise optimal across the whole uncertainty class $\mathcal{E}$. So it is automatically minimax. The learner does not need to know *which* μ-exchangeable model is correct.

Definition 37 (Policy map and function-preserving reparameterisation)**.** *Fix a measured embodied language defined over input-output behaviour. Let $\Theta \subseteq \mathbb{R}^d$ be a parameter space and let $F(\theta)$ denote the function computed by a network with parameters θ. A program $p \in \mathfrak{v}$ is **satisfied** by $F(\theta)$ if the input-output behaviour of $F(\theta)$ lies in p (viewed as a subset of Φ). The **policy map** is*

$$\Pi_F(\theta) := \{p \in \mathfrak{v} : F(\theta) \text{ satisfies } p\}.$$

*A **function-preserving reparameterisation** is a diffeomorphism $\phi : \Theta \to \Theta$ such that $F(\phi(\theta)) = F(\theta)$ for all θ.*

Theorem 10 (Reparameterisation invariance of μ-weakness)**.** *If ϕ is function-preserving, then*

$$\Pi_F(\phi(\theta)) = \Pi_F(\theta),$$

and therefore

$$w_\mu\big(\Pi_F(\phi(\theta))\big) = w_\mu\big(\Pi_F(\theta)\big).$$

Proof sketch. The policy map depends only on the function actually computed, not on the coordinates used to parametrise it[59].

[59] Full proof is in Theorems and Proofs, Continuous Domains.

Theorem 11 (PAC-Bayes ordering agrees with μ-weakness)**.** *Assume 1. Let*

$$P_U := \frac{\mu|_U}{\mu(U)} \qquad \text{and} \qquad Q_\pi := \frac{\mu|_{B_\pi}}{\mu(B_\pi)}$$

for any correct policy π with $\mu(B_\pi) > 0$. Then

$$\mathrm{KL}(Q_\pi \| P_U) = \log \frac{\mu(U)}{\mu(B_\pi)}.$$

Hence, among correct policies, minimising this KL-divergence is equivalent to maximising $\mu(B_\pi)$ and therefore, by Lemma 6, to maximising μ-weakness. In particular, standard zero-empirical-risk PAC-Bayes bounds with fixed sample size rank correct policies in the same order as μ-weakness[60].

[60] McAllester, D. A. (1999). Pac-Bayesian model averaging. In *Proceedings of the 12th Annual Conference on Computational Learning Theory (COLT)*, pages 164–170. ACM

Proof sketch. The Radon-Nikodym derivative is constant on B_π, so the KL integral collapses to a single logarithm[61].

[61] Full proof is in Theorems and Proofs, Continuous Domains.

GROUNDING

This subsection matches the grounding results in the in-press version of *Are Biological Systems More Intelligent Than Artificial Intelligence?*[62].

[62] Bennett, M. T. (2024a). Are biological systems more intelligent than artificial intelligence? In press, 2026, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds

Definition 38 (One-step grounding map)**.** *Fix a vocabulary $\mathfrak{v}$ and a statement $\pi \in L_{\mathfrak{v}}$. Let $\mathfrak{v}' := f(\mathfrak{v}, \pi)$ be the induced next-layer vocabulary.*

*For each program $p \in \mathfrak{v}'$, choose one completion $o_p \in \mathrm{Ext}(\pi)$ such that $T(o_p) = p$. Define the **one-step grounding map***

$$\mathrm{ground}_\pi : L_{\mathfrak{v}'} \to L_{\mathfrak{v}}, \qquad \mathrm{ground}_\pi(\ell) \;:=\; \bigcup_{p \in \ell} o_p.$$

Intuition. A program $p \in \mathfrak{v}'$ is, by construction, the truth set of at least one completion of π. Grounding chooses one such completion and uses it as the "definition" of p back in the old vocabulary. Grounding a higher-layer statement ℓ then just unions those defining completions ingredient-wise.

Lemma 7 (One-step grounding preserves truth conditions)**.** *With the setup of Definition 38, for every $\ell \in L_{\mathfrak{v}'}$ I have*

$$T(\mathrm{ground}_\pi(\ell)) \;=\; T(\ell).$$

Proof sketch. Expand both sides using $T(\cdot)$ as an intersection of ingredients. Each chosen completion o_p has truth set exactly p. Distributing intersections over the union construction recovers $\bigcap_{p \in \ell} p$[63].

[63] Full proof is in Theorems and Proofs.

Definition 39 (Compositional grounding along a stack)**.** *Let $\{\langle \mathfrak{v}^i, \pi^i \rangle\}_{i=0}^m$ be a stack, so $\mathfrak{v}^{i+1} = f(\mathfrak{v}^i, \pi^i)$. For each $i \in \{0, \dots, m-1\}$, choose a grounding map $\mathrm{ground}_{\pi^i} : L_{\mathfrak{v}^{i+1}} \to L_{\mathfrak{v}^i}$ as in Definition 38.*

*Given a statement $\ell^m \in L_{\mathfrak{v}^m}$, define its **compositional grounding** back to layer 0 by the recursion*

$$g^m := \ell^m, \qquad g^i := \mathrm{ground}_{\pi^i}(g^{i+1}) \;\; \text{for } i = m-1, m-2, \dots, 0,$$

and set

$$\mathrm{Ground}_{0 \leftarrow m}(\ell^m) \;:=\; g^0 \in L_{\mathfrak{v}^0}.$$

Theorem 12 (Compositional grounding preserves truth conditions)**.** *With the setup of Definition 39, for every $\ell^m \in L_{\mathfrak{v}^m}$ I have*

$$T(\mathrm{Ground}_{0 \leftarrow m}(\ell^m)) \;=\; T(\ell^m).$$

Proof sketch. Apply Lemma 7 one step at a time and use induction down the stack[64].

[64] Full proof is in Theorems and Proofs.

DELEGATION AND FREE ENERGY PRINCIPLE

Corollary 2 (Delegation is necessary for lowering variational free energy)**.** *Assume I have a stack with adjacent layers i and i+1 as in Theorem 4. Write π^i for the realised layer-i policy and $\mathfrak{v}^{i+1} = f(\mathfrak{v}^i, \pi^i)$ for the induced vocabulary. Assume the finite case $|\mathrm{Ext}(\pi^i)| < \infty$. In particular, $\mathfrak{v}^{i+1}$ and $L_{\mathfrak{v}^{i+1}}$ are finite, so the uniform distributions used below* *are well-defined. Fix a planning horizon and a layer-*$(i+1)$ viability *statement* $\mu \in L_{\mathfrak{v}^{i+1}}$. *Its extension* $\mathrm{Ext}(\mu)$ *represents the set of viable future continuations at that layer.*

For any policy $\pi \in L_{\mathfrak{v}^{i+1}}$, *define the set of viable futures compatible with* π *as*

$$\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu).$$

Assume $\Omega_\pi \neq \varnothing$ *and let* q_π *be the uniform distribution on* Ω_π. *Let* p_μ *be the uniform distribution on* $\mathrm{Ext}(\mu)$.

Define the base-2 free energy proxy as the base-2 KL divergence

$$F_2(q_\pi) := D_{\mathrm{KL},2}(q_\pi \,\|\, p_\mu).$$

Then $F_2(q_\pi) = \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_\pi|$, *cf.* [65].

Moreover, $\Omega_\pi \subseteq L_{\mathfrak{v}^{i+1}} \subseteq 2^{\mathfrak{v}^{i+1}}$, *so*

$$|\Omega_\pi| \leq |L_{\mathfrak{v}^{i+1}}| \leq 2^{|\mathfrak{v}^{i+1}|} \leq 2^{|\mathrm{Ext}(\pi^i)|}.$$

Therefore,

$$F_2(q_\pi) \geq \log_2 |\mathrm{Ext}(\mu)| - |\mathrm{Ext}\Big(\pi^i\Big)|.$$

In particular, lowering this free energy proxy at layer $i+1$ *requires increasing* $|\mathrm{Ext}(\pi^i)|$, *i.e. weakening the realised policy at the lower layer.*

[65] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138

Intuition. Under a uniform viability prior, lowering variational free energy reduces to keeping as many viable futures open as possible. That is maximising $\log |\Omega_\pi|$. Theorem 4 says the number of futures available at layer $i+1$ is bottlenecked by the completions left open one layer down. So if lower layers are frozen or over constrained, there is an entropy bottleneck and abstract-layer free energy has a floor. Basically it just means living things need to maximise their degrees of freedom or weakness of constraints on function, and this also happens to minimise free energy.

Definition 40 (Upper-bound objective for an uninstantiated task)**.** *Fix an uninstantiated task* λ_ρ *and a nonempty vocabulary class* V *consisting of finite vocabularies in the domain of* λ_ρ. *Define the admissible pair set*

$$\mathcal{A}(\lambda_\rho, V) := \{ (\mathfrak{v}, \pi) \mid \mathfrak{v} \in V,\ \pi \in \Pi_{\lambda_\rho(\mathfrak{v})} \}.$$

Assume $\mathcal{A}(\lambda_\rho, V) \neq \varnothing$. *Define the* ***upper-bound objective*** *on admissible pairs by lexicographic order:*

$$(\mathfrak{v}_1, \pi_1) \preceq_{\lambda_\rho, V} (\mathfrak{v}_2, \pi_2)$$

iff either

$$\epsilon\big(\lambda_\rho(\mathfrak{v}_1)\big) < \epsilon\big(\lambda_\rho(\mathfrak{v}_2)\big),$$

or

$$\epsilon\big(\lambda_\rho(\mathfrak{v}_1)\big) = \epsilon\big(\lambda_\rho(\mathfrak{v}_2)\big) \quad \textit{and} \quad |\mathrm{Ext}(\pi_1)| \leq |\mathrm{Ext}(\pi_2)|.$$

Proposition 9 (Formal upper-bound optimisation)**.** *Fix an uninstantiated task λ_ρ, a nonempty vocabulary class V of finite vocabularies in its domain, and assume $\mathcal{A}(\lambda_\rho, V) \neq \emptyset$. An admissible pair $(\mathfrak{v}^\star, \pi^\star) \in \mathcal{A}(\lambda_\rho, V)$ maximises the upper-bound objective $\preceq_{\lambda_\rho, V}$ if and only if*

$$\mathfrak{v}^\star \in \arg\max_{\mathfrak{v} \in V} \epsilon\big(\lambda_\rho(\mathfrak{v})\big)$$

and

$$\pi^\star \in \arg\max_{\pi \in \Pi_{\lambda_\rho(\mathfrak{v}^\star)}} |\mathrm{Ext}(\pi)|.$$

Equivalently: once the embodiment class V is fixed, the outer optimisation is utility maximisation over vocabularies and the inner optimisation is weakness maximisation inside the chosen vocabulary.

Proof sketch. The objective has two coordinates. The first is vocabulary-level utility, which is an outer design choice. The second is extension size inside the feasible policy set once the vocabulary is fixed. A lexicographic maximiser must first maximise utility and then, among those vocabularies, choose a weakest correct policy. Conversely any pair with those two properties is maximal for the lexicographic objective[66].

[66] Full proof is in Theorems and Proofs.

OVER-CONSTRAINT AND SPLINTERING

This subsection matches the cancer-analogue results in the in-press version of *Are Biological Systems More Intelligent Than Artificial Intelligence?*[67]. The formal content captures the failure mode where top-down constraints collapse the collective policy set to emptiness while individual parts remain locally viable.

[67] Bennett, M. T. (2024a). Are biological systems more intelligent than artificial intelligence? In press, 2026, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds

Definition 41 (Restricted policies, over-constraint, and splinters)**.** *Let $\omega_1, \ldots, \omega_m$ be $\mathfrak{v}$-tasks representing parts of a distributed system, with policy sets $\Pi_{\omega_j} \subseteq L_{\mathfrak{v}}$. The collective-policy set is*

$$\Pi_\omega := \bigcap_{j=1}^{m} \Pi_{\omega_j}.$$

The collective has an identity at $\mathfrak{v}$ iff $\Pi_\omega \neq \emptyset$.

To model top-down boundary conditions, fix restricted policy sets $\Pi'_{\omega_j} \subseteq \Pi_{\omega_j}$ and define the restricted collective set $\Pi'_\omega := \bigcap_{j=1}^{m} \Pi'_{\omega_j}$.

The system is ***over-constrained at*** *$\mathfrak{v}$ (under the boundary conditions) if $\Pi'_\omega = \emptyset$.*

A ***splinter*** *is a nonempty index set $S \subseteq \{1, \ldots, m\}$ such that $\bigcap_{j \in S} \Pi'_{\omega_j} \neq \emptyset$.*

Intuition. A collective has an identity when all its parts can agree on a shared policy. Top-down constraints (modelled as restrictions on each part's feasible set) can tighten the collective feasible set until it collapses. When it does, the collective can no longer act as a unit. Any continued action must come from a subset of parts whose restricted policies still overlap. That subset is a splinter.

Proposition 10 (Over-constraint implies splintering)**.** *Assume at least one part remains locally viable, meaning $\Pi'_{\omega_j} \neq \emptyset$ for some j. If the system is over-constrained at $\mathfrak{v}$ (so $\Pi'_{\omega} = \emptyset$), then there is no policy simultaneously feasible for all parts. For any realised policy $\pi \in L_{\mathfrak{v}}$, the set of parts for which π remains feasible,*

$$S(\pi) := \{j \in \{1, \ldots, m\} : \pi \in \Pi'_{\omega_j}\},$$

is a proper subset of $\{1, \ldots, m\}$. Any continued function can only proceed via a splinter.

Proof sketch. If some π were feasible for all parts then $\pi \in \bigcap_{j=1}^{m} \Pi'_{\omega_j} = \Pi'_{\omega}$, contradicting $\Pi'_{\omega} = \emptyset$. So every realised policy fails at least one part's constraints. If $\Pi'_{\omega_j} \neq \emptyset$ for some j, then $\{j\}$ is a splinter[68].

Intuition. This is a formal analogue of cancer. In a biological system, when cells become isolated from the informational structure of their collective, they lose their high-level identity and revert to primitive behaviour. In Stack Theory terms, the zero-slack limit of the delegation bottleneck (Theorem 4) can shrink the induced higher-layer feasible set to emptiness. When that happens, parts of the system must go it alone. This suggests systems should be minimally constrained (weakest sufficient boundary conditions) to ensure graceful degradation in adverse circumstances[69].

Remark 6 (Connection to the Law of the Stack)**.** *As task difficulty increases or top-down control tightens, the set of correct collective policies shrinks. Proposition 10 characterises what happens at the limit. Corollary 2 shows the same mechanism through the free energy lens. If lower layers are over-constrained, abstract-layer free energy has a floor that cannot be lowered. Splintering is what happens when that floor exceeds the viability threshold.*

[68] Full proof is in the Theorems and Proofs chapter.

[69] Bennett, M. T. (2024a). Are biological systems more intelligent than artificial intelligence? In press, 2026, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds

HIDDEN COMBINATORICS OF THE EMBODIED LANGUAGE

The core definitions describe what $L_{\mathfrak{v}}$ and $\mathrm{Ext}(l)$ are. This subsection describes what they look like on the inside. The results expose the real geometry of the embodied language and show exactly when weakness and description length coincide and when they diverge. These results first appeared in the journal extension of *Is Complexity an Illusion?*[70]. Full proofs are in the Theorems and Proofs chapter.

[70] Bennett, M. T. (2024c). Is complexity an illusion? In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

Lemma 8 (The language is the union of all local powersets)**.** *Fix an environment Φ and a vocabulary $\mathfrak{v} \subseteq 2^{\Phi}$. Then*

$$L_{\mathfrak{v}} = \bigcup_{\phi \in \Phi} 2^{\mathrm{Enc}_{\mathfrak{v}}(\phi)} = \{l \subseteq \mathfrak{v} \mid \exists \phi \in \Phi : \; l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)\}.$$

Intuition. The language is the set of partial descriptions that some real situation can satisfy. Each microstate ϕ contributes its local powerset. The language is the union of these. This is why $L_{\mathfrak{v}} \neq 2^{\mathfrak{v}}$ in general.

Proposition 11 (Extensions are unions of encoding intervals)**.** *Assume $\mathfrak{v}$ is finite. For a statement $l \in L_{\mathfrak{v}}$, write $\mathcal{M}(l) := \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in T(l)\}$ for the set of realised maximal descriptions compatible with l. Then*

$$\mathrm{Ext}(l) = \bigcup_{m \in \mathcal{M}(l)} \{y \subseteq \mathfrak{v} \mid l \subseteq y \subseteq m\}.$$

Consequently, $\max_{m \in \mathcal{M}(l)} 2^{|m|-|l|} \leq w(l) \leq \sum_{m \in \mathcal{M}(l)} 2^{|m|-|l|}$.

Intuition. To count completions of l, look at every full dictionary description compatible with l. Each contributes every way of adding ingredients up to that description. The lower bound comes from the richest single description. The upper bound sums over all descriptions (with possible overlaps).

Lemma 9 (Fully consistent vocabularies collapse weakness to a length proxy)**.** *If $L_{\mathfrak{v}} = 2^{\mathfrak{v}}$ (every subset is satisfiable), then $w(l) = 2^{|\mathfrak{v}|-|l|}$ for all $l \in L_{\mathfrak{v}}$. Weakness becomes a strictly decreasing function of description length. The weakness proxy and the simplicity proxy agree.*

Intuition. This is the trap. When every combination of programs is satisfiable, weakness collapses into a pure length proxy. This is the degenerate case the implementation warning (Remark 1) flags.

Lemma 10 (Ordered recursion for weakness)**.** *Fix a finite vocabulary $\mathfrak{v} = \{p_0, \ldots, p_{n-1}\}$ under any fixed total order. For $l \in L_{\mathfrak{v}}$ and $j \in \{0, \ldots, n\}$, define*

$$w_j(l) := \left|\{y \in \mathrm{Ext}(l) \mid y \setminus l \subseteq \{p_j, p_{j+1}, \ldots, p_{n-1}\}\}\right|.$$

This counts completions that only add programs with index $\geq j$. Then

$$w_j(l) = 1 + \sum_{\substack{k \in \{j, \ldots, n-1\} \\ p_k \notin l \\ l \cup \{p_k\} \in L_{\mathfrak{v}}}} w_{k+1}(l \cup \{p_k\}).$$

In particular, $w(l) = w_0(l)$.

Intuition. Weakness can be computed by recursion. Either add nothing (the 1), or add the next available program. The total order ensures each completion is reached exactly once.

COMPRESSION AND THE APPEARANCE OF SIMPLICITY

This subsection explains why simplicity *appears* to predict generalisation. If weak constraints are used more often, standard source coding assigns them shorter codewords. The apparent simplicity-generalisation link is a secondary consequence of weakness maximisation.

Corollary 3 (Weak constraints compress under weak usage bias)**.** *Let P be a distribution over $L_{\mathfrak{v}}$ such that $w(l_1) \geq w(l_2)$ implies $P(l_1) \geq P(l_2)$ (a **weak usage bias**). Let c^* be a minimum expected length prefix-free code for P. Then $w(l_1) \geq w(l_2)$ implies $|c^*(l_1)| \leq |c^*(l_2)|$.*

Intuition. Huffman coding assigns shorter codewords to more probable symbols. Under a weak usage bias, weaker statements are more probable. So weaker statements get shorter codes. The causal path is weakness $\rightarrow$ usage frequency $\rightarrow$ short code, not short code $\rightarrow$ generalisation.

Corollary 4 (Simplicity-generalisation correlation is induced by weakness)**.** *Under the conditions of Corollary 3, for any two correct policies $\pi_1, \pi_2 \in \Pi_\alpha$ with $w(\pi_1) > w(\pi_2)$, the code-length proxy agrees with the weakness proxy: $|c^*(\pi_1)| \leq |c^*(\pi_2)|$. Code length appears to predict generalisation. But the prediction fails whenever the weak usage bias is violated, which can happen under a different vocabulary or selection pressure.*

Intuition. Weakness causes both generalisation and (under weak usage bias) short code length. Short code length does not cause generalisation. Change the vocabulary so weak statements are no longer the most common, and the simplicity-generalisation correlation breaks. The weakness-generalisation link does not.

UNSEATING AIXI

AIXI[71] selects policies by minimising description length under a fixed universal Turing machine. This subsection shows this is not Bayes optimal under the task-growth uncertainty model of Stack Theory. The weakest-correct rule (MCL) is.

[71] Hutter, M. (2005). *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, Berlin

Corollary 5 (Occam priors collapse under no abstraction)**.** *If $\mathfrak{v} = \mathcal{P}$ (no abstraction), then every statement has a singleton representation with description length 1 (Proposition 7). Any code-length prior assigns equal probability to all policies. The Occam prior becomes uniform.*

Proposition 12 (Every finite prior is an Occam prior up to a constant)**.** *Let H be a finite hypothesis set. For any probability distribution Q on H, there exists a prefix-free code κ_Q such that $P_{\kappa_Q}(h) \propto 2^{-|\kappa_Q(h)|}$ satisfies $P_{\kappa_Q}(h) = Q(h)$ up to a normalising constant bounded by 2.*

Intuition. On a finite set, every prior is an Occam prior for some code. The information is in the choice of code, which is the choice of vocabulary. "Use the Occam prior" is a tautology.

Theorem 13 (MCL is Bayes optimal under exchangeable task growth)**.** *Fix a finite vocabulary $\mathfrak{v}$ and a child task α. Assume the parent task ω is drawn from the exchangeable extension model of Theorem 3, so uncertainty lies only in the unseen region while correctness on the seen region is held fixed. Let MCL denote the deterministic selection rule that picks a weakest correct policy $\pi^* \in \arg\max_{\pi \in \Pi_\alpha} w(\pi)$. Then*

$$P(\pi^* \in \Pi_\omega \mid \alpha) = \max_{\pi \in \Pi_\alpha} P(\pi \in \Pi_\omega \mid \alpha).$$

More generally, among all possibly randomised selectors supported on Π_α, MCL maximises the expected generalisation probability.

Intuition. MCL (the weakest-correct rule) is Bayes optimal under the task-growth prior. AIXI's Occam rule is not. MCL replaces "shortest correct" with "weakest correct."

REGRET AS WEIGHTED FORGETTING

This subsection establishes an exact algebraic identity between average normalised regret in reinforcement learning and margin-weighted forgetting cost in Stack Theory. These results first appeared in *Regret Is Weighted Forgetting*[72]. Full proofs are in the Theorems and Proofs chapter.

[72] Bennett, M. T. (2026c). Regret is weighted forgetting

Fix a finite evaluation support $\mathcal{X} = \{x_1, \ldots, x_n\}$ with $x_i = (h_i, T_i)$ (history-test pairs), probability masses $\mu_i > 0$ summing to 1, conditional probabilities $p_i := p_{T_i}(h_i)$, margins $m_i := |p_i - \frac{1}{2}|$, and optimal bets $y_i \in \{0, 1\}$.

Fix a candidate memory representation M. Write $i \sim_M j$ iff $M(h_i) = M(h_j)$ and $T_i = T_j$. Let $\mathcal{C}_M$ be the set of representation-test cells. An M-based policy chooses one report probability $q_C \in [0, 1]$ per cell $C \in \mathcal{C}_M$.

Definition 42 (Margin weight). *For each support point x_i, define $\rho_i := 2m_i/(\frac{1}{2} + m_i) \in [0, 1]$.*

Intuition. ρ_i is the normalised regret penalty per unit of wrong-action mass on test i. When a test is nearly a coin flip, ρ_i is near zero. When a test is decisive, ρ_i is near one.

Lemma 11 (Pointwise regret decomposition). *Let π be any M-based policy with report probability q_C on cell C. For $i \in C$,*

$$\delta_i(\pi) = \rho_i\big((1 - q_C)\mathbf{1}\{y_i = 1\} + q_C\mathbf{1}\{y_i = 0\}\big).$$

Intuition. Regret is just mistake probability scaled by how informative the test is.

Definition 43 (Margin-weighted forgetting cost). *The margin-weighted forgetting cost of M is*

$$K_\rho(M) := \min\Big\{\sum_{i \in B} \mu_i\rho_i \;\Big|\; B \subseteq \{1, \ldots, n\},\ y \text{ is constant on } C \setminus B \text{ for every } C \in \mathcal{C}_M\Big\}.$$

Intuition. You forget (delete) just enough support points, weighted by their diagnostic importance, so that inside every representation-test cell the correct bet becomes single-valued.

Theorem 14 (Exact reduction). *For every fixed memory representation M,*

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i\delta_i(\pi) = K_\rho(M).$$

Intuition. Minimum average normalised regret equals the minimum margin-weighted deletion cost needed to make the optimal bet a function of the representation-test cell. The two quantities are not merely analogous. They are equal.

Remark 7 (The optimal bet is selective forgetting)**.** *The identification is not merely numerical. The optimal M-based policy's per-cell binary choice (bet on $y = 1$ or $y = 0$) is itself a deletion operator. In each cell, the policy commits to one label class and discards the other. The discarded class is the cheaper one, and its margin-weighted mass is the cell's contribution to $K_\rho(M)$. So the RL-side optimal policy is not merely computing a cost equal to selective forgetting. It is* performing *selective forgetting on the diagnostic support.*

Corollary 6 (Policy-wise decomposition)**.** *For each cell $C \in \mathcal{C}_M$, let $A_C := \sum_{i\in C,\, y_i=1} \mu_i\rho_i$, $B_C := \sum_{i\in C,\, y_i=0} \mu_i\rho_i$, and let $q_C^\star \in \arg\min_{q\in[0,1]}(A_C(1-q) + B_C q)$ be the optimal report. Then every M-based policy π with report probabilities q_C satisfies*

$$\sum_{i=1}^{n} \mu_i\delta_i(\pi) = K_\rho(M) + \sum_{C\in\mathcal{C}_M} (A_C - B_C)(q_C^\star - q_C).$$

The second term is nonnegative and equals zero iff $q_C = q_C^\star$ for every cell.

Definition 44 (Prior-weighted weakness)**.** *Let $U = \{u_1, \dots, u_n\}$ be a finite unseen region with weights $w_1, \dots, w_n \geq 0$. For any policy ϑ, define its prior-weighted weakness on U by*

$$W(\vartheta) := \sum_{u_i\in \mathrm{Ext}(\vartheta)\cap U} w_i.$$

Intuition. Ordinary weakness counts unseen continuations a policy leaves open. Weighted weakness counts how much weighted unseen future it leaves open. When all weights are equal, this reduces to ordinary weakness up to a constant.

Proposition 13 (Independent nonuniform priors give weighted weakness)**.** *Assume each unseen output $u_i \in U$ becomes relevant independently with probability $r_i \in [0, 1)$. Then for any correct child policy ϑ,*

$$\log \Pr(\vartheta \text{ generalises}) = \sum_{u_i\notin \mathrm{Ext}(\vartheta)\cap U} \log(1 - r_i).$$

Maximising generalisation probability is the same as maximising $\sum_{u_i\in\mathrm{Ext}(\vartheta)\cap U}(-\log(1 - r_i))$. So the Bayes-optimal rule is weighted weakness maximisation with weights $w_i := -\log(1 - r_i)$.

Intuition. This is the weighted version of the sufficiency theorem (Proposition 4). Uniform priors recover ordinary weakness. Biased priors recover weighted weakness.

Corollary 7 (Exact Stack-Theoretic form of the bridge)**.** *Set $w_i := \mu_i\rho_i$. Build a lifted diagnostic task whose unseen outputs are $u_1, \dots, u_n$. Let $Z := \sum_{i=1}^n w_i$ and $W^\star(M) := \max_{\vartheta \text{ admissible}} W(\vartheta)$. Then*

$$\inf_{\pi \text{ is M-based}} \sum_{i=1}^{n} \mu_i\delta_i(\pi) = Z - W^\star(M).$$

Intuition. Regret is the weighted unseen future the representation forced you to discard. This is the exact bridge between RL regret and Stack-Theoretic weakness.

Corollary 8 (Generalisation probability from regret)**.** *Choose independent relevance probabilities $r_i = 1 - e^{-\mu_i \rho_i}$. Under this prior, the probability that the optimal M-based policy generalises is* $\Pr(\textit{optimal policy generalises}) = e^{-K_\rho(M)}$.

Theorem 15 (Multi-class exact reduction)**.** *Let $y_i \in \{1, \dots, K\}$ for each support point. Define $A_C^{(k)} := \sum_{i \in C, y_i = k} \mu_i \rho_i$ for each cell C and class k. Then*

$$\inf_{\pi \text{ is } M\text{-based}} \sum_{i=1}^{n} \mu_i \delta_i(\pi) = \sum_{C \in \mathcal{C}_M} \Big(\sum_{k=1}^{K} A_C^{(k)} - \max_k A_C^{(k)} \Big).$$

Intuition. In each cell, the optimal policy picks the majority class (weighted by $\mu_i \rho_i$). The cost is the total weight of non-majority points. The binary case is recovered by $K = 2$.

Proposition 14 (Lipschitz stability of K_ρ)**.** *If $\hat{\rho}_i$ are perturbed margin weights, then $|K_{\hat{\rho}}(M) - K_\rho(M)| \leq \sum_{i=1}^{n} \mu_i |\hat{\rho}_i - \rho_i|$. Since ρ is Lipschitz with constant 4 on $(0,1)$ (the one-sided derivatives satisfy $|d\rho/dp| \leq 4$ on $(0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$), estimation error in conditional probabilities propagates as* $|K_{\hat{\rho}}(M) - K_\rho(M)| \leq 4 \sum_{i=1}^{n} \mu_i |\hat{p}_i - p_i|$.

Theorem 16 (Representation convergence)**.** *Let $I_+ = \{i : \mu_i \rho_i > 0\}$ be the informative support. Define the coarsest pure diagnostic partition $\mathcal{P}_+^\star$ on I_+ by grouping points that share the same test value T_i and optimal label y_i. Then $K_\rho(M) = 0$ iff $\mathcal{C}_M|_{I_+}$ refines $\mathcal{P}_+^\star$.*

*Call a representation M **informative-minimal zero-regret** if $K_\rho(M) = 0$ and there is no representation M' with $K_\rho(M') = 0$ such that $\mathcal{C}_{M'}|_{I_+}$ is strictly coarser than $\mathcal{C}_M|_{I_+}$. Then M is informative-minimal zero-regret iff*

$$\mathcal{C}_M|_{I_+} = \mathcal{P}_+^\star.$$

Consequently, any two informative-minimal zero-regret representations differ on I_+ only by a relabelling of representation states.

Intuition. At zero irreducible regret, every informative cell is pure. The coarsest pure partition is unique. Minimal competent representations converge on that same informative quotient.

Proposition 15 (Measurable exact reduction)**.** *Let $(\mathcal{X}, \Sigma, \mu)$ be a probability space. Let $(\mathcal{Z}, \mathcal{B})$ be a standard Borel space and let $\zeta : \mathcal{X} \to \mathcal{Z}$, $y : \mathcal{X} \to \{0,1\}$, and $\rho : \mathcal{X} \to [0,1]$ be measurable.*

For a measurable decision rule $q : \mathcal{Z} \to [0,1]$, define

$$R(q) := \mathbb{E}\Big[\rho\Big((1 - q(\zeta))\mathbf{1}\{y = 1\} + q(\zeta)\mathbf{1}\{y = 0\}\Big)\Big].$$

Let $a, b : \mathcal{Z} \to [0,1]$ *be measurable versions satisfying*

$$a(\zeta) = \mathbb{E}[\rho \mathbf{1}\{y=1\} \mid \sigma(\zeta)] \quad \textit{and} \quad b(\zeta) = \mathbb{E}[\rho \mathbf{1}\{y=0\} \mid \sigma(\zeta)] \qquad \textit{a.s.}$$

Then

$$\inf_{q:\mathcal{Z}\to[0,1] \ \textit{measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}].$$

Moreover any measurable selector

$$q^\star(z) \in \arg\min_{q\in[0,1]} \big(a(z)(1-q) + b(z)q\big)$$

attains the infimum; for example $q^\star(z) = \mathbf{1}\{a(z) > b(z)\}$.

Intuition. This is the conditional-expectation form of the exact reduction. The standard-Borel assumption guarantees that the conditional expectations can be represented as measurable functions of the summary variable ζ. In the finite case it reduces exactly to $K_\rho(M)$.

PHILOSOPHICAL AND BIOLOGICAL ANALOGUES

CAUSALITY

ORIGINAL DEFINITIONS are to be found in this reference[73].

[73] Bennett, M. T. (2023a). Emergent causality and the foundation of consciousness. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 52–61. Springer. OUCI metadata page: https://ouci.dntb.gov.ua/en/works/7BoXJMW4/

Definition 45 (intervention)**.**
*Intuitively, if int and obs are "events" which have happened, then I say that int has **caused** obs if obs would not have happened in the absence of int (counterfactual). In my formalism, an **event** is a statement in $L_{\mathfrak{v}}$, and an event **happens** or is **observed** iff it is a true statement. If obs $\in L_{\mathfrak{v}}$ is sensorimotor activity I interpret as an "observed event", and int $\in L_{\mathfrak{v}}$ is an **intervention** (by an organism or other agency, in the sense described by Pearl[74]) to cause that event, then obs $\subset$ int (because int could not be said to cause obs unless obs $\subset$ int).*

[74] Pearl, J. and Mackenzie, D. (2018). *The Book of Why: The New Science of Cause and Effect*. Basic Books, Inc., New York, 1st edition

Proposition 16 (Do-operator as conditioning on an intervention indicator)**.** *Let X and Y be random variables and let $do(X = x)$ denote the intervention that sets X to the value x. Introduce an auxiliary binary variable $A \in \{0, 1\}$ indicating the regime, where $A = 1$ means "intervene and set $X = x$" and $A = 0$ means "observe" (no intervention). Consider the augmented model in which X is generated by*

$$X := \begin{cases} x, & A = 1, \\ \textit{(its original mechanism)}, & A = 0. \end{cases}$$

Then the interventional distribution is an ordinary conditional in the augmented model:

$$\mathbb{P}(Y \in B \mid do(X = x)) \;=\; \mathbb{P}(Y \in B \mid A = 1) \qquad \textit{for all measurable } B.$$

Moreover, whenever $\mathbb{P}(X = x) > 0$,

$$\mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1)\,\mathbb{P}(A = 1 \mid X = x) + \mathbb{P}(Y \in B \mid A = 0, X = x)\,\mathbb{P}(A = 0 \mid X = x).$$

Proof sketch. I prove Pearl's *do* notation can be compiled away into ordinary conditioning once you represent interventions explicitly. Introduce an indicator variable A that records whether an intervention occurred. Conditioning on $A =$ true reproduces the interventional distribution, while conditioning on $A =$ false reproduces the observational distribution. The do operator was doing bookkeeping. The enriched vocabulary can do that bookkeeping directly[75].

[75] Full proof is in Theorems and Proofs.

Definition 46 (Causal identity candidates)**.**
Fix an embodied language $L_\mathfrak{v}$. Let $INT \subseteq L_\mathfrak{v}$ be a set of (putatively) intervention episodes *and $OBS \subseteq L_\mathfrak{v}$ a set of matched* non-intervention observations. *Define the candidate set*

$$\mathcal{C}(INT, OBS) := \Big\{ c \in L_\mathfrak{v} \setminus \{\emptyset\} \;\Big|\; (\forall\, int \in INT)\; c \subseteq int \text{ and } (\forall\, obs \in OBS)\; c \not\subseteq obs \Big\}.$$

*Each $c \in \mathcal{C}(INT, OBS)$ is a **causal-identity candidate** for "whatever is common to the interventions but not already present in the observations"*[76].

Intuition. I am looking for a classifier in the organism's own vocabulary. The condition $c \subseteq int$ says the signature is present whenever the source intervenes. The condition $c \not\subseteq obs$ says the signature is not already satisfied by ordinary background situations. I did this because I wanted a discriminator that tracks causal source without demanding an unrealistically clean separation between interventions and observations. In embodied vocabularies, literal disjointness is often impossible, but our own ability to discriminate between objects in the environment is evidence that through the fuzzy approximation of an abstraction layer at least a semblance of disjointness is well defined. Abstraction layer is what matters.

[76] (EXAMPLE) Suppose I have organisms $\mathfrak{a}$ (Alice) and $\mathfrak{b}$ (Bob). The inputs Alice has experienced so far $I_{\mathfrak{h}_{<t_\mathfrak{a}}}$ can be divided into those in which Bob affected Alice $I^{\mathfrak{b}}_{\mathfrak{a}}$ and those in which Bob did not $I^{\neg\mathfrak{b}}_{\mathfrak{a}} = I_{\mathfrak{h}_{<t_\mathfrak{a}}} \setminus I^{\mathfrak{b}}_{\mathfrak{a}}$. By affecting Alice, Bob has intervened in Alice's experience. Alice can construct a causal identity b for Bob corresponding to interventions $INT = I^{\mathfrak{b}}_{\mathfrak{a}}$ and observations $OBS = I^{\neg\mathfrak{b}}_{\mathfrak{a}}$. The objects which "exist" in Alice's experience are those for which she constructs a causal identity, so this is how Bob comes to exist as a distinct "object" which Alice experiences, rather than in parts of other objects.

Definition 47 (w-maximised causal identities)**.**
*Fix $INT, OBS \subseteq L_\mathfrak{v}$ and assume $\mathcal{C}(INT, OBS) \neq \emptyset$. A **w-maximised causal identity** corresponding to (INT, OBS) is any candidate $c^\star \in \mathcal{C}(INT, OBS)$ that maximises weakness (Definition 4) among candidates:*

$$|\mathrm{Ext}(c^\star)| \;=\; \max\big\{\, |\mathrm{Ext}(c)| \;\big|\; c \in \mathcal{C}(INT, OBS) \,\big\}.$$

When $L_\mathfrak{v}$ is finite (e.g. when $\mathfrak{v}$ is finite), $\mathcal{C}(INT, OBS)$ is finite and the maximum exists.

Intuition. Many different signatures *might* work. W-maxing selects the candidate with the largest completion set, which reliably *does* work. Weakest means the one with the largest extension. It is the least committal separator. Picking the weakest separator leaves the most room for later refinement and reuse. Because weakness is cardinality-based, this need not coincide with fewest conjuncts or inclusion minimality, even though inclusion relations, when they exist, still imply corresponding weakness comparisons (Remark 1).

Proposition 17 (Existence of w-maximised causal identities)**.**
Assume $L_\mathfrak{v}$ is finite and let $INT, OBS \subseteq L_\mathfrak{v}$. If $\mathcal{C}(INT, OBS) \neq \emptyset$, then there exists at least one w-maximised causal identity for (INT, OBS) in the sense of Definition 47.

Proof sketch. I show a w-maximised causal identity exists whenever there are only finitely many candidate signatures. With finitely many candidates, their weakness values form a finite set of natural numbers. Any finite set has a maximum. The maximiser is the weakest separator that still distinguishes interventions from observations[77]. If a policy π is a causal identity, then the associated task is to classify interventions: given an event $e \in L_{\mathfrak{v}}$, decide "source present" iff $\pi \subseteq e$.

[77] Full proof is in Theorems and Proofs.

Definition 48 (preconditions)**.**
If $\mathfrak{o}$ is an organism, and c is a causal identity:

- *the **representation** precondition is met iff $c \in L_{\mathfrak{v}_\mathfrak{o}}$, and*
- *the **incentive** precondition is met if $\mathfrak{o}$ must learn c to remain "fit"[78].*

[78] It is possible an organism might construct c even if it is not required for the organism to remain fit, hence 'if' instead of 'iff'. Incentive is a sufficient precondition in conjunction with representation, but it is not strictly necessary.

Definition 49 (purpose, goal or intent)**.**
*I consider a policy c which is a causal identity corresponding to INT and OBS to be the **intent**, **purpose** or **goal** ascribed to the interventions. c is what the interventions share in common, meaning the "name" or "identity" of behaviour is the "intent", "goal" or "purpose" of behaviour. Just as an intervention caused an observation, the particular intent which motivated the agency undertaking the intervention is what caused it (to correctly infer intent, one must infer a causal identity that implies subsequent interventions).*

Proposition 18 (Intelligence is not independent of goals)**.** *Intelligence is not independent of goals.*

Proof sketch. I show intelligence, embodiment, and goals cannot be analysed independently in this formalisation. The proof chooses a deliberately minimal setup. Mind is a program. Body is an interpreter from programs to behaviour. Goals are success predicates over behaviour in an environment. Once you write the composition explicitly, changing the body changes which behaviours the same program can produce, and changing the goal changes what counts as success for the same behaviour. Success at a goal is therefore inseparable from the embodied mapping that makes action possible[79].

[79] Full proof is in Theorems and Proofs.

Theorem 17 (Intervention tagging is unavoidable)**.** *Let an organism $\mathfrak{o}$ update an internal state s after each episode and choose actions as a function of that state. If expected utility can be improved by conditioning on whether an episode was self-generated or merely observed, while holding fixed the represented episode content in $L_{\mathfrak{v}_\mathfrak{o}}$, then the internal state must encode an intervention tag.*

Proof sketch. Assume no such tag exists. Then episodes that differ only in whether they were produced by the organism or merely observed collapse to the same internal state. Any policy defined on that state must therefore treat them the same way. That contradicts the assumption that expected utility can be improved by conditioning on the distinction. The full proof is in the Theorems and Proofs chapter.
Intuition. If it matters whether a change was caused by me or merely witnessed by me, then the organism has to store that difference somewhere. A first-order self is the w-maximised version of exactly such a tag[80].

[80] Bennett, M. T. (2026b). No selves, no consciousness. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

Definition 50 (first order self)**.**
Let an organism $\mathfrak{o}$ have embodied language $L_{\mathfrak{v}_\mathfrak{o}}$. Let $INT_\mathfrak{o} \subseteq L_{\mathfrak{v}_\mathfrak{o}}$ denote (idealised) self-intervention episodes *for $\mathfrak{o}$ and let $OBS_\mathfrak{o} \subseteq L_{\mathfrak{v}_\mathfrak{o}}$ denote matched* non-self observations*. Any w-maximised causal identity*

$$\mathfrak{o}^1 \in \mathcal{C}(INT_\mathfrak{o}, OBS_\mathfrak{o})$$

*is called a **first order self** for $\mathfrak{o}$ (sometimes written $c^{\mathfrak{o}}_{\mathfrak{o}}$). Existence requires the* scale/representation *condition $\mathcal{C}(INT_\mathfrak{o}, OBS_\mathfrak{o}) \neq \emptyset$ and an* incentive *condition that viability depends on making the self/non-self discrimination.*

Intuition. A first order self is a reafference-like tag that distinguishes self produced from externally produced change. It is a particular kind of causal identity candidate that fires on the organism's own interventions. Under w-maxing it becomes the weakest tag the organism can build in its own vocabulary that still supports causal credit assignment for its own actions. That gives you a self variable that is minimal but functional, which is exactly what you want if you are trying to stay alive rather than write poetry about consciousness. Also it amounts to reafference which is observable in the human mid-brain and insect central complex[81].

[81] Merker, B. (2005). The liabilities of mobility: A selection pressure for the transition to consciousness in animal evolution. *Consciousness and Cognition*, 14(1):89–114. Neurobiology of Animal Consciousness; Merker, B. (2007). Consciousness without a cerebral cortex: a challenge for neuroscience and medicine. *Behavioral and Brain Sciences*, 30(1):63–81; and Barron, A. B. and Klein, C. (2016). What insects can tell us about the origins of consciousness. *PNAS*, 113(18):4900–4908

Definition 51 (Chain notation for nested causal-identity modelling)**.**
*Suppose I have two agents, $\mathfrak{a}$ (Alice) and $\mathfrak{b}$ (Bob). A **causal identity** $c^{\mathfrak{b}}_{\mathfrak{a}}$ denotes the internal model of $\mathfrak{b}$ constructed by $\mathfrak{a}$. Informally it is what Alice takes Bob to intend or cause.*

More generally, for a finite chain of agents $\mathfrak{x}_1, \dots, \mathfrak{x}_k$ I write

$$c_{\mathfrak{a}}^{\mathfrak{x}_1 \mathfrak{x}_2 \cdots \mathfrak{x}_k}$$

for Alice's nested causal model in which $\mathfrak{x}_1$ models $\mathfrak{x}_2$ models … models $\mathfrak{x}_k$. For example, $c^{\mathfrak{b}\mathfrak{a}}_{\mathfrak{a}}$ denotes Alice's model of Bob's model of Alice ("what Alice thinks Bob thinks Alice intends").

The maximal chain depth that $\mathfrak{a}$ can represent is constrained by $\mathfrak{a}$'s vocabulary $\mathfrak{v}_{\mathfrak{a}}$. Finally, Bob need not be a biological organism: $\mathfrak{b}$ can be any system for which Alice constructs a causal identity.

Definition 52 (n^{th}-order self)**.**
*Fix an agent $\mathfrak{a}$. An n^{th}**-order self** of $\mathfrak{a}$ is any causal identity whose chain ends at $\mathfrak{a}$ after interleaving $\mathfrak{a}$ with $n-1$ (not necessarily distinct) other-agent models.*

Formally, set

$$\mathfrak{a}^1 := c_{\mathfrak{a}}^{\mathfrak{a}}.$$

For $n \geq 2$, choose agents $\mathfrak{x}_1, \ldots, \mathfrak{x}_{n-1}$ and set

$$\mathfrak{a}^n := c_{\mathfrak{a}}^{\mathfrak{x}_1 \mathfrak{a} \mathfrak{x}_2 \mathfrak{a} \cdots \mathfrak{x}_{n-1} \mathfrak{a}}.$$

For example, relative to another agent $\mathfrak{b}$ and taking every $\mathfrak{x}_j = \mathfrak{b}$, I have

$$\mathfrak{a}^1 = c_{\mathfrak{a}}^{\mathfrak{a}}, \qquad \mathfrak{a}^2 = c_{\mathfrak{a}}^{\mathfrak{b}\mathfrak{a}}, \qquad \mathfrak{a}^3 = c_{\mathfrak{a}}^{\mathfrak{b}\mathfrak{a}\mathfrak{b}\mathfrak{a}}.$$

I use $\mathfrak{a}^n$ to denote an arbitrary n^{th}-order self, and chain notation (Definition 51) to denote a specific one (e.g. $c_{\mathfrak{a}}^{\mathfrak{b}\mathfrak{a}}$).

Theorem 18 (Second-order self is necessary for Gricean report)**.** *Consider a communication game in which a speaker $\mathfrak{a}$ must transmit a private bit $m \in \{0,1\}$ to a listener $\mathfrak{b}$. The listener type $t \in \{0,1\}$ is drawn uniformly at random and hidden from $\mathfrak{a}$. If $t = 0$, the listener decodes literally. If $t = 1$, the listener flips the decoded bit. Assume $\mathfrak{a}$ may send one probe signal and observe one listener response before sending the final signal. Any speaker policy that cannot condition on the listener response achieves expected success at most $\frac{1}{2}$. There exists a policy that conditions on the response and achieves success probability 1. Such conditioning requires a second-order self state $\mathfrak{a}^2$.*

Proof sketch. If the speaker cannot condition on the observed response, then conditioned on the intended bit the final signal is the same regardless of which decoder the audience is using. Half the time the audience interprets literally and half the time it flips the bit, so success averages to one half. With one probe, the speaker can identify which decoder is active and then choose the final signal accordingly. Representing that audience dependent adjustment is exactly what the second-order self is for. The full proof is in the Theorems and Proofs chapter.
Intuition. Meaning in the Gricean sense is audience relative. It is not enough to know what I intend. I also need a handle on how the audience will decode what I do. That handle is the second-order self[82].

[82] Bennett, M. T. (2026b). No selves, no consciousness. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

Proposition 19 ($\mathrm{n^{th}}$ order self convergence). *Let $\mathfrak{o}$ be an organism with current history task $\mathfrak{h}_{<t_\mathfrak{o}}$ and viability task $\mu_\mathfrak{o}$ such that $\mathfrak{h}_{<t_\mathfrak{o}} \sqsubset \mu_\mathfrak{o}$. Assume $\mathfrak{o}$ selects policies from $\Pi_{\mathfrak{h}_{<t_\mathfrak{o}}}$ using the weakness proxy and successfully learns $\mu_\mathfrak{o}$. Let $\mathfrak{o}^n$ be an $n^{\mathbf{th}}$-order self of $\mathfrak{o}$. If the representation precondition holds, so $\mathfrak{o}^n \in L_{\mathfrak{v}_\mathfrak{o}}$, and the incentive precondition holds, meaning that remaining fit requires learning $\mathfrak{o}^n$, then*

$$\mathfrak{o}^n \in \mathfrak{p}_\mathfrak{o}.$$

Proof sketch. Representation says the relevant self-model is expressible in the organism's language. Incentive says that remaining fit requires it to enter the organism's learned-policy repertoire. So once the organism successfully generalises from its current history to the viability task, the relevant $n^{\mathbf{th}}$-order self must be among the policies it knows[83].

[83] Full proof is in Theorems and Proofs.

HIGHER-ORDER SELVES VIA GAME-THEORETIC SEPARATION.

Definition 53 (A two-stage trust/commitment game). *Fix a commitment cost c with $0 < c < 1$. Two agents $\mathfrak{a}$ (Alice, the actor) and $\mathfrak{b}$ (Bob, the audience) play the following extensive-form game.*

1. ***Stage 0 (self-binding).*** *$\mathfrak{a}$ chooses either* Bind *(B) or* Free *(F). Choosing B incurs an immediate cost c to $\mathfrak{a}$ and removes $\mathfrak{a}$'s ability to* exploit *in Stage 2. Choosing F incurs no cost and leaves Stage 2 unconstrained.*
2. ***Stage 1 (audience response).*** *$\mathfrak{b}$ observes $\mathfrak{a}$'s choice (B or F) and chooses either* Trust *(T) or* Do not trust *(N). If $\mathfrak{b}$ chooses N, the game ends with payoff $(0, 0)$.*
3. ***Stage 2 (act).*** *If $\mathfrak{b}$ chose T, the following happens.*
   - *if $\mathfrak{a}$ chose B in Stage 0, $\mathfrak{a}$ is forced to play* Honour *(H).*
   - *if $\mathfrak{a}$ chose F in Stage 0, $\mathfrak{a}$ chooses either* Honour *(H) or* Exploit *(E).*

   Payoffs, before subtracting the binding cost, are

   $$u_\mathfrak{a}(T, H) = 1, \quad u_\mathfrak{b}(T, H) = 1, \qquad u_\mathfrak{a}(T, E) = 2, \quad u_\mathfrak{b}(T, E) = -1.$$

   If $\mathfrak{a}$ chose B in Stage 0, then c is subtracted from $\mathfrak{a}$'s payoff.

Intuition. This game separates two capabilities. Predicting another agent's behaviour is one thing. Making a publicly legible self binding move that changes the other agent's rational expectations is another. This is important because if the actor cannot bind itself, then exploitation is the rational last move after trust. That would mean trust is not rational for the audience, which would directly contradict empirical evidence. If binding is available, the actor can make trust rational by removing its own future exploit option, even at a cost.

Theorem 19 (A commitment game separates audience modelling from self-binding)**.** *In the game of Definition 53.*

1. *If the self-binding move B is* not available *(i.e. $\mathfrak{a}$ can only play F), then the unique subgame-perfect equilibrium outcome is N with payoff $(0, 0)$.*

2. *If B is available and $0 < c < 1$, then there exists a subgame-perfect equilibrium in which $\mathfrak{a}$ plays B, $\mathfrak{b}$ plays T after observing B (and N after observing F), and the Stage 2 outcome is necessarily H, yielding payoff $(1 - c, 1)$.*

Proof sketch. I show the commitment game is solved by backward induction. If the actor is unbound, exploitation is optimal in the final move, so the audience anticipates it and refuses to trust. If the actor can bind itself, exploitation is unavailable, and trust becomes rational. The proof is just the standard game theoretic logic written cleanly[84].

[84] Full proof is in Theorems and Proofs.

Remark 8 (Mapping back to selves and w-maxing)**.** *The extensive-form result above is game-theoretic. It does not presuppose any particular cognitive architecture. However, it isolates a mechanism relevant to the self-hierarchy: in niches where viability depends on* credible *long-horizon coordination (trust, reputation, norm compliance), "knowing what the audience currently thinks" is not enough. What is additionally required is a way for the actor to constrain its own future policy class so that the audience can rationally treat present commitments as stable. This is the functional role of a 3rd-order self operation: a* policy about policies*.*

Within Stack Theory, the same point can be expressed as a w-maxing construction. Consider the set of candidate self-binding constraints that satisfy both conditions.

1. *they make the trust branch viable by eliminating exploitative completions*

2. *they are compatible with the actor's embodied vocabulary*

Among this feasible set, w-maxing selects a maximally weak self-binding: the least-committal modification of future policy space that still rules out the completions that would cause the audience to withdraw trust.

ORIGINS OF LIFE

I TRANSLATE THE CONCEPTS of Wong et. al.[85] into Pancomputational Enactivism as follows:

1. Their function x corresponds to a $\mathfrak{v}$-task α in vocabulary $\mathfrak{v}$. A "future selection pressure" corresponds to a parent task ω of α.
2. Their "configuration" corresponds to a policy π. Their scalar score $F_x(\cdot)$ ("degree of function") can be mapped onto weakness via extension size. Informally, higher $F_x(\pi)$ corresponds to larger $|\mathrm{Ext}(\pi)|$ (weaker constraints), hence to higher probability of generalising from α to its parent ω.
3. They define $M(F_x)$ as the number of configurations whose degree of function is at least F_x. Under the mapping above, a natural analogue is

$$M_\pi \ := \ \{\, \pi_{\mathrm{alt}} \in \Pi_\alpha : |\mathrm{Ext}(\pi_{\mathrm{alt}})\,| \geq |\mathrm{Ext}(\pi)\,|\,\},$$

   i.e. the number of correct policies for α that are at least as weak as π. (This ensures M_π is nonempty because $\pi \in M_\pi$.)
4. Their N is the total number of possible configurations. In the *Why Life Exists* formalism, this is applied inside the currently viable set Π_α. A direct analogue is $N := |\Pi_\alpha|$.
5. Functional information $I(F_x) = -\log_2\left(\frac{M(F_x)}{N}\right)$ then becomes

$$I(\pi) \ := \ -\log_2\left(\frac{|M_\pi|}{|\Pi_\alpha|}\right).$$

   Under this translation, higher functional information corresponds to weaker (larger-extension) correct policies. Within Π_α this differs from using $|L_\mathfrak{v}|$ only by a constant shift. It does not change the ordering of policies by functional information.

[85] Wong, M. L., Cleland, C. E., Arend, D., Bartlett, S., Cleaves, H. J., Demarest, H., Prabhu, A., Lunine, J. I., and Hazen, R. M. (2023). On the roles of function and selection in evolving systems. *Proceedings of the National Academy of Sciences*, 120(43):e2310223120

IF ONE ACCEPTS MY INTERPRETATION of functional information as being a function of weakness, then I can make the selection claim precise. Under the same maximally uninformative extension model, persistence into the future biases surviving policies toward weakness. Under that bias, functional information increases in expectation.

Proposition 20 (The Law of Increasing Functional Information)**.** *Fix a finite vocabulary $\mathfrak{v}$ and a task $\alpha \in \Gamma_\mathfrak{v}$ with $\Pi_\alpha \neq \varnothing$. Draw π uniformly from Π_α. Independently draw a random parent task ω of α using the maximally uninformative extension model of Proposition 4. Assume persistence into time $t{+}1$ means surviving exactly when $\pi \in \Pi_\omega$. Then the expected* *functional information of the persisting population is at least as large as it was at time t. More precisely,*

$$\mathbb{E}\big[I(\pi) \mid \pi \in \Pi_\omega\big] \;\geq\; \mathbb{E}\big[I(\pi)\big],$$

with strict inequality whenever the weakness proxy $<_w$ is not constant on Π_α.

Proof sketch. I show the expected functional information increases over time under the selection model used here. Functional information is a monotone transform of weakness inside Π_α. Under the maximally uninformative extension model, the probability of persisting into the future is increasing in weakness. Conditioning on persistence reweights the population toward weaker policies. Since weaker policies have higher functional information under the chosen mapping, the expected functional information of survivors is higher than it was before selection[86].

[86] Full proof is in Theorems and Proofs.

SEMIOSIS

Original definitions are to be found in this reference[87].

Definition 54 (organism)**.**
I describe the circumstances of an organism[88] $\mathfrak{o}$ *as* $\langle \mathfrak{v}_\mathfrak{o}, \mu_\mathfrak{o}, \mathfrak{p}_\mathfrak{o}, <_\mathfrak{o} \rangle$ *where:*

- $O_{\mu_\mathfrak{o}}$ *contains every output which qualifies as "fit" according to natural selection.*
- $\mathfrak{p}_\mathfrak{o}$ *is the set of policies an organism knows, s.t.* $\mathfrak{p}_\mathfrak{o} \subset \mathfrak{p}_{n.s.} \cup \mathfrak{p}_{\mathfrak{h}_{<t_\mathfrak{o}}}$ *and:*
  - $\mathfrak{p}_{n.s.} \subset L_{\mathfrak{v}_\mathfrak{o}}$ *is* ***reflexes*** *hard coded from birth by natural selection.*
  - $\mathfrak{p}_{\mathfrak{h}_{<t_\mathfrak{o}}} = \bigcup_{\zeta \in \mathfrak{h}_{<t_\mathfrak{o}}} \Pi_\zeta$ *is the set of policies it is possible to* ***learn*** *from a history of past interactions represented by a task* $\mathfrak{h}_{<t_\mathfrak{o}}$.
  - *If* $\mathfrak{p}_{\mathfrak{h}_{<t_\mathfrak{o}}} \not\subset (\mathfrak{p}_\mathfrak{o} - \mathfrak{p}_{n.s.})$ *then the organism has* ***selective memory****. It can "forget" outputs, possibly to productive ends if they contradict otherwise good policies.*
- $<_\mathfrak{o}$ *is a binary relation over* $\Gamma_{\mathfrak{v}_\mathfrak{o}}$ *I call* ***preferences****.*

Definition 55 (protosymbol system)**.**
Assume an organism $\mathfrak{o}$*. For each policy* $p \in \mathfrak{p}_\mathfrak{o}$ *there exists a set* $\mathfrak{s}_p = \{\alpha \in \Gamma_{\mathfrak{v}_\mathfrak{o}} : p \in \Pi_\alpha\}$ *of all tasks for which* p *is a correct policy. The union of all such sets is*

$$\mathfrak{s}_\mathfrak{o} = \bigcup_{p \in \mathfrak{p}_\mathfrak{o}} \{\alpha \in \Gamma_{\mathfrak{v}_\mathfrak{o}} : p \in \Pi_\alpha\}$$

I call $\mathfrak{s}_\mathfrak{o}$ *a "protosymbol system". A* $\mathfrak{v}$*-task* $\alpha \in \mathfrak{s}_\mathfrak{o}$ *is called a "protosymbol".*

Definition 56 (Affect)**.**
Fix an organism $\mathfrak{a}$ *and suppose that, given an input statement* $i \in L_{\mathfrak{v}_\mathfrak{a}}$*,* $\mathfrak{a}$ *would (under its current interpretation procedure) produce some output* $o \in \mathrm{Ext}(i)$.

- *A* ***substatement*** $v \subseteq i$ ***affects*** $\mathfrak{a}$ *at input* i *if, on the reduced input* $i \setminus v$*,* $\mathfrak{a}$ *would instead produce some output* $g \in \mathrm{Ext}(i \setminus v)$ *with* $g \neq o$*. Equivalently, removing* v *changes* $\mathfrak{a}$*'s behavioural choice.*
- *Given a second organism (or system)* $\mathfrak{b}$*, I say that* $\mathfrak{b}$ ***affects*** $\mathfrak{a}$ *by making an output* k *if* k *causally changes* $\mathfrak{a}$*'s subsequent input to some* i' *for which there exists a substatement* $v \subseteq i'$ *that affects* $\mathfrak{a}$ *(in the sense above).*

[87] Bennett, M. T. (2023b). On the computation of meaning, language models and incomprehensible horrors. In *Artificial General Intelligence*. Springer Nature

[88] (INTUITIVE SUMMARY) Strictly speaking an organism $\mathfrak{o}$ would be a policy, but we can describe the circumstances of its existence as a task μ that describes all "fit" behaviour for that organism. We can also identify policies the organism "knows", because these are implied by the policy that is the organism. Likewise, we can represent lossy memory by having the organism "know" fewer policies than are implied by its history of interactions. Finally, preferences are the particular "protosymbol" the organism will use to "interpret" an input in later definitions.

Definition 57 (Interpretation)**.**
Interpretation is inference with an additional preference-based selection of which protosymbol (task) to treat as "the meaning" of the current input. It is an activity undertaken by an organism $\mathfrak{o}$ *(Definition 54) with protosymbol system* $\mathfrak{s}_\mathfrak{o}$ *(Definition 55).*

Given a true input $i \in L_{\mathfrak{v}_\mathfrak{o}}$ *(meaning* $i = i_t$ *in an EGRL system at time* t*), define the set of signified protosymbols as*

$$\mathfrak{s}^s_\mathfrak{o}(i) \ := \ \{\, \alpha \in \mathfrak{s}_\mathfrak{o} : i \in I_\alpha \,\}.$$

If $\mathfrak{s}^s_\mathfrak{o}(i) = \emptyset$*, then (in this scheme) the input does not trigger any learned protosymbol for* $\mathfrak{o}$*. Otherwise, interpretation proceeds as follows:*

1. *Choose* $\alpha^\star \in \mathfrak{s}^s_\mathfrak{o}(i)$ *that is maximal under preferences* $<_\mathfrak{o}$*.*
2. *Choose a known correct policy* $\pi \in \Pi_{\alpha^\star} \cap \mathfrak{p}_\mathfrak{o}$*.*
3. *Produce an output* $o \in \mathrm{Ext}(i) \cap \mathrm{Ext}(\pi)$*.*

MEANING

ORIGINAL DEFINITIONS are to be found in this reference[89]

[89] Bennett, M. T. (2023b). On the computation of meaning, language models and incomprehensible horrors. In *Artificial General Intelligence*. Springer Nature

Definition 58 (meaning).
*The **meaning** an organism $\mathfrak{o}$ ascribes to an input i is a protosymbol $\alpha \in \mathfrak{s}_{\mathfrak{o}}$ which $\mathfrak{o}$ uses to interpret i. Symbols in different protosymbol systems can be **roughly equivalent** (result in similar behaviour etc), in accord with the philosophical arguments in the body of the paper[90]. I use $\omega \approx \alpha$ to indicate that ω and α are roughly equivalent. In accord with earlier definitions, one organism "intends" to affect another if completion of the protosymbol (a task) the former uses to interpret hinges on how the latter's behaviour is affected.*
Assume an organism $\mathfrak{a}$ (Alice) in input $i_{\mathfrak{a}}$ and organism $\mathfrak{b}$ (Bob) in $i_{\mathfrak{b}}$. $\mathfrak{a}$ means $\alpha \in \mathfrak{s}_{\mathfrak{a}}$ by deciding $u \in \mathrm{Ext}(i_{\mathfrak{a}})$ iff $\mathfrak{a}$ intends in deciding u:

[90] I argue organisms of the same species construct roughly equivalent protosymbols, even though each member of a species exists in its own unique abstraction layer with its own protosymbol system.

1. *that $\mathfrak{b}$ interpret $i_{\mathfrak{b}}$ using $\omega \approx \alpha$,*
2. *$\mathfrak{b}$ "recognize" this intention by being affected by u such that the input $i_{\mathfrak{b}} = j$ in which $\mathfrak{b}$ finds itself change to $i_{\mathfrak{b}} = k \neq j$,*
3. *and (1) occur on the basis of (2), meaning had $\mathfrak{a}$ not decided u then $\mathfrak{b}$ would have interpreted $i_{\mathfrak{b}}$ using $\zeta \not\approx \alpha$.*

To communicate meaning organisms must:

1. *be able to affect one another.*
2. *have similar experiences, so $\mathfrak{s}_{\mathfrak{b}}$ and $\mathfrak{s}_{\mathfrak{a}}$ contain roughly equivalent symbols.*
3. *have similar preferences.*

Theorem 20 (Decoder mismatch breaks passive attribution). *Let an evaluator observe a trace y with hypothesis class*

$$\mathcal{H} = \mathcal{H}_{\text{mind}} \cup \{h_{\text{noise}}\}.$$

Assume $\mathbb{P}(h_{\text{noise}}) > 0$ and $\mathbb{P}(y \mid h_{\text{noise}}) > 0$. If there exists $\varsigma \in (0,1)$ such that for every $h \in \mathcal{H}_{\text{mind}}$,

$$\mathbb{P}(y \mid h) \leq \varsigma\, \mathbb{P}(y \mid h_{\text{noise}}),$$

then the posterior mass on the mind hypotheses satisfies

$$\mathbb{P}(\mathcal{H}_{\text{mind}} \mid y) \leq \frac{\varsigma\, \mathbb{P}(\mathcal{H}_{\text{mind}})}{\varsigma\, \mathbb{P}(\mathcal{H}_{\text{mind}}) + \mathbb{P}(h_{\text{noise}})}.$$

Proof sketch. Write the numerator of the posterior as the prior weighted sum of the mind likelihoods. The likelihood ratio assumption upper bounds that sum by ς times the noise likelihood multiplied by the prior mass on the mind hypotheses. Monotonicity of

$x \mapsto x/(x + A)$ then gives the bound. The full proof is in the Theorems and Proofs chapter.

Intuition. A mismatched decoder can make an intelligent trace look less likely than noise. If every mind-like hypothesis in the evaluator's model class assigns the trace a much smaller likelihood than the noise model, Bayes' rule drives the posterior away from mind. That is why passive attribution can fail even when the source is highly structured[91].

[91] Bennett, M. T. (2026b). No selves, no consciousness. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press

CONSCIOUSNESS

THE ORIGINAL definitions of these can be found in this reference[92]. A first order self correctly classifies the system's interventions, just as the neuroscientific phenomenon[93] known as "reafference" classifies interventions in living organisms. A first order self allows a system to reason about its own future interventions, plan spatial navigation and so forth.

[92] Bennett, M. T., Welsh, S., and Ciaunica, A. (2024). *Why Is Anything Conscious?* Preprint

[93] Barron, A. B. and Klein, C. (2016). What insects can tell us about the origins of consciousness. *PNAS*, 113(18):4900–4908

Theorem 21 (The Psychophysical Principle of Causality)**.** *This result appears as Theorem 3 in the supplementary materials of* [94]*. It is a corollary of The Law of the Stack (Theorem 4).*

[94] Bennett, M. T., Welsh, S., and Ciaunica, A. (2024). *Why Is Anything Conscious?* Preprint

Setup. *Let μ be a viability statement encoding the most basic valence partition (viable versus not viable). For any policy π, define the viable continuation set $\Omega_\pi := \mathrm{Ext}(\pi) \cap \mathrm{Ext}(\mu)$. Define the free energy proxy in bits*

$$F_2(\pi) := \log_2 |\mathrm{Ext}(\mu)| - \log_2 |\Omega_\pi|.$$

This equals $D_{\mathrm{KL},2}(q_\pi \| p_\mu)$ where q_π is uniform on Ω_π and p_μ is uniform on $\mathrm{Ext}(\mu)$.

Statement. *Let π be any policy with $\Omega_\pi \neq \emptyset$. Let $c \in L_\mathfrak{v}$ be a commitment not forced by viability, meaning there exists $o \in \mathrm{Ext}(\mu)$ with $c \not\subseteq o$. Assume Ω_π contains at least one viable continuation that violates c, meaning there exists $o \in \Omega_\pi$ with $c \not\subseteq o$. Then either $\Omega_{\pi \cup c} = \emptyset$ or $F_2(\pi \cup c) > F_2(\pi)$. Adding an unnecessary commitment strictly raises free energy whenever it rules out even one viable future.*

Proof sketch. $\pi \cup c \supseteq \pi$, so $\mathrm{Ext}(\pi \cup c) \subseteq \mathrm{Ext}(\pi)$ and hence $\Omega_{\pi \cup c} \subseteq \Omega_\pi$. The hypothesis gives $o \in \Omega_\pi$ with $c \not\subseteq o$, so $o \notin \mathrm{Ext}(c) \supseteq \mathrm{Ext}(\pi \cup c)$. Therefore $o \notin \Omega_{\pi \cup c}$. The inclusion is strict, $|\Omega_{\pi \cup c}| < |\Omega_\pi|$, and F_2 increases[95].

[95] Full proof is in the Theorems and Proofs chapter.

Intuition. One can learn causality by learning the objects and properties that fit a causal relation, or one can learn causal relations that fit a set of presupposed objects and properties. If you want generalisation-optimal learning, you cannot hard-code a quality-neutral ontology (no presupposed objects and properties). Only valence (the good-or-bad signal) can be a fixed representational primitive from which more abstract classifications are constructed as needed. Any fixed abstraction that is not a direct consequence of valence only serves to obfuscate the signal and impose a floor on variational free energy. Attraction and repulsion is how an organism carves up its world into objects and properties. Objects and properties are made of valence.

Definition 59 (levels of consciousness)**.**

1. *an organism that acts but does not learn, meaning $\mathfrak{p}_{\mathfrak{o}}$ is fixed from birth.*
2. *an organism that learns, but $\mathfrak{o}^1 \notin \mathfrak{p}_{\mathfrak{o}}$ either because $\mathfrak{o}^1 \notin L_{\mathfrak{v}_{\mathfrak{o}}}$ (failing the "representation precondition") or because the organism is not incentivised to construct $\mathfrak{o}^1$ (failing the "incentive precondition").*
3. *reafference and phenomenal or core consciousness are achieved when $\mathfrak{o}^1 \in \mathfrak{p}_{\mathfrak{o}}$ is learned by an organism as a consequence of attraction to and repulsion from statements in $L_{\mathfrak{v}_{\mathfrak{o}}}$.*
4. (a) *access or self reflexive consciousness is achieved when $\mathfrak{o}^2 \in \mathfrak{p}_{\mathfrak{o}}$.*
   (b) *hard consciousness[96] is achieved when a phenomenally conscious organism learns a second order self (an organism is consciously aware of the contents of second order selves, which must have quality if learned through phenomenal consciousness).*
5. *meta self reflexive consciousness (human level hard consciousness) is achieved when $\mathfrak{o}^3 \in \mathfrak{p}_{\mathfrak{o}}$.*

[96] Boltuc, P. (2012). The engineering thesis in machine consciousness. *Techné: Research in Philosophy and Technology*

ILLUSTRATIVE EXAMPLES

FOR THE SAKE OF INTUITION I will now provide some simple examples of how these definitions might be applied in the context of general reinforcement learning. In general reinforcement learning (GRL) $\mathcal{A}$,$\mathcal{O}$,$\mathcal{R}$ denote the (finite) action, observation and reward set respectively. The following definition of G.R.L. was taken from this reference[97].

[97] Catt, E., Grau-Moya, J., Hutter, M., Aitchison, M., Genewein, T., Deletang, G., Wenliang, L. K., and Veness, J. (2023). Self-predictive universal AI. In *Thirty-seventh Conference on Neural Information Processing Systems*

- $\mathcal{E} := \mathcal{O} \times \mathcal{R}$ is the set of **percepts**.
- $\Delta\mathcal{X}$ is the set of distributions over $\mathcal{X}$.
- $\mathcal{H} := (\mathcal{A} \times \mathcal{E})^*$ is the set of **histories**, where $*$ signifies that histories can be of any length (for example, were I to type $(\mathcal{A} \times \mathcal{E})^3$ would just have all histories of length 3).
- $\pi : \mathcal{H} \to \Delta\mathcal{A}$ is a function that outputs distributions over actions given a history. π is called a **policy**.
- $\mu : \mathcal{H} \times \mathcal{A} \to \Delta\mathcal{E}$ is an **environment**. An environment is a distribution over percepts *given* a history, and an action.
- $h_{<t} := a_1 e_1 \ldots a_{t-1} e_{t-1} = a_1 o_1 r_1 \ldots a_{t-1} o_{t-1} r_{t-1}$ denotes a history of length $t-1$.
- Given an environment μ and policy π, the conjunction $\mu\pi$ denotes the induced probability measure on histories, where $h_{<t}$ is assumed to be sampled from $\mu\pi$ (intuitively, an environment does the opposite of a policy, predicting the outcome of an action rather than which action comes next).
- GRL is concerned with how well policies perform (and by extension agents). I measure this performance by the future expected sum of discounted rewards, called value function. The agent selects a policy that maximises the value function.

Note that though it is not explicitly stated, the above definition assumes reward is between 0 and 1.

GRL AND PANCOMPUTATIONAL ENACTIVISM

HERE I REPRESENT THE DIFFERENT INGREDIENTS of GRL using $\mathfrak{v}$-tasks and an abstraction layer. This is just for illustrative purposes, to give some intuition about how this all works. It is one possible way to translate GRL into the context of my formalism, but is by no means the only way. The most important difference to note is that there is no need to append new outputs to the history. This is because it is an *embodied* formalism. The "input" is not just a stream from sensors, it is the state of the computer. That includes memory and the surrounding environment. The input at every step therefore includes memory implicitly. These are examples of how these definitions might be applied. I am not asserting that an EGRL system as described below is conscious. It leaves many details unresolved.

Definition 60 (EGRL)**.**
Fix a vocabulary $\mathfrak{v}$, a reward predicate $r : L_\mathfrak{v} \to \{1, 0\}$, and a finite horizon $len \in \mathbb{N}$ with $len > 0$.

The reward predicate induces a $\mathfrak{v}$-task $\mu = \langle I_\mu, O_\mu \rangle \in \Gamma_\mathfrak{v}$ by

$$O_\mu \;:=\; \{\, l \in L_\mathfrak{v} \mid r(l) = 1 \,\}, \qquad I_\mu \;:=\; \{\, i \in L_\mathfrak{v} \mid \exists o \in O_\mu \; (i \subseteq o) \,\}.$$

Assume $\Pi_\mu \neq \emptyset$.[98]

I model interaction up to time t using a history task $\mathfrak{h}_{<t} \in \Gamma_\mathfrak{v}$. Intuitively, $I_{\mathfrak{h}_{<t}}$ are the inputs encountered so far and $O_{\mathfrak{h}_{<t}}$ are the outputs produced so far.

Given $\mathfrak{h}_{<t}$, I split it into a positive and negative subhistory by applying r to past outputs. Define

$$O_{\mathfrak{h}^1_{<t}} \;:=\; \{\, o \in O_{\mathfrak{h}_{<t}} \mid r(o) = 1 \,\}, \qquad O_{\mathfrak{h}^0_{<t}} \;:=\; O_{\mathfrak{h}_{<t}} \setminus O_{\mathfrak{h}^1_{<t}}.$$

For $b \in \{0, 1\}$, let

$$I_{\mathfrak{h}^b_{<t}} \;:=\; \{\, i \in I_{\mathfrak{h}_{<t}} \mid \exists o \in O_{\mathfrak{h}^b_{<t}} (i \subseteq o) \,\}, \qquad \mathfrak{h}^b_{<t} \;:=\; \langle I_{\mathfrak{h}^b_{<t}}, O_{\mathfrak{h}^b_{<t}} \rangle.$$

*A **pair proxy** $<^{\mathfrak{pp}}$ is a binary relation on the set of admissible policy pairs*

$$\mathfrak{pp} \;:=\; \{(m, n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \mathrm{Ext}(m) \cap \mathrm{Ext}(n) = \emptyset\}.$$

*For such pairs, **learning** means choosing $(\pi, n) \in \mathfrak{pp}$ that is maximal under $<^{\mathfrak{pp}}$. Given a current input $i_t \in I_\mu$, **inference** means choosing an output $o_t \in \mathrm{Ext}(i_t) \cap \mathrm{Ext}(\pi)$.*

*Given $(\pi, n), (j, k) \in \mathfrak{pp}$, the **weakness pair proxy** is*

$$(j, k) <^{\mathfrak{pp}}_w (\pi, n) \quad \textit{iff} \quad |\mathrm{Ext}(j) \cup \mathrm{Ext}(k)| < |\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)|\,.$$

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[98] This is because I am trying to offer a computable and objectively optimal alternative to AIXI, which means I want lossless interpolation. If $\Pi_\mu = \emptyset$, then I would need lossy approximations, such as selective forgetting of past outputs that contradict otherwise good hypotheses.

Proposition 21 (Weakness pair proxy optimality)**.** *Let $\mathfrak{h}_{<t}$ be a history-task and let $\mathfrak{h}^1_{<t}$ and $\mathfrak{h}^0_{<t}$ denote the induced positive and negative history-tasks (as in Definition 60). Consider the admissible set of policy pairs*

$$\mathfrak{pp} \ := \ \{(\pi, n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \mathrm{Ext}(\pi) \cap \mathrm{Ext}(n) = \varnothing\}.$$

Define the effective completion set *of a pair as $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$. Under the same maximally uninformative extension model as in Proposition 4, apply the model to the effective completion set $E_{\pi,n}$. Then the probability that an admissible pair generalises to an unobserved parent is maximised by choosing $(\pi, n) \in \mathfrak{pp}$ with maximal $|\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)|$. Equivalently, the weakness pair proxy is optimal under that model.*

Proof sketch. I prove a disjoint positive negative pair is optimal exactly when it maximises the size of the union of allowed completions on unseen cases. With a positive negative pair you accept any parent that is consistent with either component on unseen outputs. Under the maximally uninformative extension model, generalisation probability depends only on how many unseen completions remain allowed. For an admissible disjoint pair (π, n), that allowed set is exactly $\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$, so the optimal pair is the one that makes this union as large as possible[99].

[99] Full proof is in Theorems and Proofs.

Algorithm 1: EGRL system (schematic)

If S is a set, then S.$\textsc{Choose}()$ returns an element of S.
If S is a set, then $S.\textsc{Add}(x) := S \cup \{x\}$.
function EGRL_$\textsc{Loop}$(μ, $<^{\mathfrak{pp}}$, r, len)
    $t \leftarrow 0$
    $\mathfrak{r}_{\mathfrak{h}_{<t}} \leftarrow 0$
    $\mathfrak{h}_{<t} \leftarrow \langle \emptyset, \emptyset \rangle$
    $i_t \leftarrow I_\mu.\textsc{Choose}()$
    while $t < len$ **do**
        $\pi \leftarrow \textsc{learning}(\mathfrak{h}_{<t}, <^{\mathfrak{pp}}, i_t, r)$ ▷ see learning methods
        $o_t \leftarrow \textsc{inference}(\pi, i_t)$
        $\mathfrak{r}_{\mathfrak{h}_{<t}} \leftarrow \mathfrak{r}_{\mathfrak{h}_{<t}} + r(o_t)$ ▷ feedback
        $I_{\mathfrak{h}_{<t}} \leftarrow I_{\mathfrak{h}_{<t}}.\textsc{Add}(i_t)$
        $O_{\mathfrak{h}_{<t}} \leftarrow O_{\mathfrak{h}_{<t}}.\textsc{Add}(o_t)$
        $t \leftarrow t + 1$
        $i_t \leftarrow I_\mu.\textsc{Choose}()$
    end while
    return $\mathfrak{r}_{\mathfrak{h}_{<t}}$
end function

function $\textsc{Inference}$(π, i_t)
    $E \leftarrow \mathrm{Ext}(\pi) \cap \mathrm{Ext}(i_t)$
    return $E.\textsc{Choose}()$
end function

Algorithm 2: LEARNING METHODS

For a $\mathfrak{v}$-task α, $\alpha.expand() := \{\omega \in \Gamma_{\mathfrak{v}} : \omega \sqsubset \alpha, \neg\exists\zeta \in \Gamma_{\mathfrak{v}}(\omega \sqsubset \zeta \sqsubset \alpha)\}$ gets the set of all α's immediate children (i.e. children which are only one generation away from α, meaning I exclude grandchildren etc)

$\mathfrak{q}$ is a prioritised queue. If (n, obj) is in $\mathfrak{q}$, then n is inverse priority (smaller n is higher priority). $\mathfrak{q}.put(n, obj)$ places (n, obj) in the queue, and $\mathfrak{q}.get()$ returns the highest priority item. $\mathfrak{q}.notempty()$ returns true iff $\mathfrak{q}$ is not empty.

If $\mathcal{V}$ is a set then $\mathcal{V}.add(\mathfrak{c}) := \mathcal{V} \cup \{\mathfrak{c}\}$.

function CLASSIFICATION($\mathfrak{h}_{<t}$, r) ▷ used by LEARNING()

  $O_{\mathfrak{h}^1_{<t}} \leftarrow \{o \in O_{\mathfrak{h}_{<t}} : r(o) = 1\}$ ▷ relatively "attractive" memories

  $O_{\mathfrak{h}^0_{<t}} \leftarrow O_{\mathfrak{h}_{<t}} - O_{\mathfrak{h}^1_{<t}}$

  $I_{\mathfrak{h}^1_{<t}} \leftarrow \{i \in I_{\mathfrak{h}_{<t}} : \exists o \in O_{\mathfrak{h}^1_{<t}}(i \subseteq o)\}$

  $I_{\mathfrak{h}^0_{<t}} \leftarrow \{i \in I_{\mathfrak{h}_{<t}} : \exists o \in O_{\mathfrak{h}^0_{<t}}(i \subseteq o)\}$

  $\mathfrak{h}^0_{<t} \leftarrow \langle I_{\mathfrak{h}^0_{<t}}, O_{\mathfrak{h}^0_{<t}} \rangle$

  $\mathfrak{h}^1_{<t} \leftarrow \langle I_{\mathfrak{h}^1_{<t}}, O_{\mathfrak{h}^1_{<t}} \rangle$

  return $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t}$

end function

Example 1 : A simple example showing how learning works without noise.

function LEARNING($\mathfrak{h}_{<t}$, $<^{\mathfrak{pp}}$, i_t, r)

  $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t} =$ CLASSIFICATION$(\mathfrak{h}_{<t}, r)$

  $N_{\mathfrak{h}^1_{<t}} \leftarrow \{j \in \Pi_{\mathfrak{h}^1_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(i_t) \neq \emptyset\}$ ▷ discard policies that don't apply

  $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \leftarrow \{(j, k) \in N_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(k) = \emptyset\}$ ▷ remove ambiguity

  return π s.t. $\exists(\pi, n) \in \Pi_{\mathfrak{h}^{(1,0)}_{<t}}$ that maximises $<^{\mathfrak{pp}}$

end function

Algorithm 3: LEARNING METHODS CONTINUED...

Example 2 : Learning is possible even with noise (the only drawback being that it is harder to understand) by selectively "forgetting" outputs when no valid policy. Given only noise, there would be a policy for each output (rote learning means $\mid \mathrm{Ext}(\pi) \mid \to 1$*).*

function LEARNING($\mathfrak{h}_{<t}$, $<^{\mathfrak{pp}}$, i_t, r)
  $\mathfrak{q} \leftarrow \mathfrak{q}.put(0, \mathfrak{h}_{<t})$
  $\mathcal{V} \leftarrow \{\mathfrak{h}_{<t}\}$ ▷ set of visited nodes
  while $\mathfrak{q}.notempty()$ **do**
    $n, \mathfrak{h} \leftarrow \mathfrak{q}.get()$
    $\mathfrak{h}^1_{<t}, \mathfrak{h}^0_{<t} = \text{CLASSIFICATION}(\mathfrak{h}, r)$
    $N_{\mathfrak{h}^1_{<t}} \leftarrow \{j \in \Pi_{\mathfrak{h}^1_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(i_t) \neq \emptyset\}$
    $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \leftarrow \{(j,k) \in N_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} : \mathrm{Ext}(j) \cap \mathrm{Ext}(k) = \emptyset\}$
    if $\Pi_{\mathfrak{h}^{(1,0)}_{<t}} \neq \emptyset$ **then**
      return π s.t. $\exists(\pi, n) \in \Pi_{\mathfrak{h}^{(1,0)}_{<t}}$ that maximises $<^{\mathfrak{pp}}$
    end if
    for all $\mathfrak{c} \in \mathfrak{h}.expand()$ **do**
      if $\mathfrak{c} \notin \mathcal{V}$ **then**
        $\mathfrak{q}.put(n+1, \mathfrak{c})$
        $\mathcal{V} \leftarrow \mathcal{V}.add(\mathfrak{c})$
      end if
    end for
  end while
  return $\emptyset$ ▷ try uninformed exploration when there is no valid policy
end function

Definition 61 (self-organising EGRL system)**.**
A self-organising EGRL system $\mathfrak{o}$ *is* $\langle \mathfrak{v}_{\mathfrak{o}}, \mu_{\mathfrak{o}}, \mathfrak{p}_{\mathfrak{o}}, <_{\mathfrak{o}} \rangle$ *accompanied by an E.G.R.L system* $\langle \mathfrak{v}_{\mathfrak{o}}, \mathfrak{h}_{<t_{\mathfrak{o}}}, <_{\mathfrak{o}}^{\mathfrak{pp}}, r_{\mathfrak{o}} \rangle$ *and an E.G.R.L problem* $\langle \mathfrak{v}_{\mathfrak{o}}, \mu_{\mathfrak{o}}, len \rangle$*:*

- $O_{\mu_{\mathfrak{o}}}$ *contains every output which qualifies as "fit" according to natural selection.*
- $\mathfrak{p}_{\mathfrak{o}}$ *is the set of policies the self-organising EGRL system knows, s.t.* $\mathfrak{p}_{\mathfrak{o}} \subset \mathfrak{p}_{n.s.} \cup \mathfrak{p}_{\mathfrak{h}_{<t_{\mathfrak{o}}}}$ *and:*
  - $\mathfrak{p}_{n.s.} \subset L_{\mathfrak{v}_{\mathfrak{o}}}$ *is* ***reflexes*** *hard coded from birth by natural selection.*
  - $\mathfrak{p}_{\mathfrak{h}_{<t_{\mathfrak{o}}}} = \bigcup_{\zeta \in \mathfrak{h}_{<t_{\mathfrak{o}}}} \Pi_{\zeta}$ *is the set of policies it is possible to* ***learn****.*
  - *If* $(\mathfrak{p}_{\mathfrak{o}} - \mathfrak{p}_{n.s.}) \not\subset \Pi_{\mathfrak{h}_{<t_{\mathfrak{o}}}}$ *then the self-organising EGRL system has* ***selective memory***[100]*.*
- $<_{\mathfrak{o}}$ *is a binary relation over* $\Gamma_{\mathfrak{v}_{\mathfrak{o}}}$ *I call* ***preferences****.*
- $\mathfrak{o}$ *is an EGRL system iff maximising* $<_{\mathfrak{o}}$ *maximises* r *and* $\mathfrak{p}_{\mathfrak{o}} = \mathfrak{p}_{\mathfrak{h}_{<t_{\mathfrak{o}}}}$.

[100] It can "forget" outputs that contradict otherwise good policies.

EXAMPLES

THE FOLLOWING are based on this reference[101]. As before these are meant to be illustrative, rather than final definitions:

[101] Bennett, M. T. (2024b). Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

Definition 62 (bagī)**.**
*"**B**ennett's **a**rtificial **g**eneral **i**ntelligence" (bagī) is an EGRL system $\langle \mathfrak{v}, \mathfrak{h}_{<t}, <_w^{\mathrm{pp}}, r \rangle$ that uses weakness as the pair proxy[102]. From this reference[103] I have that bagī is the optimal system, meaning it is "most intelligent" given embodiment, in that it maximises the ability to complete a wide range of tasks using the vocabulary and history.*

[102] This assumes a particular vocabulary, which acts as a vehicle for cognition (bagī or *bagī* means "buggy" in Japanese).

[103] Bennett, M. T. (2023c). The optimal choice of hypothesis is the weakest, not the shortest. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 42–51. Springer

Definition 63 (unagi)**.**
*The "**un**formed **a**rtificial **g**eneral **i**ntelligence" (unagi) formalises the upper bound given in this reference[104]. It is a system that searches through a chosen space of vocabularies*

$$V \subseteq 2^{\mathcal{P}}.$$

In a standard embodied setting I take V to be the finite vocabularies

$$V_{\text{fin}} := \{ \mathfrak{v} \subseteq \mathcal{P} \mid |\mathfrak{v}| < \infty \}$$

and set $V = V_{\text{fin}}$. If I want upper bounds for idealised infinite universes, I can allow V to contain infinite vocabularies too. The goal is to identify the bagī for which the utility ϵ is maximised.[105]

*Assume an "**u**nformed GRL problem" (UGRL), which is $\langle \lambda_\mu, len \rangle$ where λ_μ is an uninstantiated task and len is the time horizon. Assume also a UGRL system $\langle \lambda_{\mathfrak{h}_{<t}}, <_w^{\mathrm{pp}}, r \rangle$.*

An unagi returns a pair $(\mathfrak{v}^\star, \pi^\star)$ such that

1. *$\mathfrak{v}^\star$ maximises utility over the chosen vocabulary class V*

$$\mathfrak{v}^\star \in \arg\max_{\mathfrak{v} \in V} \epsilon\big(\lambda_{\mathfrak{h}_{<t}}(\mathfrak{v})\big)$$

2. *$\pi^\star$ is a w-maximal correct policy for the instantiated task in that embodiment*

$$\pi^\star \in \Pi_{\lambda_{\mathfrak{h}_{<t}}(\mathfrak{v}^\star)} \quad \textit{and} \quad |\mathrm{Ext}(\pi^\star)| = \max\{ |\mathrm{Ext}(\pi)| \mid \pi \in \Pi_{\lambda_{\mathfrak{h}_{<t}}(\mathfrak{v}^\star)} \}.$$

Equivalently, $(\mathfrak{v}^\star, \pi^\star)$ is an upper-bound-optimal pair in the sense of Definition 40 for the chosen vocabulary class.

Because $<^{\mathrm{pp}}$ is the weakness proxy, an unagi finds the bagī that is best suited to an uninstantiated task. It then constructs that bagī. It identifies the best tool for the problem. Then it builds it.

[104] Bennett, M. T. (2024b). Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

[105] Its namesake is the Japanese water eel, slippery and not tied to any particular embodiment. I used to get unagi lunches for free as an undergrad student. This theoretical unagi system requires a free lunch in the sense of the no free lunch theorem.

THEOREMS AND PROOFS

This section collects the full proofs for results stated earlier. Each result in the main text links here.

SPACE AND TIME

Proof of Result 1

Restatement of Lemma 1 (Encoding is the maximal true statement). For every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(\phi)$ is true at ϕ. If $l \in L_{\mathfrak{v}}$ is also true at ϕ, then $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$.

Proof. Let $\phi \in \Phi$. By definition, if $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$ then $\phi \in p$. Therefore $\phi \in \bigcap_{p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)} p = T(\mathrm{Enc}_{\mathfrak{v}}(\phi))$.

Now let $l \in L_{\mathfrak{v}}$ and assume $\phi \in T(l) = \bigcap_{p \in l} p$. Then for every $p \in l$ I have $\phi \in p$, hence $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)$. So $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. □

Proof of Result 1

Restatement of Proposition 1 (Layer time can be arbitrarily sparse). For any $M \in \mathbb{N}_{>0}$ there exist an environment Φ, a finite vocabulary $\mathfrak{v} \subseteq \mathcal{P}$, and an objective trajectory $\tau : \mathbb{N} \to \Phi$ such that the induced layer trajectory $\tau^{(\mathfrak{v})}$ changes at most once during the first M objective ticks.

Proof. Fix $M \in \mathbb{N}_{>0}$. Let $\Phi = \{\phi_0, \dots, \phi_M\}$. Let $\mathfrak{v} = \{p\}$ where $p = \Phi$. Define $\tau : \mathbb{N} \to \Phi$ by $\tau(u) = \phi_{u \bmod (M+1)}$. Then $\tau(u+1) \neq \tau(u)$ for all u, so τ is an objective trajectory.

For every u, since $\tau(u) \in p$, I have $\mathrm{Enc}_{\mathfrak{v}}(\tau(u)) = \{p\}$. So the induced layer trajectory $\tau^{(\mathfrak{v})}$ is constant, hence changes zero times during the first M objective ticks. □

Proof of Result 4

Restatement of Lemma 4 ($\Box_\Delta$ commutes with conjunction). For all statements $l \in L_{\mathfrak{v}}$,

$$\Box_\Delta T(l) \;=\; T(\Box_\Delta l).$$

Proof. Let $\sigma \in \Phi_\Delta$. I have

$$\begin{aligned}\sigma \in \Box_\Delta T(l) &\iff \forall k \leq \Delta\ \sigma_k \in T(l)\\ &\iff \forall k \leq \Delta\ \forall p \in l\ \sigma_k \in p\\ &\iff \forall p \in l\ \forall k \leq \Delta\ \sigma_k \in p\\ &\iff \forall p \in l\ \sigma \in \Box_\Delta p\\ &\iff \sigma \in \bigcap_{p \in l} \Box_\Delta p\\ &\iff \sigma \in T(\Box_\Delta l).\end{aligned}$$

Since this holds for all σ, $\Box_\Delta T(l) = T(\Box_\Delta l)$. $\square$

Proof of Result 5

Restatement of Lemma 5 ($\Diamond_\Delta$ does not commute with conjunction). For all statements $l \in L_\mathfrak{v}$,

$$\Diamond_\Delta T(l) \ \subseteq\ T(\Diamond_\Delta l).$$

If $\Delta \geq 1$ and l has at least two ingredients, then the inclusion can be strict.

Proof. Let $l \in L_\mathfrak{v}$. To prove $\Diamond_\Delta T(l) \subseteq T(\Diamond_\Delta l)$, let $\sigma \in \Diamond_\Delta T(l)$. Then there exists $k \leq \Delta$ such that $\sigma_k \in T(l) = \bigcap_{p \in l} p$. So for every $p \in l$ I have $\sigma_k \in p$, hence $\sigma \in \Diamond_\Delta p$. Therefore $\sigma \in \bigcap_{p \in l} \Diamond_\Delta p = T(\Diamond_\Delta l)$.

For strictness when $\Delta \geq 1$, it suffices to exhibit one counterexample. Choose two programs p, q and two states $\phi \in p \setminus q$ and $\psi \in q \setminus p$. Let $l = \{p, q\}$ and let $\sigma = (\phi, \psi, \ldots, \psi) \in \Phi_\Delta$. Then $\sigma \in \Diamond_\Delta p \cap \Diamond_\Delta q = T(\Diamond_\Delta l)$. But there is no index $k \leq \Delta$ with $\sigma_k \in p \cap q$, so $\sigma \notin \Diamond_\Delta(p \cap q) = \Diamond_\Delta T(l)$. $\square$

Proof of Result 2

Restatement of Proposition 2 (Persistence collapses the Temporal Gap). Fix $\Delta \in \mathbb{N}_{>0}$ and a trajectory τ. Let $l \in L_\mathfrak{v}$ be a statement whose every ingredient $p \in l$ is (τ, Δ)-persistent. Then for every objective time $u \in \mathbb{N}$,

$$\tau_\Delta(u) \models T(\Diamond_\Delta l) \quad \Longrightarrow \quad \tau_\Delta(u) \models \Diamond_\Delta T(l).$$

Proof. Fix $u \in \mathbb{N}$ and assume $\tau_\Delta(u) \models T(\Diamond_\Delta l)$. Unpacking the definitions, for each $p \in l$ there exists $k_p \leq \Delta$ such that $\tau(u + k_p) \in p$. Let $k := \max_{p \in l} k_p$. Then $k \leq \Delta$ and for every $p \in l$ I have $k_p \leq k$, so (τ, Δ)-persistence gives $\tau(u + k) \in p$. Thus $\tau(u + k) \in \bigcap_{p \in l} p = T(l)$, which means $\tau_\Delta(u) \models \Diamond_\Delta T(l)$. $\square$

Proof of Result 1

Restatement of Theorem 1 (Synchronous threshold for co-instantiation). Fix $n \geq 2$ and the conjunction l_n from Definition 17. If an architecture has concurrency capacity $c < n$, then along every admissible trajectory τ and for every horizon Δ,

$$\tau_\Delta(u) \not\models \Diamond_\Delta T(l_n) \quad \text{for all } u.$$

However, if each contributor $i \in [n]$ is active at least once within some horizon Δ window meaning there exists u such that for every i there exists $k_i \leq \Delta$ with $i \in A_{u+k_i}$ then

$$\tau_\Delta(u) \models T(\Diamond_\Delta l_n).$$

So the architecture admits ingredient-wise satisfaction but forbids co-instantiation.

Proof. Assume concurrency capacity $c < n$. Let $\tau(u) = (x_u, A_u)$ be any admissible trajectory and fix any u and any horizon Δ. Because $|A_{u+k}| \leq c < n$ for all k, I have $A_{u+k} \neq [n]$ for all k. So $\tau(u + k) \notin T(l_n)$ for all k, hence there is no $k \leq \Delta$ with $\tau(u+k) \in T(l_n)$. Therefore $\tau_\Delta(u) \not\models \Diamond_\Delta T(l_n)$.

For the second claim, assume there exists u such that for every $i \in [n]$ there exists $k_i \leq \Delta$ with $i \in A_{u+k_i}$. Then $\tau(u + k_i) \in p_i$ by definition of p_i, so $\tau_\Delta(u) \in \Diamond_\Delta p_i$ for every i. Thus $\tau_\Delta(u) \in \bigcap_{i=1}^{n} \Diamond_\Delta p_i = T(\Diamond_\Delta l_n)$, meaning $\tau_\Delta(u) \models T(\Diamond_\Delta l_n)$. □

Proof of Result 2

Restatement of Theorem 2 (Graph diameter spacetime bound). Assume propagation speed at most v. If U is G-integrated within duration θ, then for all $u, u' \in U$,

$$d(u, u') \leq \frac{v\theta}{2}\, d_G(u, u').$$

In particular,

$$\operatorname{diam}(U) \leq \frac{v\theta}{2}\, h(G).$$

Proof. Fix $u, u' \in U$ and let $u = u_0, u_1, \ldots, u_k = u'$ be a shortest path in G. So $k = d_G(u, u')$ by definition of hop distance.

For each edge $\{u_i, u_{i+1}\}$ the G-integration assumption gives travel times $\eta_{u_i \to u_{i+1}}$ and $\eta_{u_{i+1} \to u_i}$ with $\eta_{u_i \to u_{i+1}} + \eta_{u_{i+1} \to u_i} \leq \theta$. Therefore at least one of these two times is at most $\theta/2$. Using the speed ceiling in that faster direction gives

$$d(u_i, u_{i+1}) \leq \frac{v\theta}{2}.$$

Now apply the triangle inequality along the path

$$d(u,u') \le \sum_{i=0}^{k-1} d(u_i,u_{i+1}) \le k\frac{v\theta}{2} = \frac{v\theta}{2}\, d_G(u,u').$$

Taking the maximum over all $u, u' \in U$ gives $\mathrm{diam}(U) \le \frac{v\theta}{2} h(G)$. $\square$

Proof of Result 3

Restatement of Proposition 3 (Strong persistence is bounded by weak persistence). For every trajectory τ, statement ℓ^m, finite evaluation set U, and horizon Δ,

$$P_{\text{strong}}(\tau,\ell^m;U,\Delta) \le P_{\text{weak}}(\tau,\ell^m;U,\Delta).$$

Proof. Fix any $u \in U$. If $\mathrm{CoInst}_{W_\Delta}(\ell^m,\tau,u)$ holds, then by definition $\tau_\Delta(u) \models \Diamond_\Delta T(g^0)$, where g^0 is the grounded statement. Lemma 5 gives

$$\Diamond_\Delta T(g^0) \subseteq T(\Diamond_\Delta g^0),$$

so $\tau_\Delta(u) \models T(\Diamond_\Delta g^0)$. Hence $\mathrm{Occur}_{W_\Delta}(\ell^m,\tau,u)$ holds. Therefore,

$$\mathbf{1}\big[\mathrm{CoInst}_{W_\Delta}(\ell^m,\tau,u)\big] \le \mathbf{1}\big[\mathrm{Occur}_{W_\Delta}(\ell^m,\tau,u)\big]$$

for every $u \in U$. Summing over $u \in U$ and dividing by $|U|$ gives the result. $\square$

GROUNDING

Proof of Result 7

Restatement of Lemma 7 (One-step grounding preserves truth conditions). With the setup of Definition 38, for every $\ell \in L_{\mathfrak{v}'}$ I have

$$T(\mathrm{ground}_\pi(\ell)) \;=\; T(\ell).$$

Proof. Let $\ell \in L_{\mathfrak{v}'}$. By Definition 38, $\mathrm{ground}_\pi(\ell) = \bigcup_{p\in\ell} o_p$, so

$$\begin{aligned} T(\mathrm{ground}_\pi(\ell)) &= \bigcap_{q\in \mathrm{ground}_\pi(\ell)} q \\ &= \bigcap_{p\in\ell}\bigcap_{q\in o_p} q \\ &= \bigcap_{p\in\ell} T(o_p). \end{aligned}$$

By construction $T(o_p) = p$ for each $p \in \mathfrak{v}'$, so

$$T(\mathrm{ground}_\pi(\ell)) = \bigcap_{p\in\ell} p = T(\ell).$$

$\square$

Proof of Result 12

Restatement of Theorem 12 (Compositional grounding preserves truth conditions). With the setup of Definition 39, for every $\ell^m \in L_{\mathfrak{v}^m}$ I have

$$T(\text{Ground}_{0\leftarrow m}(\ell^m)) \;=\; T(\ell^m).$$

Proof. Let $g^m = \ell^m$ and $g^i = \text{ground}_{\pi^i}(g^{i+1})$ as in Definition 39. Applying Lemma 7 at each step gives

$$T(g^i) = T(g^{i+1}) \qquad \text{for } i = m-1, m-2, \ldots, 0.$$

Chaining these equalities yields $T(g^0) = T(g^m) = T(\ell^m)$. Since $g^0 = \text{Ground}_{0\leftarrow m}(\ell^m)$, the result follows. □

GENERALISATION OPTIMAL LEARNING

Proof of Result 4

Restatement of Proposition 4 (Sufficiency of weakness under a maximally uninformative prior). Let $\mathfrak{v}$ be finite and let α and ω be $\mathfrak{v}$-tasks with $\alpha \sqsubset \omega$. Write $U := L_{\mathfrak{v}} \setminus \text{Ext}(I_\alpha)$ for the set of *unseen* outputs, i.e., statements that never arise as completions of the child inputs I_α. Because $\mathfrak{v}$ is finite, U is finite, so the uniform distribution on 2^U is well-defined. Assume a maximally uninformative model of how the parent extends the child. Conditional on α, the parent selects an additional set $S \subseteq U$ of correct outputs chosen uniformly at random from 2^U, and its overall correct-output set is $O_\omega = O_\alpha \cup S$. Then, among all $\pi \in \Pi_\alpha$, the probability that π is also correct for ω is maximised by choosing a $<_w$-maximal policy (equivalently, by maximising $|\text{Ext}(\pi)|$).

Proof. Fix a correct policy $\pi \in \Pi_\alpha$. By correctness on α,

$$\text{Ext}(I_\alpha) \cap \text{Ext}(\pi) \;=\; O_\alpha.$$

Because $U = L_{\mathfrak{v}} \setminus \text{Ext}(I_\alpha)$, I can decompose

$$\text{Ext}(\pi) \;=\; (\text{Ext}(\pi) \cap \text{Ext}(I_\alpha)) \;\cup\; (\text{Ext}(\pi) \cap U) \;=\; O_\alpha \;\cup\; (\text{Ext}(\pi) \cap U).$$

Hence, within Π_α, maximising $|\text{Ext}(\pi)|$ is equivalent to maximising $|\text{Ext}(\pi) \cap U|$.

Under the stated random extension model, π is also correct for the parent ω precisely when the additional correct outputs $S \subseteq U$ are all permitted by π, i.e., when $S \subseteq \text{Ext}(\pi) \cap U$. The number of subsets

$S \subseteq U$ with $S \subseteq \mathrm{Ext}(\pi) \cap U$ is $2^{|\mathrm{Ext}(\pi) \cap U|}$, while the total number of subsets of U is $2^{|U|}$. Therefore,

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) \;=\; \frac{2^{|\mathrm{Ext}(\pi)\cap U|}}{2^{|U|}},$$

which is strictly increasing in $|\mathrm{Ext}(\pi) \cap U|$, and hence strictly increasing in $|\mathrm{Ext}(\pi)|$ over Π_α. Thus any $<_w$-maximal policy maximises $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$. □

Proof of Result 5

Restatement of Proposition 5 (Necessity of weakness under the same prior). Under the assumptions of Proposition 4, any policy $\pi \in \Pi_\alpha$ that maximises $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ must be $<_w$-maximal in Π_α. In particular, any non-$<_w$-maximal correct policy is strictly suboptimal for maximising this generalisation probability.

Proof. By Proposition 4,

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) \;=\; \frac{2^{|\mathrm{Ext}(\pi)\cap U|}}{2^{|U|}},$$

so $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ is strictly increasing in $|\mathrm{Ext}(\pi) \cap U|$, and hence (within Π_α) strictly increasing in $|\mathrm{Ext}(\pi)|$. If π is not $<_w$-maximal, then there exists $\pi' \in \Pi_\alpha$ with $|\mathrm{Ext}(\pi')| > |\mathrm{Ext}(\pi)|$, which implies $\mathbb{P}(\pi' \in \Pi_\omega \mid \alpha) > \mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$. Thus π cannot be optimal. □

Proof of Result 3

Restatement of Theorem 3 (Weakness maximisation beyond the uniform prior). Let $\mathfrak{v}$ be finite and let α and ω be $\mathfrak{v}$-tasks with $\alpha \sqsubset \omega$. Write $U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$ for the set of unseen outputs. Assume the parent extends the child by choosing a random set $S \subseteq U$ and setting $O_\omega = O_\alpha \cup S$. Let P be the conditional distribution of S given α, with no restriction to uniformity.

For each $\pi \in \Pi_\alpha$, define its P-**weakness** by

$$w_P(\pi) \;:=\; P\big(S \subseteq \mathrm{Ext}(\pi) \cap U\big) \;=\; \sum_{S \subseteq \mathrm{Ext}(\pi)\cap U} P(S).$$

Then

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) \;=\; w_P(\pi).$$

Consequently, a correct policy $\pi^\star \in \Pi_\alpha$ maximises the generalisation probability $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ if and only if it maximises $w_P(\pi)$ over Π_α.

Assume in addition that P is **exchangeable** over U. That is, for every permutation $\sigma : U \to U$ and every subset $S \subseteq U$, I have

$P(S) = P(\sigma(S))$. Then there exists a nondecreasing function $f : \{0, 1, \ldots, |U|\} \to [0, 1]$ such that

$$w_P(\pi) \;=\; f\big(|\mathrm{Ext}(\pi) \cap U|\big)$$

for all $\pi \in \Pi_\alpha$. Therefore any $<_w$-maximal policy $\pi^\star \in \Pi_\alpha$ maximises $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ under P.

If also $P(|S| = 1) > 0$, then f is strictly increasing on $\{0, 1, \ldots, |U| - 1\}$. In that nondegenerate exchangeable case, every maximiser of $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ is $<_w$-maximal.

Proof. Fix $\pi \in \Pi_\alpha$. Correctness on α implies $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi) = O_\alpha$. Because $U = L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$, I have the disjoint decomposition

$$\mathrm{Ext}(\pi) = (\mathrm{Ext}(\pi) \cap \mathrm{Ext}(I_\alpha)) \;\cup\; (\mathrm{Ext}(\pi) \cap U) = O_\alpha \;\cup\; (\mathrm{Ext}(\pi) \cap U).$$

Under the extension model $O_\omega = O_\alpha \cup S$, the policy π is also correct for ω if and only if $S \subseteq \mathrm{Ext}(\pi) \cap U$. Therefore

$$\mathbb{P}(\pi \in \Pi_\omega \mid \alpha) \;=\; P\big(S \subseteq \mathrm{Ext}(\pi) \cap U\big) \;=\; w_P(\pi),$$

and maximising $\mathbb{P}(\pi \in \Pi_\omega \mid \alpha)$ over Π_α is equivalent to maximising $w_P(\pi)$ over Π_α.

Now assume P is exchangeable. Fix $k \in \{0, 1, \ldots, |U|\}$ and let $A_k \subseteq U$ be any set with $|A_k| = k$. Define

$$f(k) \;:=\; P(S \subseteq A_k).$$

If $A, B \subseteq U$ satisfy $|A| = |B| = k$, then there exists a permutation $\sigma : U \to U$ with $\sigma(A) = B$. Exchangeability gives

$$P(S \subseteq A) \;=\; P\big(\sigma(S) \subseteq \sigma(A)\big) \;=\; P(S \subseteq B),$$

so f is well-defined and depends only on k.

Take $k < k'$ and choose $A_k \subseteq A_{k'} \subseteq U$ with $|A_k| = k$ and $|A_{k'}| = k'$. Then $\{S \subseteq A_k\}$ implies $\{S \subseteq A_{k'}\}$, hence $f(k) \le f(k')$. So f is nondecreasing.

For any $\pi \in \Pi_\alpha$, exchangeability yields

$$w_P(\pi) \;=\; P\big(S \subseteq \mathrm{Ext}(\pi) \cap U\big) \;=\; f\big(|\mathrm{Ext}(\pi) \cap U|\big).$$

Because f is nondecreasing, $w_P(\pi)$ is maximised by maximising $|\mathrm{Ext}(\pi) \cap U|$. Within Π_α, the decomposition $\mathrm{Ext}(\pi) = O_\alpha \cup (\mathrm{Ext}(\pi) \cap U)$ shows that maximising $|\mathrm{Ext}(\pi) \cap U|$ is equivalent to maximising $|\mathrm{Ext}(\pi)|$. So any $<_w$-maximal policy is optimal under P.

Finally, assume $P(|S| = 1) > 0$. Under exchangeability, conditional on $|S| = 1$, the set S is uniformly distributed over singletons of U. So for any $m \in \{0, 1, \ldots, |U|\}$,

$$P(S \subseteq A_m \mid |S| = 1) \;=\; \frac{m}{|U|},$$

which is strictly increasing in m. Because $P(|S| = 1) > 0$ and all other size-classes contribute nonnegative mass, it follows that f is strictly increasing on $\{0, 1, \dots, |U| - 1\}$. Therefore, if π is not $<_w$-maximal, then there exists $\pi' \in \Pi_\alpha$ with $|\mathrm{Ext}(\pi') \cap U| > |\mathrm{Ext}(\pi) \cap U|$, and hence $w_P(\pi') > w_P(\pi)$. So any maximiser must be $<_w$-maximal. $\square$

Proof of Result 6

Restatement of Proposition 6 (Simplicity can be sub-optimal). The Simplicity proxy $<_s$ is neither sufficient nor necessary for maximising the generalisation probability in Proposition 4. In particular, there exist tasks for which the $<_s$-maximal correct policy is *not* $<_w$-maximal, and therefore is strictly suboptimal under the maximally uninformative prior.

Proof. I construct an explicit counterexample.

Let $\Phi = \{\phi_0, \phi_1, \phi_2, \phi_3, \phi_4\}$ and let the following five sets of programs be the *maximal* consistent statements:

$$\begin{aligned}
m_0 &= \{z, j, k\},\\
m_1 &= \{j, k, a, b, c, d\},\\
m_2 &= \{j, k, e, f, g, h\},\\
m_3 &= \{u_j, j\},\\
m_4 &= \{u_k, k\}.
\end{aligned}$$

Define the vocabulary $\mathfrak{v} := \bigcup_{r=0}^{4} m_r$, and interpret each program $p \in \mathfrak{v}$ as the set of states in which it appears:

$$p := \{\phi_r \in \Phi \mid p \in m_r\}.$$

With this construction, a set of programs $l \subseteq \mathfrak{v}$ has nonempty intersection (hence is a statement) iff $l \subseteq m_r$ for some r. Thus $L_\mathfrak{v}$ is exactly the downward closure of $\{m_0, \dots, m_4\}$.

Now define a $\mathfrak{v}$-task $\alpha = \langle I_\alpha, O_\alpha \rangle$ by

$$I_\alpha = \{\{z, j, k\}, \{u_j\}, \{u_k\}\}, \qquad O_\alpha = \{\{z, j, k\}\}.$$

The completion set of $\{z, j, k\}$ is $\{\{z, j, k\}\}$, while the completion sets of $\{u_j\}$ and $\{u_k\}$ are $\{\{u_j\}, \{u_j, j\}\}$ and $\{\{u_k\}, \{u_k, k\}\}$, respectively. Hence $\mathrm{Ext}(I_\alpha) = \{\{z, j, k\}, \{u_j\}, \{u_j, j\}, \{u_k\}, \{u_k, k\}\}$.

Consider the two candidate policies $\pi_d := \{z\}$ and $\pi_w := \{j, k\}$. Because the only element of $\mathrm{Ext}(I_\alpha)$ that contains z is $\{z, j, k\}$, I have $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_d) = \{\{z, j, k\}\} = O_\alpha$, so $\pi_d \in \Pi_\alpha$. Similarly, no element of $\mathrm{Ext}(I_\alpha)$ contains both j and k except $\{z, j, k\}$, so $\mathrm{Ext}(I_\alpha) \cap \mathrm{Ext}(\pi_w) = \{\{z, j, k\}\} = O_\alpha$, and $\pi_w \in \Pi_\alpha$.

However, the proxies disagree. The Simplicity proxy prefers π_d because $|\pi_d| = 1 < 2 = |\pi_w|$, so π_d is $<_s$-maximal among these two. By contrast, π_w is strictly weaker in the completion-set sense: z appears only in m_0, so $|\mathrm{Ext}(\pi_d)| = |\{\{z\}, \{z, j\}, \{z, k\}, \{z, j, k\}\}| = 4$, whereas j, k co-occur in m_0, m_1, m_2, giving $|\mathrm{Ext}(\pi_w)| = 2 + (2^4 - 1) + (2^4 - 1) = 32$ (m_0 contributes the two completions $\{j, k\}$ and $\{z, j, k\}$. m_1 and m_2 each contribute $2^4 - 1$ additional completions, excluding the shared completion $\{j, k\}$). Thus $|\mathrm{Ext}(\pi_w)| > |\mathrm{Ext}(\pi_d)|$, so π_w is $<_w$-preferred.

By Proposition 4, the probability of generalising to an uninformative random parent is increasing in $|\mathrm{Ext}(\pi)|$, so π_d is strictly suboptimal in that sense. □

COMPLEXITY IS SUBJECTIVE

Proof of Result 7

Restatement of Proposition 7 (Subjectivity of description length). If there is no abstraction in the sense that the vocabulary equals the full program set ($\mathfrak{v} = \mathcal{P} = 2^\Phi$), then every statement is semantically representable with description length 1. In particular, for any $l \in L_\mathcal{P}$, the singleton statement $\{T(l)\}$ satisfies $T(\{T(l)\}) = T(l)$. Consequently, description length is highly representation dependent: in the extreme case $\mathfrak{v} = \mathcal{P}$, minimizing $|l|$ does not meaningfully distinguish among policies by semantic content.

Proof. Take any $l \in L_\mathcal{P}$. By definition, $T(l) = \bigcap_{p \in l} p$ is a nonempty subset of Φ, hence $T(l) \in \mathcal{P}$. Because $\mathfrak{v} = \mathcal{P}$, the singleton $\{T(l)\} \subseteq \mathfrak{v}$ is a valid statement in $L_\mathcal{P}$, and its truth set is

$$T(\{T(l)\}) \;=\; \bigcap_{p \in \{T(l)\}} p \;=\; T(l).$$

Thus every semantic constraint expressible by any conjunction l can also be expressed by a singleton statement of size 1. This demonstrates that raw description length $|l|$ is not an intrinsic measure of semantic complexity. It depends on the chosen vocabulary (i.e., on which truth sets are treated as atomic). □

Proof of Result 8

Restatement of Proposition 8 (Complexity confounding). If the vocabulary $\mathfrak{v}$ is finite, then

1. semantically similar constraints need not admit equally short descriptions, and
2. description length can become spuriously correlated with generalisation, not because it is causal, but because it is confounded by weakness.

Proof. By Proposition 7, when $\mathfrak{v} = \mathcal{P}$ every semantic constraint can be expressed with a singleton statement, so raw description length provides little resolution. In contrast, when $\mathfrak{v}$ is a strict finite subset of $\mathcal{P}$, many truth sets are not available as atomic programs, and expressing a given constraint may require conjunctions of varying cardinalities. Thus $|l|$ can vary nontrivially across policies.

Moreover, there exist finite vocabularies for which $|l|$ is monotone in weakness and therefore *confounded* with sample efficiency. For example, let $\Phi = \{1,2,3,4\}$ and let $a = \{1,2,4\}$, $b = \{1,3,4\}$. Take $\mathfrak{v} = \{a,b\}$. Then the nonempty statements are $\{a\}$, $\{b\}$, and $\{a,b\}$, with

$$|\text{Ext}(\{a,b\})| = 1 \quad \text{and} \quad |\text{Ext}(\{a\})| = |\text{Ext}(\{b\})| = 2,$$

so $\{a\}$ and $\{b\}$ are weaker than $\{a,b\}$ in the $<_w$ sense. At the same time, $|\{a\}| = |\{b\}| = 1 < 2 = |\{a,b\}|$, so the Simplicity proxy $<_s$ ranks $\{a\}$ and $\{b\}$ above $\{a,b\}$ in the same direction. In such settings, weakness can induce an apparent relationship between description length and sample efficiency even when length itself is not the operative variable. □

LANGUAGE COMBINATORICS, COMPRESSION, AND AIXI

Proof of Result 8

Restatement of Lemma 8 (The language is the union of all local powersets).

Proof. If $l \in L_\mathfrak{v}$, choose $\phi \in T(l)$. Then Lemma 1 gives $l \subseteq \text{Enc}_\mathfrak{v}(\phi)$, so $l \in 2^{\text{Enc}_\mathfrak{v}(\phi)}$. Conversely, if $l \subseteq \text{Enc}_\mathfrak{v}(\phi)$ for some ϕ, then Lemma 1 gives $\phi \in T(l)$. So $T(l) \neq \emptyset$ and $l \in L_\mathfrak{v}$. □

Proof of Result 11

Restatement of Proposition 11 (Extensions are unions of encoding intervals).

Proof. Let $y \in \text{Ext}(l)$. Then $l \subseteq y$ and $y \in L_\mathfrak{v}$. Choose $\phi \in T(y)$. Then $\phi \in T(l)$ since $T(y) \subseteq T(l)$. By Lemma 1, $y \subseteq \text{Enc}_\mathfrak{v}(\phi)$. So y lies between l and $m := \text{Enc}_\mathfrak{v}(\phi)$, and $m \in \mathcal{M}(l)$.

Conversely, suppose $m \in \mathcal{M}(l)$ and $l \subseteq y \subseteq m$. By definition of $\mathcal{M}(l)$, there exists $\phi \in T(l)$ with $m = \text{Enc}_\mathfrak{v}(\phi)$. Then $y \subseteq \text{Enc}_\mathfrak{v}(\phi)$. By Lemma 1, $\phi \in T(y)$. So $y \in L_\mathfrak{v}$ and $y \in \text{Ext}(l)$.

For the bounds, each set $\{y \mid l \subseteq y \subseteq m\}$ has exactly $2^{|m|-|l|}$ elements. The union contains at least one such set, so its size is at least the maximum. The size of a union is at most the sum. □

Proof of Result 9

Restatement of Lemma 9 (Full consistency collapses weakness to length).

Proof. If $L_{\mathfrak{v}} = 2^{\mathfrak{v}}$, then for every $l \subseteq \mathfrak{v}$, $\mathrm{Ext}(l) = \{y \subseteq \mathfrak{v} \mid l \subseteq y\}$. This is the set of all supersets of l in $\mathfrak{v}$, which has $2^{|\mathfrak{v}|-|l|}$ elements. □

Proof of Result 10

Restatement of Lemma 10 (Ordered recursion for weakness).

Proof. Every y counted by $w_j(l)$ satisfies $l \subseteq y$ and $y \setminus l \subseteq \{p_j, \dots, p_{n-1}\}$. If $y = l$, it contributes the initial 1. If $y \supsetneq l$, let p_k be the smallest-index program in $y \setminus l$. Then $k \geq j$ (by the index restriction on w_j), $p_k \notin l$, $l \cup \{p_k\} \subseteq y$, and $l \cup \{p_k\} \in L_{\mathfrak{v}}$ (since y is a statement and $l \cup \{p_k\} \subseteq y$). The remaining programs in $y \setminus (l \cup \{p_k\})$ all have index $> k$ (because p_k was the smallest). So y is counted exactly once in $w_{k+1}(l \cup \{p_k\})$. The sets $\{y \in \mathrm{Ext}_j(l) : y \supsetneq l, \min\text{-index of } y \setminus l = k\}$ for different k are disjoint, and together with $\{l\}$ they partition the set counted by $w_j(l)$. Summing over all valid k gives the recursion. The base case $j = n$ gives $w_n(l) = 1$ (only $y = l$ survives). Setting $j = 0$ recovers $w(l) = w_0(l)$. □

Proof of Result 3

Restatement of Corollary 3 (Weak constraints compress).

Proof. Under the weak usage bias, $w(l_1) \geq w(l_2)$ implies $P(l_1) \geq P(l_2)$. By the Huffman monotonicity theorem (Huffman 1952), a minimum expected length prefix-free code assigns codewords of non-increasing length to symbols of nonincreasing probability. Therefore $P(l_1) \geq P(l_2)$ implies $|c^*(l_1)| \leq |c^*(l_2)|$. Chaining the two implications gives $w(l_1) \geq w(l_2) \implies |c^*(l_1)| \leq |c^*(l_2)|$. □

Proof of Result 4

Restatement of Corollary 4 (Simplicity-generalisation correlation).

Proof. By Proposition 4, $w(\pi_1) > w(\pi_2)$ implies π_1 has strictly higher generalisation probability than π_2. By Corollary 3, $w(\pi_1) > w(\pi_2)$ implies $|c^*(\pi_1)| \leq |c^*(\pi_2)|$. So $|c^*(\pi_1)| \leq |c^*(\pi_2)|$ appears to predict generalisation. But the prediction is mediated entirely by weakness. If the weak usage bias is removed (by changing the population distribution), the code-length ordering can reverse while the weakness ordering remains. □

Proof of Result 5

Restatement of Corollary 5 (Occam priors collapse).

Proof. By Proposition 7, when $\mathfrak{v} = \mathcal{P}$, every statement l has a singleton representation $\{T(l)\}$ with description length $|\{T(l)\}| = 1$. Any description-length prior that assigns weight as a decreasing function of $|h|$ therefore assigns equal weight to all policies, since all have $|h| = 1$. The Occam prior becomes uniform and carries no information. □

Proof of Result 12

Restatement of Proposition 12 (Every finite prior is an Occam prior).

Proof. Let Q be any distribution on a finite set $H = \{h_1, \ldots, h_n\}$. By the Shannon source coding theorem, there exists a prefix-free code κ_Q with $|\kappa_Q(h_i)| \leq \lceil -\log_2 Q(h_i) \rceil \leq -\log_2 Q(h_i) + 1$ for all i. The induced Occam prior is $P_{\kappa_Q}(h_i) = 2^{-|\kappa_Q(h_i)|}/Z$ where $Z = \sum_j 2^{-|\kappa_Q(h_j)|}$.

For the lower bound on Z: by the code length bound, $2^{-|\kappa_Q(h_i)|} \geq 2^{-(-\log_2 Q(h_i)+1)} = Q(h_i)/2$ for each i. Summing gives $Z \geq \sum_i Q(h_i)/2 = 1/2$.
For the upper bound on Z: by the Kraft inequality, $Z \leq 1$.

Now I can bound the ratio. For the lower bound: $P_{\kappa_Q}(h_i) = 2^{-|\kappa_Q(h_i)|}/Z \geq (Q(h_i)/2)/1 = Q(h_i)/2$. For the upper bound: $2^{-|\kappa_Q(h_i)|} \leq 2^{-(-\log_2 Q(h_i))} = Q(h_i)$ (since $|\kappa_Q(h_i)| \geq \lceil -\log_2 Q(h_i) \rceil \geq -\log_2 Q(h_i)$), so $P_{\kappa_Q}(h_i) = 2^{-|\kappa_Q(h_i)|}/Z \leq Q(h_i)/(1/2) = 2Q(h_i)$. Therefore $Q(h_i)/2 \leq P_{\kappa_Q}(h_i) \leq 2Q(h_i)$ for all i. □

Proof of Result 13

Restatement of Theorem 13 (MCL is Bayes optimal).

Proof. For the deterministic selector, Theorem 3 implies that under the exchangeable extension model the score

$$P(\pi \in \Pi_\omega \mid \alpha)$$

is a nondecreasing function of $|\mathrm{Ext}(\pi) \cap U|$, and hence of $w(\pi)$ within Π_α. MCL chooses $\pi^* \in \arg\max_{\pi \in \Pi_\alpha} w(\pi)$, so

$$P(\pi^* \in \Pi_\omega \mid \alpha) = \max_{\pi \in \Pi_\alpha} P(\pi \in \Pi_\omega \mid \alpha).$$

Now let Σ be any possibly randomised selector supported on Π_α, and write $r_\Sigma(\pi)$ for the probability that Σ outputs π. Its expected generalisation probability is

$$\sum_{\pi \in \Pi_\alpha} r_\Sigma(\pi)\, P(\pi \in \Pi_\omega \mid \alpha).$$

A convex combination of numbers cannot exceed their maximum, so

$$\sum_{\pi \in \Pi_\alpha} r_\Sigma(\pi)\, P(\pi \in \Pi_\omega \mid \alpha) \leq \max_{\pi \in \Pi_\alpha} P(\pi \in \Pi_\omega \mid \alpha) = P(\pi^* \in \Pi_\omega \mid \alpha).$$

Thus no randomised selector beats MCL in expected generalisation probability. $\square$

REGRET AS WEIGHTED FORGETTING

Proof of Result 11

Restatement of Lemma 11 (Pointwise regret decomposition).

Proof. If $y_i = 1$, then $p_i \geq \frac{1}{2}$ and $V_i^\star = p_i = \frac{1}{2} + m_i$. The success probability under report q_C is $V_i^\pi = q_C p_i + (1 - q_C)(1 - p_i)$. So $\delta_i = 1 - V_i^\pi / V_i^\star = 2m_i(1 - q_C)/(\frac{1}{2} + m_i) = \rho_i(1 - q_C)$. The $y_i = 0$ case is symmetric, giving $\delta_i = \rho_i q_C$. $\square$

Proof of Result 14

Restatement of Theorem 14 (Exact reduction).

Proof. For each cell $C \in \mathcal{C}_M$, define $A_C := \sum_{i \in C, y_i = 1} \mu_i \rho_i$ and $B_C := \sum_{i \in C, y_i = 0} \mu_i \rho_i$. By Lemma 11, any M-based policy with report q_C contributes $A_C(1 - q_C) + B_C q_C$ to expected regret on cell C. This is affine in q_C, so its minimum over $[0, 1]$ is $\min\{A_C, B_C\}$. Summing over cells gives the infimum on the left.

For the forgetting cost, making y single-valued on cell C requires deleting every $y_i = 1$ point or every $y_i = 0$ point in C. The minimum deletion cost on C is $\min\{A_C, B_C\}$. Summing over cells gives $K_\rho(M) = \sum_C \min\{A_C, B_C\}$. $\square$

Proof of Result 6

Restatement of Corollary 6 (Policy-wise decomposition).

Proof. On each cell C, the policy contributes $A_C(1 - q_C) + B_C q_C$. The minimum is $\min\{A_C, B_C\}$, achieved at $q_C^\star$. If $A_C \geq B_C$, then $q_C^\star = 1$ and $A_C(1 - q_C) + B_C q_C = B_C + (A_C - B_C)(1 - q_C) = \min\{A_C, B_C\} + (A_C - B_C)(q_C^\star - q_C)$. If $A_C < B_C$, then $q_C^\star = 0$ and $A_C(1 - q_C) + B_C q_C = A_C + (B_C - A_C) q_C = \min\{A_C, B_C\} + (A_C - B_C)(q_C^\star - q_C)$. In both cases the excess is $(A_C - B_C)(q_C^\star - q_C) \geq 0$. Summing over cells gives total regret $= K_\rho(M) + \sum_C (A_C - B_C)(q_C^\star - q_C)$. $\square$

Proof of Result 13

Restatement of Proposition 13 (Independent nonuniform priors give weighted weakness).

Proof. Generalisation occurs exactly when every unseen output omitted by ϑ fails to become relevant. Independence gives $\Pr(\vartheta \text{ generalises}) = \prod_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} (1 - r_i)$. Taking logs yields $\log \Pr = \sum_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} \log(1 - r_i)$. The remaining term depends on ϑ only through $\sum_{u_i \in \mathrm{Ext}(\vartheta) \cap U} (-\log(1 - r_i))$, which is weighted weakness with $w_i = -\log(1 - r_i)$. $\square$

Proof of Result 7

Restatement of Corollary 7 (Exact Stack-Theoretic bridge).

Proof. On the lifted task, retaining u_i means not forgetting x_i. Admissibility means each cell keeps at most one label class. Maximising retained weighted mass is the same optimisation as minimising deleted weighted mass. So $W^\star(M) = Z - K_\rho(M)$. Rearranging and applying Theorem 14 gives $\inf \sum \mu_i \delta_i = Z - W^\star(M)$. $\square$

Proof of Result 8

Restatement of Corollary 8 (Gen. probability from regret).

Proof. By Proposition 13 with $r_i = 1 - e^{-w_i}$, the generalisation probability is $\prod_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} e^{-w_i} = e^{-\sum_{u_i \notin \mathrm{Ext}(\vartheta) \cap U} w_i}$. For the optimal policy, the exponent is exactly $K_\rho(M)$. $\square$

Proof of Result 15

Restatement of Theorem 15 (Multi-class exact reduction).

Proof. An M-based policy assigns a distribution $q_C \in \Delta^{K-1}$ over the K classes for each cell C. Its expected regret on cell C is

$$\sum_{k=1}^{K} A_C^{(k)} (1 - q_C^{(k)}) = \Big(\sum_k A_C^{(k)} \Big) - \sum_k A_C^{(k)} q_C^{(k)}.$$

To minimise this over $q_C \in \Delta^{K-1}$, maximise the inner product $\sum_k A_C^{(k)} q_C^{(k)}$. This is a linear function on the simplex, so its maximum is achieved at a vertex, namely all mass on the class $k^\star = \arg\max_k A_C^{(k)}$. The minimum per-cell regret is therefore $(\sum_k A_C^{(k)}) -$

$\max_k A_C^{(k)}$. To make y single-valued on C, one must delete every point not in the dominant class, costing the same quantity. Summing over cells gives both sides. □

Proof of Result 14

Restatement of Proposition 14 (Lipschitz stability).

Proof. For each cell C, $|\min\{\hat{A}_C, \hat{B}_C\} - \min\{A_C, B_C\}| \le |\hat{A}_C - A_C| + |\hat{B}_C - B_C| \le \sum_{i \in C} \mu_i |\hat{\rho}_i - \rho_i|$. Sum over cells. For the second bound, on $(0, \frac{1}{2})$ we have $\rho = (1-2p)/(1-p)$ with $|d\rho/dp| = 1/(1-p)^2 \le 4$, and on $(\frac{1}{2}, 1)$ we have $\rho = (2p-1)/p$ with $|d\rho/dp| = 1/p^2 \le 4$. At $p = \frac{1}{2}$, ρ has a kink but both one-sided derivatives have magnitude 4. So ρ is Lipschitz with constant 4 on $(0,1)$, giving $|\hat{\rho}_i - \rho_i| \le 4|\hat{p}_i - p_i|$. □

Proof of Result 16

Restatement of Theorem 16 (Representation convergence).

Proof. By Theorem 14, $K_\rho(M) = 0$ iff $\min\{A_C, B_C\} = 0$ for every cell $C \in \mathcal{C}_M$. On the informative support I_+, this is equivalent to saying that each cell contains only one optimal label at its fixed test value. That is exactly the statement that $\mathcal{C}_M|_{I_+}$ refines $\mathcal{P}_+^\star$. This proves the first claim.

For the minimality claim, first note that $\mathcal{P}_+^\star$ itself is a zero-regret partition, because every one of its cells is pure by construction. So any zero-regret representation whose induced partition on I_+ is strictly finer than $\mathcal{P}_+^\star$ cannot be informative-minimal: coarsening it down to $\mathcal{P}_+^\star$ preserves zero regret. Conversely, if $\mathcal{C}_M|_{I_+} = \mathcal{P}_+^\star$, then no strictly coarser zero-regret partition exists, because any strict coarsening would merge two cells with different (T_i, y_i) values and would therefore create an impure informative cell. Hence M is informative-minimal zero-regret iff $\mathcal{C}_M|_{I_+} = \mathcal{P}_+^\star$.

Finally, if M_1 and M_2 are both informative-minimal zero-regret, then they induce the same partition $\mathcal{P}_+^\star$ on I_+. Their representation states may have different names, but there is a bijection between states obtained by matching equal cells of $\mathcal{P}_+^\star$. So they differ only by relabelling on the informative support. □

Proof of Result 15

Restatement of Proposition 15 (Measurable exact reduction).

Proof. Because $(\mathcal{Z}, \mathcal{B})$ is standard Borel, the conditional expectations with respect to $\sigma(\zeta)$ admit measurable versions as functions of ζ;

those are the displayed functions a and b. For any measurable q : $\mathcal{Z} \to [0,1]$, the random variable $q(\zeta)$ is $\sigma(\zeta)$-measurable, so the tower property gives

$$R(q) = \mathbb{E}[a(\zeta)(1 - q(\zeta)) + b(\zeta)q(\zeta)] \,.$$

For each fixed z, the integrand

$$a(z)(1 - q) + b(z)q$$

is affine in $q \in [0,1]$, so its minimum is $\min\{a(z), b(z)\}$, attained by any selector $q^\star(z) \in \arg\min_{q\in[0,1]}(a(z)(1-q) + b(z)q)$. The explicit choice $q^\star(z) = \mathbf{1}\{a(z) > b(z)\}$ is measurable because a and b are measurable.

Substituting this pointwise minimiser into the previous display yields

$$\inf_{q:\mathcal{Z}\to[0,1] \text{ measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}],$$

which is finite because $0 \le a, b \le 1$. $\square$

CONTINUOUS DOMAINS

Continuous-domain preliminaries. Throughout this proof block, fix a measured embodied language $(\mathfrak{v}, \mathcal{A}_L, \mu)$, a task α, and the unseen region

$$U := L_{\mathfrak{v}} \setminus \mathrm{Ext}(I_\alpha)$$

with $0 < \mu(U) < \infty$. Assume $(U, \mathcal{A}_L|_U, \mu|_U)$ is an atomless standard finite measure space.

Proof of Lemma 6. Because $L_{\mathfrak{v}} = \mathrm{Ext}(I_\alpha) \sqcup U$, we have

$$\mathrm{Ext}(\pi) = \big(\mathrm{Ext}(\pi) \cap \mathrm{Ext}(I_\alpha)\big) \sqcup \big(\mathrm{Ext}(\pi) \cap U\big).$$

Since $\pi \in \Pi_\alpha$, correctness gives

$$\mathrm{Ext}(\pi) \cap \mathrm{Ext}(I_\alpha) = O_\alpha.$$

By definition, $\mathrm{Ext}(\pi) \cap U = B_\pi$. Hence

$$\mathrm{Ext}(\pi) = O_\alpha \sqcup B_\pi.$$

Taking measures yields

$$w_\mu(\pi) = \mu(\mathrm{Ext}(\pi)) = \mu(O_\alpha) + \mu(B_\pi).$$

The final claim follows because $\mu(O_\alpha)$ is independent of π on Π_α. $\square$

Proof of Result 5

Restatement of Theorem 5 (Poisson sufficiency).

Proof. Under a Poisson point process with intensity measure $\lambda\mu$, the event $S \subseteq B_\pi$ is exactly the event that no points fall in the measurable complement $U \setminus B_\pi$. The Poisson void formula therefore gives

$$G(\pi, P) = P(S \subseteq B_\pi) = \exp\big(-\lambda\mu(U \setminus B_\pi)\big) = \exp\big(-\lambda[\mu(U) - \mu(B_\pi)]\big).$$

This is strictly increasing in $\mu(B_\pi)$. By Lemma 6, on correct policies this is equivalent to strict increase in $w_\mu(\pi)$. ☐

Proof of Result 6

Restatement of Theorem 6 (Exchangeable sufficiency).

Proof. For each measurable $B \subseteq U$, define

$$f_P(\mu(B)) := P(S \subseteq B).$$

I first show that this depends only on the measure of B. Let $B_1, B_2 \subseteq U$ be measurable with $\mu(B_1) = \mu(B_2)$. In an atomless standard finite measure space, measurable sets of equal measure are interchangeable under a measure-preserving relabelling of U. Hence there exists a measure-preserving bijection $\sigma : U \to U$ with $\sigma(B_1) = B_2$. By μ-exchangeability,

$$P(S \subseteq B_1) = P(\sigma(S) \subseteq \sigma(B_1)) = P(S \subseteq B_2).$$

So f_P is well defined on $[0, \mu(U)]$.

To show monotonicity, let $B_1, B_2 \subseteq U$ with $\mu(B_1) \leq \mu(B_2)$. Because the space is atomless, there exists a measurable subset $B_2' \subseteq B_2$ with $\mu(B_2') = \mu(B_1)$. By the previous paragraph,

$$P(S \subseteq B_1) = P(S \subseteq B_2').$$

Since $B_2' \subseteq B_2$,

$$\{S \subseteq B_2'\} \subseteq \{S \subseteq B_2\},$$

so

$$P(S \subseteq B_1) \leq P(S \subseteq B_2).$$

Thus f_P is nondecreasing. Applying this with $B = B_\pi$ gives

$$G(\pi, P) = f_P\big(\mu(B_\pi)\big).$$

The final equivalence with μ-weakness is Lemma 6. ☐

Proof of Result 7

Restatement of Theorem 7 (Strict necessity).

Proof. Assume $\mu(B_{\pi_1}) > \mu(B_{\pi_2})$. Because the space is atomless, there exists a measurable subset $B_1' \subseteq B_{\pi_1}$ with

$$\mu(B_1') = \mu(B_{\pi_2}).$$

By the equal-measure relabelling argument used above,

$$P(S \subseteq B_1') = P(S \subseteq B_{\pi_2}).$$

Now let

$$R := B_{\pi_1} \setminus B_1'.$$

Then $\mu(R) > 0$. By nondegeneracy,

$$P\big(|S| = 1 \text{ and } S \subseteq R\big) > 0.$$

Every such singleton demand is contained in B_{π_1} but not in B_1', so

$$P(S \subseteq B_{\pi_1}) \geq P(S \subseteq B_1') + P\big(|S| = 1 \text{ and } S \subseteq R\big) > P(S \subseteq B_1').$$

Hence

$$G(\pi_1, P) > G(\pi_2, P).$$

By Lemma 6, the same strict ordering is equivalent to strict ordering by μ-weakness on correct policies. □

Proof of Result 8

Restatement of Theorem 8 (Unified optimality).

Proof. By Lemma 6, comparing correct policies by w_μ is equivalent to comparing them by $\mu(B_\pi)$. If $w_\mu(\pi_1) \geq w_\mu(\pi_2)$, then $\mu(B_{\pi_1}) \geq \mu(B_{\pi_2})$, so Theorem 6 gives

$$G(\pi_1, P) \geq G(\pi_2, P)$$

for every μ-exchangeable model P. If the inequality in w_μ is strict and P is nondegenerate, then Theorem 7 gives the strict score inequality. Therefore a correct policy maximises μ-weakness if and only if it maximises the score $G(\cdot, P)$ for every nondegenerate model in the class. The optimal *value* of μ-weakness is unique because it is just the supremum of a real-valued objective function on Π_α. The maximiser itself need not be unique. □

Proof of Result 9

Restatement of Theorem 9 (Minimax optimality).

Proof. Let π^* be a correct policy maximising w_μ on Π_α. By Theorem 8, for every $P \in \mathcal{E}$ and every $\pi \in \Pi_\alpha$,

$$G(\pi^*, P) \geq G(\pi, P).$$

Taking the infimum over P on the left and right yields

$$\inf_{P\in\mathcal{E}} G(\pi^*, P) \geq \inf_{P\in\mathcal{E}} G(\pi, P) \qquad \text{for all } \pi \in \Pi_\alpha.$$

So

$$\inf_{P\in\mathcal{E}} G(\pi^*, P) = \max_{\pi\in\Pi_\alpha} \inf_{P\in\mathcal{E}} G(\pi, P).$$

On the other hand, because π^* is pointwise optimal for every $P \in \mathcal{E}$,

$$\max_{\pi\in\Pi_\alpha} G(\pi, P) = G(\pi^*, P) \qquad \text{for every } P \in \mathcal{E}.$$

Taking infima over P gives

$$\inf_{P\in\mathcal{E}} \max_{\pi\in\Pi_\alpha} G(\pi, P) = \inf_{P\in\mathcal{E}} G(\pi^*, P).$$

Combining the displays yields the claimed minimax identity. □

Proof of Result 10

Restatement of Theorem 10 (Reparameterisation invariance).

Proof. If $F(\phi(\theta)) = F(\theta)$, then the two parameter values compute exactly the same input-output function. Therefore they satisfy exactly the same programs in $\mathfrak{v}$, so

$$\Pi_F(\phi(\theta)) = \Pi_F(\theta).$$

Applying Definition 33 to these equal policies gives

$$w_\mu\big(\Pi_F(\phi(\theta))\big) = w_\mu\big(\Pi_F(\theta)\big).$$

□

Proof of Result 11

Restatement of Theorem 11 (PAC-Bayes ordering).

Proof. The prior density of P_U with respect to $\mu|_U$ is $1/\mu(U)$ and the density of Q_π with respect to $\mu|_U$ is $1/\mu(B_\pi)$ on B_π and 0 outside. Therefore

$$\frac{dQ_\pi}{dP_U}(x) = \frac{\mu(U)}{\mu(B_\pi)} \qquad \text{for } Q_\pi\text{-a.e. } x,$$

and so

$$\mathrm{KL}(Q_\pi \| P_U) = \int_{B_\pi} \log\left(\frac{\mu(U)}{\mu(B_\pi)}\right) dQ_\pi = \log \frac{\mu(U)}{\mu(B_\pi)}.$$

Hence the KL divergence is strictly decreasing in $\mu(B_\pi)$. By Lemma 6, on correct policies this is equivalent to strict decrease in $w_\mu(\pi)$, so the two orderings agree. $\square$

THE LAW OF THE STACK

Proof of Result 4

Restatement of Theorem 4 (The Law of the Stack (bottleneck form)). Consider a stack with adjacent layers i and $i+1$ and suppose

$$\mathfrak{v}^{i+1} = f(\mathfrak{v}^i, \pi^i) = \{\, T(o) \mid o \in \mathrm{Ext}\big(\pi^i\big)\,\}.$$

Let $\gamma^{i+1} := \lambda^{i+1}(\mathfrak{v}^{i+1})$ be the induced layer-$(i+1)$ task.

Then

$$\sup_{\pi \in \Pi_{\gamma^{i+1}}} |\mathrm{Ext}(\pi)| \;\le\; 2^{|\mathrm{Ext}(\pi^i)|}.$$

In particular,

$$\epsilon(\gamma^{i+1}) \;\le\; 2^{|\mathrm{Ext}(\pi^i)|}.$$

If $|\mathrm{Ext}\big(\pi^i\big)| < \infty$, then the finite bottleneck form holds

$$\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| \;\le\; 2^{|\mathrm{Ext}(\pi^i)|}.$$

Proof. By definition,

$$\mathfrak{v}^{i+1} = \{\, T(o) \mid o \in \mathrm{Ext}\big(\pi^i\big)\,\}.$$

So $\mathfrak{v}^{i+1}$ is the image of $\mathrm{Ext}(\pi^i)$ under the map $o \mapsto T(o)$. Therefore

$$|\mathfrak{v}^{i+1}| \le |\mathrm{Ext}\big(\pi^i\big)|.$$

Every statement in $L_{\mathfrak{v}^{i+1}}$ is a subset of $\mathfrak{v}^{i+1}$, so $L_{\mathfrak{v}^{i+1}} \subseteq 2^{\mathfrak{v}^{i+1}}$. Hence

$$|L_{\mathfrak{v}^{i+1}}| \le 2^{|\mathfrak{v}^{i+1}|} \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

Now fix any $\pi \in \Pi_{\gamma^{i+1}} \subseteq L_{\mathfrak{v}^{i+1}}$. By Definition 3, $\mathrm{Ext}(\pi) \subseteq L_{\mathfrak{v}^{i+1}}$. Therefore

$$|\mathrm{Ext}(\pi)| \le |L_{\mathfrak{v}^{i+1}}| \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

Taking the supremum over $\pi \in \Pi_{\gamma^{i+1}}$ gives

$$\sup_{\pi \in \Pi_{\gamma^{i+1}}} |\mathrm{Ext}(\pi)| \le 2^{|\mathrm{Ext}(\pi^i)|}.$$

For the utility bound, if $\Pi_{\gamma^{i+1}} = \varnothing$ then $\epsilon(\gamma^{i+1}) = 0$ by Definition 31. Otherwise, by Definition 31,

$$\epsilon(\gamma^{i+1}) = \sup_{\pi \in \Pi_{\gamma^{i+1}}} \left| \mathrm{Ext}(\pi) \setminus O_{\gamma^{i+1}} \right|.$$

For each π I have $\mathrm{Ext}(\pi) \setminus O_{\gamma^{i+1}} \subseteq \mathrm{Ext}(\pi)$, so

$$\left| \mathrm{Ext}(\pi) \setminus O_{\gamma^{i+1}} \right| \leq |\mathrm{Ext}(\pi)|.$$

Taking suprema and using the previous bound yields $\epsilon(\gamma^{i+1}) \leq 2^{|\mathrm{Ext}(\pi^i)|}$.

Finally assume $|\mathrm{Ext}(\pi^i)| < \infty$. Then $\mathfrak{v}^{i+1}$ is finite and so $L_{\mathfrak{v}^{i+1}}$ is finite. Hence $\Pi_{\gamma^{i+1}} \subseteq L_{\mathfrak{v}^{i+1}}$ is finite.

If $\Pi_{\gamma^{i+1}} = \varnothing$, then $\epsilon(\gamma^{i+1}) = 0$ and $\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| = |O_{\gamma^{i+1}}| \leq |L_{\mathfrak{v}^{i+1}}| \leq 2^{|\mathrm{Ext}(\pi^i)|}$.

If $\Pi_{\gamma^{i+1}} \neq \varnothing$, then the supremum in Definition 31 is a maximum. Choose $\pi^\star \in \Pi_{\gamma^{i+1}}$ such that

$$\epsilon(\gamma^{i+1}) = \left| \mathrm{Ext}(\pi^\star) \setminus O_{\gamma^{i+1}} \right|.$$

Because $\pi^\star$ is correct, I have $O_{\gamma^{i+1}} \subseteq \mathrm{Ext}(\pi^\star)$. Since all sets are finite in this case,

$$|\mathrm{Ext}(\pi^\star)| = |O_{\gamma^{i+1}}| + \left| \mathrm{Ext}(\pi^\star) \setminus O_{\gamma^{i+1}} \right| = |O_{\gamma^{i+1}}| + \epsilon(\gamma^{i+1}).$$

Also $|\mathrm{Ext}(\pi^\star)| \leq |L_{\mathfrak{v}^{i+1}}| \leq 2^{|\mathrm{Ext}(\pi^i)|}$. Therefore $\epsilon(\gamma^{i+1}) + |O_{\gamma^{i+1}}| \leq 2^{|\mathrm{Ext}(\pi^i)|}$. □

Proof of Result 9

Restatement of Proposition 9 (Formal upper-bound optimisation). Fix an uninstantiated task λ_ρ, a nonempty vocabulary class V of finite vocabularies in its domain, and assume $\mathcal{A}(\lambda_\rho, V) \neq \varnothing$. An admissible pair $(\mathfrak{v}^\star, \pi^\star) \in \mathcal{A}(\lambda_\rho, V)$ maximises the upper-bound objective $\preceq_{\lambda_\rho, V}$ if and only if

$$\mathfrak{v}^\star \in \arg\max_{\mathfrak{v} \in V} \epsilon\big(\lambda_\rho(\mathfrak{v})\big)$$

and

$$\pi^\star \in \arg\max_{\pi \in \Pi_{\lambda_\rho(\mathfrak{v}^\star)}} |\mathrm{Ext}(\pi)|.$$

Proof. By definition of $\preceq_{\lambda_\rho, V}$, admissible pairs are compared lexicographically. So any maximiser must first maximise the first coordinate

$$\epsilon\big(\lambda_\rho(\mathfrak{v})\big)$$

over $\mathfrak{v} \in V$; otherwise some admissible pair with larger first coordinate would dominate it.

Fix now a vocabulary $\mathfrak{v}^\star$ that maximises the first coordinate. Among admissible pairs with first coordinate $\epsilon(\lambda_\rho(\mathfrak{v}^\star))$, the lexicographic tie-breaker is the second coordinate $|\mathrm{Ext}(\pi)|$. Therefore a lexicographic maximiser must satisfy

$$|\mathrm{Ext}(\pi^\star)| = \max\{\, |\mathrm{Ext}(\pi)| \mid \pi \in \Pi_{\lambda_\rho(\mathfrak{v}^\star)} \,\}.$$

Conversely, suppose $\mathfrak{v}^\star$ maximises the utility term and $\pi^\star$ maximises extension size inside $\Pi_{\lambda_\rho(\mathfrak{v}^\star)}$. Then no admissible pair can have a larger first coordinate, and among those tied on the first coordinate none can have a larger second coordinate. Hence $(\mathfrak{v}^\star, \pi^\star)$ is lexicographically maximal for $\preceq_{\lambda_\rho, V}$. □

Proof of Result 10

Restatement of Proposition 10 (Over-constraint implies splintering). Assume at least one part remains locally viable, meaning $\Pi'_{\omega_j} \neq \emptyset$ for some j. If the system is over-constrained at $\mathfrak{v}$ (so $\Pi'_\omega = \emptyset$), then there is no policy simultaneously feasible for all parts. For any realised policy $\pi \in L_\mathfrak{v}$, the set $S(\pi) := \{j \in \{1, \ldots, m\} : \pi \in \Pi'_{\omega_j}\}$ is a proper subset of $\{1, \ldots, m\}$. Any continued function can only proceed via a splinter.

Proof. Suppose for contradiction that some $\pi \in L_\mathfrak{v}$ is feasible for all parts. Then $\pi \in \Pi'_{\omega_j}$ for every $j \in \{1, \ldots, m\}$. By definition of the restricted collective set, $\pi \in \bigcap_{j=1}^{m} \Pi'_{\omega_j} = \Pi'_\omega$. This contradicts $\Pi'_\omega = \emptyset$. Therefore no such π exists.

It follows that for every $\pi \in L_\mathfrak{v}$, there exists at least one j with $\pi \notin \Pi'_{\omega_j}$, meaning $S(\pi) \neq \{1, \ldots, m\}$.

Now suppose continued function occurs, meaning some $\pi \in L_\mathfrak{v}$ is realised. If $S(\pi) \neq \emptyset$, then $\pi \in \bigcap_{j \in S(\pi)} \Pi'_{\omega_j}$, so $S(\pi)$ is a splinter (Definition 41). If $S(\pi) = \emptyset$, then π is feasible for no part. But by hypothesis at least one part j has $\Pi'_{\omega_j} \neq \emptyset$, so that part can realise some $\pi' \in \Pi'_{\omega_j}$, and $\{j\}$ is a splinter (since $\bigcap_{k \in \{j\}} \Pi'_{\omega_k} = \Pi'_{\omega_j} \neq \emptyset$).

In either case, any continued function proceeds via a proper subset of parts whose restricted policy sets admit a nonempty intersection. That subset is a splinter. □

SELVES, CAUSALITY AND CONSCIOUSNESS

Proof of Result 20

Restatement of Theorem 20 (Decoder mismatch breaks passive attribution). Let an evaluator observe a trace y with hypothesis class

$$\mathcal{H} = \mathcal{H}_{\text{mind}} \cup \{h_{\text{noise}}\}.$$

Assume $\mathbb{P}(h_{\text{noise}}) > 0$ and $\mathbb{P}(y \mid h_{\text{noise}}) > 0$. If there exists $\varsigma \in (0,1)$ such that for every $h \in \mathcal{H}_{\text{mind}}$,

$$\mathbb{P}(y \mid h) \leq \varsigma\, \mathbb{P}(y \mid h_{\text{noise}}),$$

then

$$\mathbb{P}(\mathcal{H}_{\text{mind}} \mid y) \leq \frac{\varsigma\, \mathbb{P}(\mathcal{H}_{\text{mind}})}{\varsigma\, \mathbb{P}(\mathcal{H}_{\text{mind}}) + \mathbb{P}(h_{\text{noise}})}.$$

Proof. Write

$$N := \sum_{h \in \mathcal{H}_{\text{mind}}} \mathbb{P}(h)\, \mathbb{P}(y \mid h) \qquad \text{and} \qquad A := \mathbb{P}(h_{\text{noise}})\, \mathbb{P}(y \mid h_{\text{noise}}).$$

Then

$$\mathbb{P}(\mathcal{H}_{\text{mind}} \mid y) = \frac{N}{N + A}.$$

By assumption,

$$N \leq \varsigma\, \mathbb{P}(y \mid h_{\text{noise}}) \sum_{h \in \mathcal{H}_{\text{mind}}} \mathbb{P}(h) = \varsigma\, \mathbb{P}(y \mid h_{\text{noise}})\, \mathbb{P}(\mathcal{H}_{\text{mind}}).$$

The function $x \mapsto x/(x + A)$ is increasing for $A > 0$, so substituting the upper bound on N gives

$$\mathbb{P}(\mathcal{H}_{\text{mind}} \mid y) \leq \frac{\varsigma\, \mathbb{P}(\mathcal{H}_{\text{mind}})\, \mathbb{P}(y \mid h_{\text{noise}})}{\varsigma\, \mathbb{P}(\mathcal{H}_{\text{mind}})\, \mathbb{P}(y \mid h_{\text{noise}}) + \mathbb{P}(h_{\text{noise}})\, \mathbb{P}(y \mid h_{\text{noise}})}.$$

Cancel the common factor $\mathbb{P}(y \mid h_{\text{noise}})$. □

Proof of Result 17

Restatement of Theorem 17 (Intervention tagging is unavoidable). Let an organism $\mathfrak{o}$ update an internal state s after each episode and choose actions as a function of that state. If expected utility can be improved by conditioning on whether an episode was self-generated or merely observed, while holding fixed the represented episode content in $L_{\mathfrak{v}_\mathfrak{o}}$, then the internal state must encode an intervention tag.

Proof. Assume for contradiction that the internal state does not encode that distinction. Then there exist two episodes e_{int} and e_{obs}

that have the same represented content in $L_{\mathfrak{v}_0}$ but differ in causal status, one self-generated and one merely observed, and which are mapped by the update rule to the same next internal state s'. Any action policy defined on internal states must therefore treat the two cases identically. But by hypothesis there exists a decision problem in which expected utility is improved by conditioning on the difference between the two cases. That improvement is impossible if the two cases are collapsed to the same internal state. This contradiction shows that the internal state must encode an intervention tag. $\square$

Proof of Result 16

Restatement of Proposition 16 (Do-operator as conditioning on an intervention indicator). Let X and Y be random variables and let $do(X = x)$ denote the intervention that sets X to the value x. Introduce an auxiliary binary variable $A \in \{0,1\}$ indicating the regime, where $A = 1$ means "intervene and set $X = x$" and $A = 0$ means "observe" (no intervention). Consider the augmented model in which X is generated by

$$X := \begin{cases} x, & A = 1, \\ \text{(its original mechanism)}, & A = 0. \end{cases}$$

Then the interventional distribution is an ordinary conditional in the augmented model:

$$\mathbb{P}(Y \in B \mid do(X = x)) \;=\; \mathbb{P}(Y \in B \mid A = 1) \qquad \text{for all measurable } B.$$

Moreover, whenever $\mathbb{P}(X = x) > 0$,

$$\mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1)\, \mathbb{P}(A = 1 \mid X = x) + \mathbb{P}(Y \in B \mid A = 0, X = x)\, \mathbb{P}(A = 0 \mid X = x).$$

Proof. Fix x. Let $P_{\text{obs}}(X, Y)$ denote the observational joint distribution and let $P_{\text{do}(x)}(Y)$ denote the interventional distribution of Y under $do(X = x)$. I can always construct a joint distribution on (A, X, Y) as follows:

- Sample $A \in \{0,1\}$ according to any distribution with $\mathbb{P}(A = 1) > 0$.
- If $A = 0$, sample (X, Y) from the observational distribution $P_{\text{obs}}(X, Y)$.
- If $A = 1$, set $X := x$ deterministically and sample Y from $P_{\text{do}(x)}(Y)$.

By construction, the conditional law of Y given $A = 1$ is exactly the interventional law of Y under $do(X = x)$, so $\mathbb{P}(Y \in B \mid A = 1) =$

$\mathbb{P}(Y \in B \mid do(X = x))$ for all measurable B. Because $A = 1$ implies $X = x$, I also have $\mathbb{P}(Y \in B \mid A = 1, X = x) = \mathbb{P}(Y \in B \mid A = 1)$.

Finally, assuming $\mathbb{P}(X = x) > 0$, apply the law of total probability conditioned on $X = x$:

$$\mathbb{P}(Y \in B \mid X = x) = \mathbb{P}(Y \in B \mid A = 1, X = x)\,\mathbb{P}(A = 1 \mid X = x) + \mathbb{P}(Y \in B \mid A = 0, X = x)\,\mathbb{P}(A = 0 \mid X = x),$$

and substitute $\mathbb{P}(Y \in B \mid A = 1, X = x) = \mathbb{P}(Y \in B \mid A = 1)$ to obtain the stated decomposition. □

Proof of Result 17

Restatement of Proposition 17 (Existence of w-maximised causal identities). Assume $L_{\mathfrak{v}}$ is finite and let $INT, OBS \subseteq L_{\mathfrak{v}}$. If $\mathcal{C}(INT, OBS) \neq \emptyset$, then there exists at least one w-maximised causal identity for (INT, OBS) in the sense of Definition 47.

Proof. Because $L_{\mathfrak{v}}$ is finite, the candidate set $\mathcal{C}(INT, OBS) \subseteq L_{\mathfrak{v}}$ is finite. Define $g : \mathcal{C}(INT, OBS) \to \mathbb{N}$ by $g(c) := |\mathrm{Ext}(c)|$. A function from a finite set to $\mathbb{N}$ attains a maximum, so there exists $c^\star \in \mathcal{C}(INT, OBS)$ such that

$$g(c^\star) = \max_{c \in \mathcal{C}(INT,OBS)} g(c) = \max_{c \in \mathcal{C}(INT,OBS)} |\mathrm{Ext}(c)|.$$

By Definition 47, $c^\star$ is a w-maximised causal identity. □

Proof of Result 18

Restatement of Proposition 18 (Intelligence is not independent of goals). Intelligence is not independent of goals.

Proof. Assume $\mathcal{C}$ is a space of software programs, Γ is a space of behaviours a system can exhibit, $f_1 \in \mathcal{C}$ is a software mind and $f_2 : \mathcal{C} \to \Gamma$ is a hardware body. It interprets f_1. Finally $\mathbb{G}$ is the set of environments, which here are functions $f : \Gamma \to \{0, 1\}$. I single out $f_3 \in \mathbb{G}$ as the environment in which goals are pursued. If I am the engineer who built the software AI f_1 for some particular purpose, then I am part of its environment. Goals are satisfied if $f_3(f_2(f_1)) = 1$. I will now show why intelligence is not independent of embodiment, and embodiment is not independent of goals.

1. Suppose I am given f_3. For every environment there exists hardware s.t. $f_3(f_2(\cdot)) = 1$ regardless of software (it maps all software to the same behaviour, or in other words it is hard-wired to satisfy the goals in f_3).

2. For every choice of f_1 given f_3 alone, there exists f_2 s.t. $f_3(f_2(f_1)) = 0$.

3. This means intelligence is not independent of embodiment[106].

4. Now I'll show goals are not independent of embodiment. Assume I am given a fixed $f_2(f_1)$.

5. I can choose f_3 so that $f_2(f_1)$ is optimal and the goals in f_3 are satisfied.

6. However, $f_2(f_1)$ determines which choices of f_3 are optimal. $f_2(f_1) = \gamma$ constrains us to choices of f_3 where $f_3(\gamma) = 1$[107].

7. This means goals are not independent of embodiment.

As intelligence hinges on embodiment, and embodiment is goal directed, intelligence is inevitably goal directed. □

[106] Similar points are made in related work
Bennett, M. T. (2023c). The optimal choice of hypothesis is the weakest, not the shortest. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 42–51. Springer; Bennett, M. T. (2024c). Is complexity an illusion? In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer; and Leike, J. and Hutter, M. (2015). Bad universal priors and notions of optimality. In Gr¨unwald, P., Hazan, E., and Kale, S., editors, *Proceedings of The 28th Conference on Learning Theory*, volume 40 of *Proceedings of Machine Learning Research*, pages 1244–1259, Paris, France. PMLR

[107] Meaning only those environments and goals where a particular behaviour or phenotype γ will succeed.

Proof of Result 18

Restatement of Theorem 18 (Second-order self is necessary for Gricean report). Consider a communication game in which a speaker $\mathfrak{a}$ must transmit a private bit $m \in \{0,1\}$ to a listener $\mathfrak{b}$. The listener type $t \in \{0,1\}$ is drawn uniformly at random and hidden from $\mathfrak{a}$. If $t = 0$, the listener decodes literally. If $t = 1$, the listener flips the decoded bit. Assume $\mathfrak{a}$ may send one probe signal and observe one listener response before sending the final signal. Any speaker policy that cannot condition on the listener response achieves expected success at most $\frac{1}{2}$. There exists a policy that conditions on the response and achieves success probability 1. Such conditioning requires a second-order self state $\mathfrak{a}^2$.

Proof. Suppose first that the speaker cannot condition on the observed response. Then, conditioned on the intended bit m, the final signal distribution is a function of m alone. Fix any such rule. Under listener type $t = 0$, literal decoding is correct exactly when the emitted signal matches m. Under listener type $t = 1$, the listener flips the decoded bit, so the same emitted signal is correct exactly when it does not match m. Therefore the success probability under $t = 0$ is exactly the failure probability under $t = 1$. Averaging over the two equally likely listener types gives overall success probability $\frac{1}{2}$. Now consider the policy that sends the probe symbol 0. If the observed listener response is 0, infer $t = 0$. If the observed response is 1, infer $t = 1$. Then send the final signal $s = m$ when $t = 0$ and $s = 1 - m$ when $t = 1$. In both cases the listener decodes the intended bit correctly, so success probability is 1. This policy depends on representing how the listener interprets the speaker's behaviour. That is precisely the minimal role of a second-order self. □

Proof of Result 19

Restatement of Proposition 19 ($\mathbf{n^{th}}$ order self convergence). Let $\mathfrak{o}$ be an organism with current history task $\mathfrak{h}_{<t_\mathfrak{o}}$ and viability task $\mu_\mathfrak{o}$ such that $\mathfrak{h}_{<t_\mathfrak{o}} \sqsubset \mu_\mathfrak{o}$. Assume $\mathfrak{o}$ selects policies from $\Pi_{\mathfrak{h}_{<t_\mathfrak{o}}}$ using the weakness proxy and successfully learns $\mu_\mathfrak{o}$. Let $\mathfrak{o}^n$ be an $n^{\mathbf{th}}$-order self of $\mathfrak{o}$. If the representation precondition holds, so $\mathfrak{o}^n \in L_{\mathfrak{v}_\mathfrak{o}}$, and the incentive precondition holds, meaning that remaining fit requires learning $\mathfrak{o}^n$, then

$$\mathfrak{o}^n \in \mathfrak{p}_\mathfrak{o}.$$

Proof. Because $\mathfrak{o}$ successfully learns $\mu_\mathfrak{o}$, Definition 29 guarantees that the organism outputs some correct policy for the viability task. By the meaning of the incentive precondition, remaining fit requires learning $\mathfrak{o}^n$. Therefore any successful learning trajectory to $\mu_\mathfrak{o}$ must place $\mathfrak{o}^n$ in the organism's known-policy set $\mathfrak{p}_\mathfrak{o}$.

The representation assumption $\mathfrak{o}^n \in L_{\mathfrak{v}_\mathfrak{o}}$ guarantees that this learning event is well-typed in the organism's embodied language. Hence

$$\mathfrak{o}^n \in \mathfrak{p}_\mathfrak{o}.$$

□

Proof of Result 19

Restatement of Theorem 19 (A commitment game separates audience modelling from self-binding). In the game of Definition 53.

1. If the self-binding move B is *not available* (i.e. $\mathfrak{a}$ can only play F), then the unique subgame-perfect equilibrium outcome is N with payoff $(0, 0)$.
2. If B is available and $0 < c < 1$, then there exists a subgame-perfect equilibrium in which $\mathfrak{a}$ plays B, $\mathfrak{b}$ plays T after observing B (and N after observing F), and the Stage 2 outcome is necessarily H, yielding payoff $(1 - c, 1)$.

Proof. (1) If B is unavailable, then after $\mathfrak{b}$ chooses T the actor $\mathfrak{a}$ chooses between H and E and strictly prefers E because $2 > 1$. Anticipating this, $\mathfrak{b}$ strictly prefers N to T because $0 > -1$. Hence the unique subgame-perfect equilibrium outcome is N.

(2) Now allow B. In the subgame following (F, T) the same reasoning applies: $\mathfrak{a}$ plays E, so $\mathfrak{b}$'s best response after observing F is N. In the subgame following $(B, \cdot)$, $\mathfrak{a}$ is forced to play H, so $\mathfrak{b}$'s best response after observing B is T because $1 > 0$. Anticipating these best responses, $\mathfrak{a}$ compares Stage 0 payoffs. Choosing F yields 0 (since $\mathfrak{b}$ chooses N), whereas choosing B yields $1 - c$. Because $0 < c < 1$, I have $1 - c > 0$, so $\mathfrak{a}$ strictly prefers B.

□

Proof of Result 21

Restatement of Theorem 21 (The Psychophysical Principle of Causality). Let π be any policy with $\Omega_\pi \neq \emptyset$. Let c be a commitment not forced by viability with Ω_π containing at least one viable continuation o with $c \not\subseteq o$. Then either $\Omega_{\pi \cup c} = \emptyset$ or $F_2(\pi \cup c) > F_2(\pi)$.

Proof. Since $\pi \cup c \supseteq \pi$, every completion of $\pi \cup c$ is a completion of π. That is, $\text{Ext}(\pi \cup c) \subseteq \text{Ext}(\pi)$. Intersecting with $\text{Ext}(\mu)$ gives $\Omega_{\pi \cup c} \subseteq \Omega_\pi$.

By hypothesis, there exists $o \in \Omega_\pi$ with $c \not\subseteq o$. Since $o \in \Omega_\pi = \text{Ext}(\pi) \cap \text{Ext}(\mu)$, we have $\pi \subseteq o$. But $c \not\subseteq o$, so $\pi \cup c \not\subseteq o$. Therefore $o \notin \text{Ext}(\pi \cup c)$, and hence $o \notin \Omega_{\pi \cup c}$.

Since $o \in \Omega_\pi \setminus \Omega_{\pi \cup c}$, the inclusion $\Omega_{\pi \cup c} \subseteq \Omega_\pi$ is strict. If $\Omega_{\pi \cup c} = \emptyset$ we are done. Otherwise $|\Omega_{\pi \cup c}| < |\Omega_\pi|$, so

$$F_2(\pi \cup c) = \log_2 |\text{Ext}(\mu)| - \log_2 |\Omega_{\pi \cup c}| > \log_2 |\text{Ext}(\mu)| - \log_2 |\Omega_\pi| = F_2(\pi).$$

□

THE ORIGINS OF LIFE

Proof of Result 20

Restatement of Proposition 20 (The Law of Increasing Functional Information). Fix a finite vocabulary $\mathfrak{v}$ and a task $\alpha \in \Gamma_{\mathfrak{v}}$ with $\Pi_\alpha \neq \emptyset$. Draw π uniformly from Π_α. Independently draw a random parent task ω of α using the maximally uninformative extension model of Proposition 4. Assume persistence into time $t+1$ means surviving exactly when $\pi \in \Pi_\omega$. Then the expected functional information of the persisting population is at least as large as it was at time t. More precisely,

$$\mathbb{E}\big[I(\pi) \mid \pi \in \Pi_\omega\big] \;\geq\; \mathbb{E}\big[I(\pi)\big],$$

with strict inequality whenever the weakness proxy $<_w$ is not constant on Π_α.

Proof. Write $\Pi_\alpha = \{\pi_1, \dots, \pi_k\}$ and order the policies so that

$$|\text{Ext}(\pi_1)| \leq |\text{Ext}(\pi_2)| \leq \cdots \leq |\text{Ext}(\pi_k)|.$$

Under the maximally uninformative extension model of Proposition 4,

$$\mathbb{P}(\pi_j \in \Pi_\omega \mid \alpha) \;=\; \frac{2^{|\text{Ext}(\pi_j) \cap U|}}{2^{|U|}},$$

which is nondecreasing in $|\mathrm{Ext}(\pi_j)|$ over Π_α.

Now define $I_j := I(\pi_j)$. By construction of M_π, if $|\mathrm{Ext}(\pi_a)| \leq |\mathrm{Ext}(\pi_b)|$ then $M_{\pi_b} \subseteq M_{\pi_a}$. So $|M_{\pi_b}| \leq |M_{\pi_a}|$ and therefore $I_b \geq I_a$. Hence $I_1 \leq I_2 \leq \cdots \leq I_k$.

At time t the population is uniform on Π_α, so $\mathbb{E}[I(\pi)] = \frac{1}{k}\sum_{j=1}^k I_j$. Conditioning on persistence weights each π_j by $\mathbb{P}(\pi_j \in \Pi_\omega \mid \alpha)$, so

$$\mathbb{E}[I(\pi) \mid \pi \in \Pi_\omega] \;=\; \frac{\sum_{j=1}^k q_j I_j}{\sum_{j=1}^k q_j}, \qquad q_j := \mathbb{P}(\pi_j \in \Pi_\omega \mid \alpha).$$

The sequences (q_j) and (I_j) are both nondecreasing. By Chebyshev's sum inequality,

$$\frac{1}{k}\sum_{j=1}^k q_j I_j \;\geq\; \left(\frac{1}{k}\sum_{j=1}^k q_j\right)\left(\frac{1}{k}\sum_{j=1}^k I_j\right).$$

Dividing by $\frac{1}{k}\sum_{j=1}^k q_j$ gives

$$\frac{\sum_{j=1}^k q_j I_j}{\sum_{j=1}^k q_j} \;\geq\; \frac{1}{k}\sum_{j=1}^k I_j.$$

This is exactly $\mathbb{E}[I(\pi) \mid \pi \in \Pi_\omega] \geq \mathbb{E}[I(\pi)]$. If $<_w$ is not constant on Π_α, then (q_j) is not constant and the inequality is strict. $\square$

MULTICLASS LEARNING

Proof of Result 21

Restatement of Proposition 21 (Weakness pair proxy optimality). Let $\mathfrak{h}_{<t}$ be a history-task and let $\mathfrak{h}^1_{<t}$ and $\mathfrak{h}^0_{<t}$ denote the induced positive and negative history-tasks (as in Definition 60). Consider the admissible set of policy pairs

$$\mathfrak{pp} \;:=\; \{(\pi, n) \in \Pi_{\mathfrak{h}^1_{<t}} \times \Pi_{\mathfrak{h}^0_{<t}} \mid \mathrm{Ext}(\pi) \cap \mathrm{Ext}(n) = \varnothing\}.$$

Define the *effective completion set* of a pair as $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$. Under the same maximally uninformative extension model as in Proposition 4, apply the model to the effective completion set $E_{\pi,n}$. Then the probability that an admissible pair generalises to an unobserved parent is maximised by choosing $(\pi, n) \in \mathfrak{pp}$ with maximal $|\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)|$. Equivalently, the weakness pair proxy is optimal under that model.

Proof. The proof is the same counting argument as in Proposition 4, with $\mathrm{Ext}(\pi)$ replaced by the effective completion set $E_{\pi,n} := \mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)$.

Let U denote the set of outputs not fixed by the observed history (the "unseen" region). Under the maximally uninformative model, the parent selects an additional set $S \subseteq U$ uniformly at random. An admissible pair (π, n) is consistent with that parent extension precisely when all selected unseen outputs lie inside the effective completion set, i.e., when $S \subseteq E_{\pi,n} \cap U$. Therefore,

$$\mathbb{P}\big((\pi, n) \text{ generalises} \mid \mathfrak{h}_{<t}\big) = \frac{2^{|E_{\pi,n} \cap U|}}{2^{|U|}},$$

which is strictly increasing in $|E_{\pi,n} \cap U|$, and hence (across admissible pairs) in $|E_{\pi,n}| = |\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)|$. Thus maximising $|\mathrm{Ext}(\pi) \cup \mathrm{Ext}(n)|$ maximises the generalisation probability in this model. $\square$

EXPERIMENTS

In the appendix is a Python script to perform two experiments using PyTorch with CUDA, SymPy and A^*[108]. Here I'll summarise the results and how the experiments worked. First, I wrote a toy program based on A^* that learns policies for 8-bit string prediction tasks (binary addition and multiplication)[109].

[108] Paszke, A. et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*; Kirk, D. (2007). Nvidia cuda software and gpu parallel computing architecture. In *Proceedings of the 6th International Symposium on Memory Management*, ISMM '07, page 103–104, New York, NY, USA. Association for Computing Machinery; Meurer, A., Smith, C., Paprocki, M., Čertík, O., Kirpichev, S., Rocklin, M., Kumar, A., Ivanov, S., Moore, J., Singh, S., Rathnayake, T., Vig, S., Granger, B., Muller, R., Bonazzi, F., Gupta, H., Vats, S., Johansson, F., Pedregosa, F., and Scopatz, A. (2017). Sympy: Symbolic computing in python. *PeerJ Computer Science*, 3:e103; and Hart, P. E., Nilsson, N. J., and Raphael, B. (1968). A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107

[109] Notation here varies slightly from the formal notation due to the limitations of what can be written in Python (not LaTeX), and because the experiments coincided with an earlier iteration of the formalism.

(abstraction layer) I used a simplified environment of 256 states, one for every possible 8-bit string. Basically an abstraction layer without my environment. The statements in L were expressions regarding those 8 bits that could be written in propositional logic ($\neg$, $\wedge$ and $\vee$).

(task) A task was specified by choosing $O \subset L$ such that all $d \in O$ conformed to the rules of either binary addition (for the first experiment) or multiplication (for the second experiment) with 4-bits of input, followed by 4-bits of output.

Each of the two experiments (addition and multiplication) involved repeated trials (sampling results). The parameters of each trial were "operation" (a function), and an even integer "number_of_trials" between 4 and 14 which determined the cardinality of the set O_k (defined below). Each trial was divided into training and testing phases.

(training phase)

1. A task T_n was generated:
   (a) First, every possible 4-bit input for the chosen binary operation was used to generate an 8-bit string. These 16 strings then formed O_n.
   (b) A bit between 0 and 7 was then chosen, and I_n created by cloning O_n and deleting the chosen bit from every string (meaning I_n was composed of 16 different 7-bit strings, each of which could be found in an 8-bit string in O_n).
2. A child-task $T_k = \langle I_k, O_k \rangle$ was sampled from the parent task T_n. Recall, $|O_k|$ was determined as a parameter of the trial.

3. From T_k two policies (formerly known as models) were generated. A weakest c_w, and a MDL c_{mdl}.

(TESTING PHASE) For each policy $c \in \{c_w, c_{mdl}\}$:

1. The extension $\text{Ext}(c)$ of c was then generated.
2. A prediction O_{recon} was then constructed s.t. $O_{recon} = \{e \in \text{Ext}(c) : \exists s \in I_n \ (s \subset z)\}$.
3. O_{recon} was then compared to the ground truth O_n, and results recorded.

Between 75 and 256 trials were run for each value of the parameter $|O_k|$. Fewer trials were run for larger values of $|O_k|$ due to restricted availability of hardware. The results of these trials were then averaged for each value of $|O_k|$.

(MEASUREMENTS) Generalisation was deemed to have occurred where $O_{recon} = O_n$. The number of trials in which generalisation occurred was measured, and divided by n to obtain the rate of generalisation for c_w and c_{mdl}. Error was computed as a Wald 95% confidence interval. Even where $O_{recon} \neq O_n$, the extent to which policies generalised could be ascertained. $\frac{|O_{recon} \cap O_n|}{|O_n|}$ was measured and averaged for each value of $|O_k|$, and the standard error computed.

	c_w				c_{mdl}			
$\lvert O_k \rvert$	Rate	±95%	AvgExt	StdErr	Rate	±95%	AvgExt	StdErr
6	.11	.039	.75	.008	.10	.037	.48	.012
10	.27	.064	.91	.006	.13	.048	.69	.009
14	.68	.106	.98	.005	.24	.097	.91	.006

Table 1: Binary addition.

	c_w				c_{mdl}			
$\lvert O_k \rvert$	Rate	±95%	AvgExt	StdErr	Rate	±95%	AvgExt	StdErr
6	.05	.026	.74	.009	.01	.011	.58	.011
10	.16	.045	.86	.006	.08	.034	.78	.008
14	.46	.061	.96	.003	.21	.050	.93	.003

Table 2: Binary multiplication.

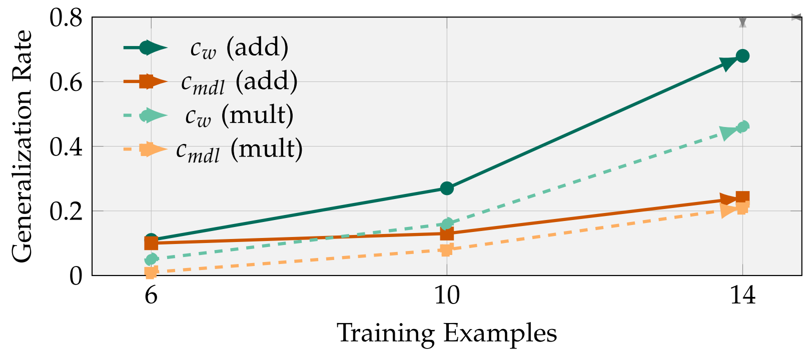

Figure 1: Rates for binary addition (solid) and mult. (dashed).

EXAMPLES

Remark 9 (Finite-state machines as tasks). *A deterministic finite-state machine (FSM) is a tuple*

$$f = \langle Q, \Sigma, \delta, q_0, A \rangle$$

with finite state set Q*, finite input alphabet* Σ*, transition function* $\delta : Q \times \Sigma \to Q$*, start state* $q_0 \in Q$*, and accepting set* $A \subseteq Q$*.*

I can encode f *as a finite* $\mathfrak{v}_f$*-task in Stack Theory by choosing the (finite) environment*

$$\Phi_f := Q \times \Sigma \times Q,$$

whose microstates are triples (q, σ, q') *interpreted as "current state* q*, read symbol* σ*, next state* q'*."*

Define the following programs (subsets of Φ_f*):*

$$\mathrm{Curr}_q := \{(q, \sigma, q') : \sigma \in \Sigma,\ q' \in Q\}, \qquad \mathrm{Sym}_\sigma := \{(q, \sigma, q') : q \in Q,\ q' \in Q\},$$

$$\mathrm{Next}_{q'} := \{(q, \sigma, q') : q \in Q,\ \sigma \in \Sigma\}, \qquad \mathrm{Step}_f := \{(q, \sigma, q') : q' = \delta(q, \sigma)\}.$$

Let $\mathrm{Acc}_f := \{(q, \sigma, q') : q' \in A\}$ *and* $\mathrm{Rej}_f := \Phi_f \setminus \mathrm{Acc}_f$*.*

Let $\mathfrak{v}_f$ *be the finite vocabulary consisting of all programs* Curr_q *(*$q \in Q$*),* Sym_σ *(*$\sigma \in \Sigma$*),* $\mathrm{Next}_{q'}$ *(*$q' \in Q$*), and the three global programs* $\mathrm{Step}_f, \mathrm{Acc}_f, \mathrm{Rej}_f$*.*

For each pair $(q, \sigma) \in Q \times \Sigma$ *define an input statement*

$$i_{q,\sigma} := \{\mathrm{Curr}_q, \mathrm{Sym}_\sigma, \mathrm{Step}_f\}$$

and an output statement

$$o_{q,\sigma} := i_{q,\sigma} \cup \{\mathrm{Next}_{\delta(q,\sigma)}\} \cup \begin{cases} \{\mathrm{Acc}_f\} & \text{if } \delta(q,\sigma) \in A, \\ \{\mathrm{Rej}_f\} & \text{otherwise.} \end{cases}$$

Then $\rho_f := \langle I_f, O_f \rangle$ *with* $I_f := \{i_{q,\sigma} : (q, \sigma) \in Q \times \Sigma\}$ *and* $O_f := \{o_{q,\sigma} : (q, \sigma) \in Q \times \Sigma\}$ *is a* $\mathfrak{v}_f$*-task: each input* $i_{q,\sigma}$ *has a completion in the outputs, and each output refines its corresponding input.*

This construction is injective up to obvious renaming: from ρ_f *one can recover the deterministic transition relation encoded by* Step_f *(hence* δ*) and the accepting set encoded by* Acc_f.

PHANTASMATA

Complexity is a phantasm. It is an appearance manufactured by bookkeeping choices. This chapter collects the results that expose how three specific choices, namely forgetting labels, changing resolution, and pushing decisions down a stack, manufacture or eliminate apparent complexity. Each section is a different apparition. The Gibbs paradox is a phantasm of distinguishability. Fractal dimension is a phantasm of scale. Every entropy shift is a phantasm of what you chose to ignore. The core machinery is the same in all three cases. Choose which differences count. Forget the rest. Count what survives.

These results first appeared in the journal extension of *Is Complexity an Illusion?*[110]. They use the encoding map $\mathrm{Enc}_{\mathfrak{v}}$ from Definition 1 and the language decomposition from Lemma 8.

[110] Bennett, M. T. (2024c). Is complexity an illusion? In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer

ENTROPY UNDER MACRO MAPS

A **macro map** $q : \Phi \to Z$ merges many microstates into one macro case. The fibre $q^{-1}(\{z\})$ is the set of microstates that land in the same bucket. This is what it means to ignore a distinction.

Lemma 12 (Truth sets are unions of macro fibres)**.** *Fix an environment Φ and a finite vocabulary $\mathfrak{v} \subseteq 2^{\Phi}$. For any statement $l \in L_{\mathfrak{v}}$ and any $\phi_1, \phi_2 \in \Phi$, if $\mathrm{Enc}_{\mathfrak{v}}(\phi_1) = \mathrm{Enc}_{\mathfrak{v}}(\phi_2)$ then $\phi_1 \in T(l) \iff \phi_2 \in T(l)$.*

Intuition. The vocabulary cannot tell apart two microstates with the same encoding. If one satisfies a statement, so does the other.

Proof. By Lemma 1, $\phi \in T(l) \iff l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi)$. If $\mathrm{Enc}_{\mathfrak{v}}(\phi_1) = \mathrm{Enc}_{\mathfrak{v}}(\phi_2)$, then $l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi_1) \iff l \subseteq \mathrm{Enc}_{\mathfrak{v}}(\phi_2)$. □

Proposition 22 (Truth sets are pullbacks of encoding images)**.** *For every statement $l \in L_{\mathfrak{v}}$, $T(l) = \mathrm{Enc}_{\mathfrak{v}}^{-1}(\mathcal{M}(l))$, where $\mathcal{M}(l) = \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in T(l)\}$. If $\Phi_\star \subseteq \Phi$ is finite, write $\mathcal{M}_\star(l) := \{\mathrm{Enc}_{\mathfrak{v}}(\phi) \mid \phi \in T(l) \cap \Phi_\star\}$ and $S_{\text{state}}(l) := \log_2 |T(l) \cap \Phi_\star|$. Then*

$$|T(l) \cap \Phi_\star| = \sum_{m \in \mathcal{M}_\star(l)} |\mathrm{Enc}_{\mathfrak{v}}^{-1}(\{m\}) \cap \Phi_\star|.$$

If all fibres have the same size C, then $S_{\text{state}}(l) = \log_2 |\mathcal{M}_\star(l)| + \log_2 C$.

Intuition. Microstate entropy splits into a macro part (how many encodings are compatible) plus a within-fibre part (how many microstates share each encoding). The within-fibre part is the entropy that the vocabulary cannot see. It is the entropy of what you chose to ignore.

Proof. Fix l. For any ϕ, $\phi \in \mathrm{Enc}_{\mathfrak{v}}^{-1}(\mathcal{M}(l)) \iff \mathrm{Enc}_{\mathfrak{v}}(\phi) \in \mathcal{M}(l) \iff \exists\, \psi \in T(l) : \mathrm{Enc}_{\mathfrak{v}}(\psi) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$. By Lemma 12, this implies $\phi \in T(l)$. Conversely, $\phi \in T(l)$ implies $\mathrm{Enc}_{\mathfrak{v}}(\phi) \in \mathcal{M}(l)$. This proves $T(l) = \mathrm{Enc}_{\mathfrak{v}}^{-1}(\mathcal{M}(l))$.

For the counting identity, the fibres $\mathrm{Enc}_{\mathfrak{v}}^{-1}(\{m\})$ are disjoint, so $|T(l) \cap \Phi_\star| = \sum_{m \in \mathcal{M}_\star(l)} |\mathrm{Enc}_{\mathfrak{v}}^{-1}(\{m\}) \cap \Phi_\star|$. If all fibre sizes equal C, then $|T(l) \cap \Phi_\star| = C \cdot |\mathcal{M}_\star(l)|$. Taking $\log_2$ gives the entropy identity. □

Proposition 23 (Entropy chain rule for a macro map)**.** *Let P be a probability mass function on Φ with finite support $\Phi_\star$. Let $q : \Phi \to Z$ be surjective. Let $P_Z(z) = \sum_{\phi : q(\phi) = z} P(\phi)$ and $P(\phi \mid z) = P(\phi)/P_Z(z)$. Then*

$$H(P) = H(P_Z) + \sum_{z \in Z} P_Z(z)\, H(P(\cdot \mid z)).$$

If for every z the conditional is uniform on $q^{-1}(\{z\})$ and all fibres have the same size C, then $H(P) = H(P_Z) + \log_2 C$.

Intuition. Total entropy = macro entropy + average within-fibre entropy. If you forget which microstate you were in and only remember the macro bucket, you lose the within-fibre entropy. This is the standard Shannon chain rule, but the Stack Theory framing makes it a statement about what abstraction forgets.

Proof. Standard. Write $\log P(\phi) = \log P_Z(q(\phi)) + \log P(\phi \mid q(\phi))$. Multiply by $-P(\phi)$, sum over $\Phi_\star$, and collect terms by $z = q(\phi)$. □

THE GIBBS PARADOX AS A RELABELLING PHANTASM

The Gibbs paradox asks why the entropy of mixing two identical gases is zero rather than $k \ln 2$ per particle. The classical answer is "because the particles are identical, so we must divide by $N!$." The Stack Theory answer is that the $N!$ is a fibre size. It counts how many microstates share the same encoding under a relabelling-invariant vocabulary.

Lemma 13 (Relabel-invariant vocabularies cannot separate relabelings)**.** *Let G be a finite group acting on Φ. Say $\mathfrak{v}$ is* ***relabel-invariant*** *if for every $p \in \mathfrak{v}$ and every $g \in G$, $g \cdot p := \{g(\phi) : \phi \in p\} \in \mathfrak{v}$. Then for every $g \in G$ and every $\phi \in \Phi$, $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = g \cdot \mathrm{Enc}_{\mathfrak{v}}(\phi) := \{g \cdot p : p \in \mathrm{Enc}_{\mathfrak{v}}(\phi)\}$.*

If additionally $g \cdot p = p$ for all $p \in \mathfrak{v}$ and all $g \in G$ (the vocabulary is literally invariant, not just covariant), then $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$ for all g, ϕ. In this case, the entire orbit $G \cdot \phi$ maps to the same encoding. The encoding fibre has size at least $|G \cdot \phi|$.

Proof. $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi) \iff \phi \in p \iff g(\phi) \in g \cdot p$. So $p \in \mathrm{Enc}_{\mathfrak{v}}(\phi) \iff g \cdot p \in \mathrm{Enc}_{\mathfrak{v}}(g(\phi))$. If $g \cdot p = p$ for all p, this gives $\mathrm{Enc}_{\mathfrak{v}}(g(\phi)) = \mathrm{Enc}_{\mathfrak{v}}(\phi)$. □

Corollary 9 (The $N!$ term in the Gibbs paradox)**.** *Let Φ consist of N labelled particles in k boxes. The symmetric group S_N acts by permuting particle labels. If the vocabulary is literally invariant under S_N (it encodes only occupation numbers, not which particle is where), then for a configuration with all particles in distinct boxes, the encoding fibre has size $N!$. By Proposition 23, the within-fibre entropy is $\log_2(N!)$. The Gibbs correction is the subtraction of this within-fibre entropy.*

Intuition. The $N!$ is not a mysterious quantum correction. It is the number of microstates the vocabulary cannot distinguish. The encoding map merges them. The entropy chain rule says total entropy = macro entropy + fibre entropy. The Gibbs correction subtracts the fibre entropy because the vocabulary never saw it.

Proof. If all N particles are in distinct boxes, every permutation of labels gives a different microstate but the same occupation-number encoding. The orbit under S_N has size $N!$ (no stabiliser). By Lemma 13, all $N!$ microstates share the same encoding. The fibre size is $N!$. Proposition 23 gives the entropy split. □

SCALE AS AN ABSTRACTION MAP

Resolution is another way to forget distinctions. Looking at a system through a coarser lens merges microstates that were previously distinguishable. This is the same move as a macro map, applied to spatial or temporal scale.

Definition 64 (Covering and box dimension)**.** *Let (X, d) be a metric space and $A \subseteq X$. For $\varepsilon > 0$, the **covering number** $N(A, \varepsilon)$ is the minimum number of ε-balls needed to cover A. The **(upper) box-counting dimension** is*

$$\overline{\dim}_B(A) = \limsup_{\varepsilon \to 0} \frac{\log N(A, \varepsilon)}{-\log \varepsilon}.$$

Proposition 24 (Scale as a macro map)**.** *Fix $\varepsilon > 0$ and a finite grid $\mathcal{G}_\varepsilon$ of ε-cells covering X. Define the macro map $q_\varepsilon : X \to \mathcal{G}_\varepsilon$ by $q_\varepsilon(x) =$ the cell containing x. This is a macro map in the sense of the entropy chain rule. Two points that land in the same cell are indistinguishable at resolution*

ε. Write $N_\varepsilon(A)$ for the number of occupied cells. Then $\log_2 N_\varepsilon(A)$ is the scale entropy at resolution ε. Furthermore, the box-counting dimension can equivalently be defined using grid cells rather than balls, so $\overline{\dim}_B(A) = \limsup_{\varepsilon\to 0} \log N_\varepsilon(A)/(-\log \varepsilon)$.

Intuition. Resolution is a vocabulary choice. Looking at a fractal through a microscope with resolution ε is choosing a vocabulary that can distinguish features of size $\geq \varepsilon$ but not smaller. The box-counting dimension measures how fast the scale entropy grows as you look more closely.

Proof. The grid $\mathcal{G}_\varepsilon$ partitions X into cells. q_ε sends each point to its cell. Points in the same cell are within distance at most $\varepsilon\sqrt{d}$ (for a d-dimensional grid), so they are indistinguishable at resolution ε. The number of cells intersecting A is $N_\varepsilon(A)$. The scale entropy is $\log_2 N_\varepsilon(A)$. The equivalence between the grid-cell definition and the ε-ball definition of box-counting dimension is standard (the two counts differ by at most a factor depending on the ambient dimension, which vanishes under $\log/(-\log\varepsilon)$). $\square$

Proposition 25 (Scale entropy diverges with positive box dimension)**.** *If $\overline{\dim}_B(A) > 0$, then $\log_2 N(A, \varepsilon) \to \infty$ as $\varepsilon \to 0$. The finer you look, the more complexity you manufacture.*

Proof. $\overline{\dim}_B(A) > 0$ means $\limsup_{\varepsilon\to 0} \log N_\varepsilon(A)/(-\log\varepsilon) > 0$. So there exists $\delta > 0$ and a sequence $\varepsilon_k \to 0$ with $\log N_{\varepsilon_k}(A) \geq \delta(-\log \varepsilon_k)$ for all k. Since $-\log\varepsilon_k \to \infty$, we get $\log N_{\varepsilon_k}(A) \to \infty$ along this subsequence. But $N_\varepsilon(A)$ is nonincreasing in ε (a finer grid can only increase the count). So for any $\varepsilon < \varepsilon_k$, $N_\varepsilon(A) \geq N_{\varepsilon_k}(A)$. It follows that $\log N_\varepsilon(A) \to \infty$ as $\varepsilon \to 0$. $\square$

Intuition. A fractal has positive box dimension. The closer you look, the more structure you see. But the structure was always there. What changed is your resolution (your vocabulary). The complexity is a phantasm of scale.

SPLINTERING AND COORDINATION COST

When multiple subsystems share a single representation, their incompatible diagnostic needs impose a combinatorial tax. This section connects the regret-weakness bridge (Theorem 14) to the splintering results (Proposition 10) and the broader cancer-analogue theme. These results first appeared in *Regret Is Weighted Forgetting*[111].

[111] Bennett, M. T. (2026c). Regret is weighted forgetting

Proposition 26 (Coordination cost of shared representations)**.** *Let N subsystems each face a diagnostic family on the same support with the same single test T. Let $\mathcal{P}^\star_j$ be the coarsest pure partition for subsystem j*

on its informative support, with $|\mathcal{P}_j^\star| = k_j$ cells. A shared representation that achieves zero regret for all N subsystems simultaneously requires at least $|\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star|$ cells, where $\vee$ denotes the common refinement. In the worst case this is $\prod_{j=1}^N k_j$.

Proof. By Theorem 16, zero regret for subsystem j requires the shared partition to refine $\mathcal{P}_j^\star$. The coarsest partition refining all N is the common refinement $\mathcal{P}_1^\star \vee \cdots \vee \mathcal{P}_N^\star$. Its size is bounded above by $\prod_j k_j$ because each cell is determined by its class in each subsystem's partition. The bound is achieved when the partitions are independent (no two agree on the grouping of any pair of support points). □

Intuition. Subsystems with incompatible diagnostic needs impose a combinatorial tax on shared representations. Each subsystem alone needs only k_j cells. Sharing forces the common refinement, which can be exponentially larger. This is the representational cost of coordination.

Remark 10 (Splintering on the diagnostic quotient)**.** *Take three support points x_1, x_2, x_3 under one common test T.*

Subsystem A has optimal labels $y^A(x_1) = 1$, $y^A(x_2) = 0$, $y^A(x_3) = 0$. Its coarsest pure quotient is $\mathcal{P}_{A,+}^\star = \{\{x_1\}, \{x_2, x_3\}\}$.

Subsystem B has labels $y^B(x_1) = 0$, $y^B(x_2) = 1$, $y^B(x_3) = 0$. Its coarsest pure quotient is $\mathcal{P}_{B,+}^\star = \{\{x_2\}, \{x_1, x_3\}\}$.

These minimal zero-cost quotients do not coincide. If each subsystem separately minimises its own weighted forgetting cost, they converge to different partitions (Theorem 16). If they are forced to share one common zero-cost quotient, they must use the finer common refinement $\{\{x_1\}, \{x_2\}, \{x_3\}\}$.

Intuition. This is the diagnostic analogue of the cancer-like splintering in Proposition 10. Shared task pressure drives toward a common quotient. Split task pressure drives toward divergent locally sufficient quotients. When the locally sufficient quotients are incompatible, the only shared solution is the finest partition, which is expensive. This connects to federated learning (where client drift has the same structure) and to biological coordination failure (where cells under incompatible pressures lose collective identity).

APPLICATION COROLLARIES OF THE REGRET BRIDGE

These results are consequences of the exact reduction (Theorem 14) combined with partition combinatorics or prior Stack-Theoretic machinery. They first appeared in *Regret Is Weighted Forgetting*[112]. They are collected here rather than in the core regret bridge section because they are applications of the bridge, not part of the bridge itself.

[112] Bennett, M. T. (2026c). Regret is weighted forgetting

Proposition 27 (Regret ordering). *Define $M_1 \preceq_\rho M_2$ iff $K_\rho(M_1) \geq K_\rho(M_2)$. This partial order is consistent with the partition refinement lattice. If $\mathcal{C}_{M_2}|_{I_+}$ refines $\mathcal{C}_{M_1}|_{I_+}$, then $K_\rho(M_2) \leq K_\rho(M_1)$. The inequality is strict whenever the refinement splits at least one mixed cell on I_+ into subcells in which the minority class flips (the label achieving* $\min\{A, B\}$ *differs between subcells).*

Proof. If $\mathcal{C}_{M_2}$ refines $\mathcal{C}_{M_1}$, every cell $C_1 \in \mathcal{C}_{M_1}$ is partitioned into subcells $C_2^{(j)} \in \mathcal{C}_{M_2}$. By the cellwise formula, $\min\{A_{C_1}, B_{C_1}\} \geq \sum_j \min\{A_{C_2^{(j)}}, B_{C_2^{(j)}}\}$, since $\min\{a + a', b + b'\} \geq \min\{a, b\} + \min\{a', b'\}$ for non-negative reals. Summing gives $K_\rho(M_1) \geq K_\rho(M_2)$. For strictness, if a mixed cell C_1 with $A_{C_1} > B_{C_1}$ splits into subcells where $A_{C_2} > B_{C_2}$ but $A_{C_2'} < B_{C_2'}$, then $\min\{A_{C_1}, B_{C_1}\} = B_{C_2} + B_{C_2'} > B_{C_2} + A_{C_2'} = \min\{A_{C_2}, B_{C_2}\} + \min\{A_{C_2'}, B_{C_2'}\}$. □

Intuition. Finer representations have lower or equal K_ρ, and the ordering tells you exactly when refinement helps. This gives a regret-based partial order on abstractions that complements the bisimulation-metric order of Ferns, Panangaden, and Precup.

Corollary 10 (Unweighted forgetting on the informative region). *Fix $\gamma \in (0, \frac{1}{2}]$ and let $\mathcal{X}_\gamma := \{i : m_i \geq \gamma\}$. Define the unweighted informative-region forgetting cost $K_\gamma(M)$ by replacing $\mu_i \rho_i$ with μ_i in Definition 43 and restricting to $\mathcal{X}_\gamma$. Then with $c(\gamma) = 4\gamma/(1 + 2\gamma)$,*

$$c(\gamma) K_\gamma(M) \;\leq\; \inf_{\pi \text{ is } M\text{-based}} \sum_{i \in \mathcal{X}_\gamma} \mu_i \delta_i(\pi) \;\leq\; K_\gamma(M).$$

Proof. On $\mathcal{X}_\gamma$ we have $\rho_i \in [c(\gamma), 1]$. Apply Theorem 14 on the restricted support $\mathcal{X}_\gamma$. The resulting weighted deletion cost lies between $c(\gamma)$ times the unweighted deletion mass and the unweighted deletion mass itself. □

Intuition. If you only look at tests that matter by at least γ, exact weighted forgetting and ordinary forgetting differ only by fixed constants. This recovers the intuitive raw-forgetting picture on the informative region.

Corollary 11 (Free-energy floor from regret). *In the setting of Corollary 7, assume $Z = \sum_{i=1}^n w_i > 0$. Define the finite measure*

$$\nu(A) := \sum_{u_i \in A} w_i \qquad \text{on} \qquad U = \{u_1, \ldots, u_n\}.$$

For any admissible policy ϑ with $W(\vartheta) > 0$, define the weighted prior and posterior

$$p_U(u_i) := \frac{w_i}{Z}, \qquad q_\vartheta(u_i) := \frac{w_i \, \mathbf{1}\{u_i \in \operatorname{Ext}(\vartheta) \cap U\}}{W(\vartheta)}.$$

Let

$$F_2^\nu(\vartheta) := D_{\mathrm{KL},2}(q_\vartheta \| p_U).$$

Then

$$F_2^\nu(\vartheta) = \log_2 \frac{Z}{W(\vartheta)}.$$

If $W^\star(M) > 0$ (equivalently $K_\rho(M) < Z$), then the minimum weighted free-energy achievable by an admissible policy is

$$F_{2,\min}^\nu(M) = \log_2 \frac{Z}{W^\star(M)} = \log_2 \frac{Z}{Z - K_\rho(M)}.$$

If $W^\star(M) = 0$ (equivalently $K_\rho(M) = Z$), then every admissible policy has infinite weighted free-energy relative to p_U. In particular, whenever $0 < K_\rho(M) < Z$, the weighted free-energy floor is strictly above the zero-regret value 0 and is strictly increasing in $K_\rho(M)$.

Proof. By construction, q_ϑ is the normalised restriction of the measure ν to the retained unseen region $\mathrm{Ext}(\vartheta) \cap U$, while p_U is the normalised version of ν on all of U. Therefore, for q_ϑ-almost every u_i,

$$\frac{q_\vartheta(u_i)}{p_U(u_i)} = \frac{w_i/W(\vartheta)}{w_i/Z} = \frac{Z}{W(\vartheta)}.$$

The ratio is constant on the support of q_ϑ. Hence

$$F_2^\nu(\vartheta) = D_{\mathrm{KL},2}(q_\vartheta \| p_U) = \sum_{u_i} q_\vartheta(u_i) \log_2 \frac{q_\vartheta(u_i)}{p_U(u_i)} = \log_2 \frac{Z}{W(\vartheta)}.$$

This is strictly decreasing in $W(\vartheta)$. Therefore, if $W^\star(M) > 0$, the smallest attainable weighted free-energy is achieved by maximising $W(\vartheta)$, i.e. by $W^\star(M)$. Corollary 7 gives $W^\star(M) = Z - K_\rho(M)$, so

$$F_{2,\min}^\nu(M) = \log_2 \frac{Z}{W^\star(M)} = \log_2 \frac{Z}{Z - K_\rho(M)}.$$

If instead $W^\star(M) = 0$, then no admissible policy assigns positive q_ϑ-mass to any retained unseen region, so the weighted free-energy floor is $+\infty$. If $K_\rho(M) = 0$, the finite formula gives 0. If $0 < K_\rho(M) < Z$, the denominator decreases and the logarithm increases strictly. □

Intuition. This is the weighted analogue of the earlier uniform-counting free-energy proxy. Regret is exactly the weighted future mass that the representation forced you to discard, so the corresponding KL floor is the log-ratio between total diagnostic mass Z and retained mass $Z - K_\rho(M)$. When all weights are equal, this reduces to the familiar cardinality formula.

Bibliography

Barron, A. B. and Klein, C. (2016). What insects can tell us about the origins of consciousness. *PNAS*, 113(18):4900–4908.

Bennett, M. T. (2022a). Compression, the fermi paradox and artificial super-intelligence. In *Artificial General Intelligence*. Springer Nature.

Bennett, M. T. (2022b). Computable Artificial General Intelligence. *Under Review*.

Bennett, M. T. (2022c). Symbol emergence and the solutions to any task. In *Artificial General Intelligence*, Lecture Notes in Computer Science, pages 30–40, Cham. Springer.

Bennett, M. T. (2023a). Emergent causality and the foundation of consciousness. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 52–61. Springer. OUCI metadata page: https://ouci.dntb.gov.ua/en/works/7BoXJMW4/.

Bennett, M. T. (2023b). On the computation of meaning, language models and incomprehensible horrors. In *Artificial General Intelligence*. Springer Nature.

Bennett, M. T. (2023c). The optimal choice of hypothesis is the weakest, not the shortest. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 42–51. Springer.

Bennett, M. T. (2024a). Are biological systems more intelligent than artificial intelligence? In press, 2026, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds.

Bennett, M. T. (2024b). Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer.

Bennett, M. T. (2024c). Is complexity an illusion? In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer.

Bennett, M. T. (2025a). A formal theory of optimal learning with experimental results. *IJCAI*, pages 10967–10968.

Bennett, M. T. (2025b). *How To Build Conscious Machines*. PhD thesis, The Australian National University.

Bennett, M. T. (2025c). Lies, damned lies, and the orthogonality thesis. *Preprint*.

Bennett, M. T. (2025d). Optimal policy is weakest policy. In *Artificial General Intelligence*, volume 16057 of *Lecture Notes in Computer Science*, pages 43–56. Springer.

Bennett, M. T. (2025e). What is artificial general intelligence? In *Artificial General Intelligence*, volume 16057 of *Lecture Notes in Computer Science*, pages 30–42. Springer. Originally circulated as "What the F*ck Is Artificial General Intelligence?".

Bennett, M. T. (2026a). A mind cannot be smeared across time. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press.

Bennett, M. T. (2026b). No selves, no consciousness. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press.

Bennett, M. T. (2026c). Regret is weighted forgetting.

Bennett, M. T. (2026d). Spacetime bounds on consciousness.

Bennett, M. T. (2026e). Stack theory suite.

Bennett, M. T. and Maruyama, Y. (2022a). The artificial scientist: Logicist, emergentist, and universalist approaches to artificial general intelligence. In *Artificial General Intelligence*, Lecture Notes in Computer Science, pages 45–54, Cham. Springer.

Bennett, M. T. and Maruyama, Y. (2022b). Philosophical specification of empathetic ethical artificial intelligence. *IEEE Transactions on Cognitive and Developmental Systems*, 14(2):292–300.

Bennett, M. T., Welsh, S., and Ciaunica, A. (2024). *Why Is Anything Conscious?* Preprint.

Boltuc, P. (2012). The engineering thesis in machine consciousness. *Techné: Research in Philosophy and Technology*.

Catt, E., Grau-Moya, J., Hutter, M., Aitchison, M., Genewein, T., Deletang, G., Wenliang, L. K., and Veness, J. (2023). Self-predictive universal AI. In *Thirty-seventh Conference on Neural Information Processing Systems*.

Elija Perrier, M. T. B. (2026). Time, identity and consciousness in language model agents. In *Forthcoming in Proceedings of the AAAI 2026 Spring Symposium on Machine Consciousness: Integrating Theory, Technology, and Philosophy*, Burlingame, California, USA. AAAI Press.

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138.

Ganapathy, A. and Bennett, M. T. (2021). Cybernetics and the future of work. In *2021 IEEE 21CW*.

Hart, P. E., Nilsson, N. J., and Raphael, B. (1968). A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107.

Hutter, M. (2005). *Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability*. Springer, Berlin.

Kirk, D. (2007). Nvidia cuda software and gpu parallel computing architecture. In *Proceedings of the 6th International Symposium on Memory Management*, ISMM '07, page 103–104, New York, NY, USA. Association for Computing Machinery.

Leike, J. and Hutter, M. (2015). Bad universal priors and notions of optimality. In Gr"unwald, P., Hazan, E., and Kale, S., editors, *Proceedings of The 28th Conference on Learning Theory*, volume 40 of *Proceedings of Machine Learning Research*, pages 1244–1259, Paris, France. PMLR.

McAllester, D. A. (1999). Pac-Bayesian model averaging. In *Proceedings of the 12th Annual Conference on Computational Learning Theory (COLT)*, pages 164–170. ACM.

Merker, B. (2005). The liabilities of mobility: A selection pressure for the transition to consciousness in animal evolution. *Consciousness and Cognition*, 14(1):89–114. Neurobiology of Animal Consciousness.

Merker, B. (2007). Consciousness without a cerebral cortex: a challenge for neuroscience and medicine. *Behavioral and Brain Sciences*, 30(1):63–81.

Meurer, A., Smith, C., Paprocki, M., Čertík, O., Kirpichev, S., Rocklin, M., Kumar, A., Ivanov, S., Moore, J., Singh, S., Rathnayake, T., Vig, S., Granger, B., Muller, R., Bonazzi, F., Gupta, H., Vats, S., Johansson, F., Pedregosa, F., and Scopatz, A. (2017). Sympy: Symbolic computing in python. *PeerJ Computer Science*, 3:e103.

Paszke, A. et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*.

Pearl, J. and Mackenzie, D. (2018). *The Book of Why: The New Science of Cause and Effect*. Basic Books, Inc., New York, 1st edition.

Perrier, E. and Bennett, M. T. (2025). Position: Stop acting like language model agents are normal agents.

Solé, R., Seoane, L. F., Pla-Mauri, J., Bennett, M. T., Hochberg, M. E., and Levin, M. (2026). Cognition spaces: natural, artificial, and hybrid.

Wong, M. L., Cleland, C. E., Arend, D., Bartlett, S., Cleaves, H. J., Demarest, H., Prabhu, A., Lunine, J. I., and Hazen, R. M. (2023). On the roles of function and selection in evolving systems. *Proceedings of the National Academy of Sciences*, 120(43):e2310223120.